

An exact gamma constant for an Apéry-like sequence

A279619: three independent proofs, all-orders asymptotics,
and a recessive companion

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Merged from two independent manuscripts of batch 73O2 of ProveIt's incoming-reports intake (manuscripts 58 and 56), with a third derivation from manuscript 32 of batch 77P5 (Sections 15 and 16) and Hankel-positivity results from manuscript 07 of batch 97 (Section 17)

$$a_n \sim \frac{\sqrt{3\pi}}{\Gamma(1/7)\Gamma(2/7)\Gamma(4/7)} \frac{27^n}{n^{3/2}}.$$

The leading constant is evaluated exactly, the displayed OEIS asymptotic is proved, and the expansion is extended to arbitrary fixed order with a rigorous remainder. A rationally normalized companion exposes an exponentially smaller alternating scale.

Abstract. We study the level-seven Apéry-like sequence A279619, defined by a second-order polynomial recurrence. Its OEIS entry displays a conjectural asymptotic with a numerical constant. We prove that asymptotic and identify the constant as a reciprocal product of three gamma values. Three independent proofs of the constant are printed. The first: a rational parameter removes the square roots in the known hypergeometric generating function, making its singular connection coefficient explicit, and classical elliptic-period and Ramanujan identities reduce this coefficient to its gamma-product form. The second: the boundary value of the generating function is a Dedekind eta product at the Fricke fixed point $i/\sqrt{7}$, evaluated by the Chowla–Selberg formula at discriminant -7 , and the singular coefficient is then read off from Hirschhorn's constant for the square sequence A183204. The third, from a manuscript added later, differentiates the Fricke involution of the weight-one theta series at its fixed point to obtain the singular coefficient directly, and evaluates the same eta value from an Epstein zeta function with an explicit conductor-two factor. A functional equation gives a second exact sum, at $z = 1/125$. Frobenius expansion and singularity analysis give all algebraic correction terms, with rational coefficients and fixed-order error bounds, together with ratio, root and critical-tail expansions. We also construct a companion continued fraction, prove exact rational bracketing, and determine the amplitude and every correction term of its recessive solution. Applications include convolution powers, A183204, eventual log-convexity, and Lambert- W inversion with a separate treatment of integer rounding. A fourth manuscript, added later, proves strict log-convexity at every index and strict total positivity of order exactly five of the Hankel matrix (a_{i+j}) ; the report adds that no tail of the sequence is a Stieltjes moment sequence. Exact symbolic certificates and rational enclosures accompany numerical experiments. Classical inputs, previously calculated terms, sequence-specific deductions, and unsettled questions are explicitly distinguished.

Shipped with this report: the programs of manuscripts 58, 56, 32 and 07, their recorded outputs, rational and polynomial certificates, numerical tables, and the source audits of manuscripts 58 and 07 (listed in the report's `README.md`). These are conventional mathematical proofs with computer-algebra checks, not Lean-certified theorems. Global priority is not claimed.

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1 The target and the scope of the contribution

The sequence begins

$$1, 2, 22, 336, 6006, 117348, 2428272, 52303680, 1160427510, \dots$$

and satisfies

$$(n+1)^2 a_{n+1} = (26n^2 + 13n + 2)a_n + 3(3n-1)(3n-2)a_{n-1}, \quad n \geq 1, \quad (1)$$

with $a_0 = 1, a_1 = 2$. (Manuscript 56 writes the last coefficient as $27n^2 - 27n + 6$.) Throughout, set

$$A(z) = \sum_{n \geq 0} a_n z^n, \quad \rho = 1/27, \quad \mathcal{G}_7 = \Gamma(1/7)\Gamma(2/7)\Gamma(4/7). \quad (2)$$

This is A279619; $A(z)^2$ is the generating function of A183204 [1, 2]. Its connections with modular forms, hypergeometric functions, and another Apéry-like sequence make it a structured target rather than an isolated numerical curiosity.

1.1 What is already known

The OEIS entry consulted on October 1, 2026 labels its leading asymptotic as conjectural and supplies a numerical constant. O’Brien’s thesis [3, §8.4, pp. 63–67] obtained the recurrence and the level-seven modular parametrization, estimates that constant and formally calculates the first three correction terms. Those three terms are *not new here*. The hypergeometric representation is also recorded in OEIS in code attributed to Václav Kotešovec in 2018 [1]. We prove and rationalize that representation before using it analytically. The leading asymptotic of the square sequence A183204, with a complete expansion, is Hirschhorn’s theorem [4]; it is used, and its leading constant re-derived, in the second proof (Section 6).

The contribution is not a new general method of asymptotic analysis. It is an exact connection-constant evaluation, a rigorous all-orders fixed-order expansion, and an explicit recessive companion with a proved error amplitude. Standard machinery is used to close the displayed asymptotic question.

Item	Status in this report
Recurrence and hypergeometric function	Existing input; the latter receives a rational differential-equation certificate.
Leading scale and first three corrections	Previously conjectured or formally calculated; now given a remainder proof.
Gamma-product constant	Exact sequence-specific evaluation deduced from classical special values, by two independent routes.
Sum at $z = 1/125$	Deduced, in manuscript 56, from the constant and a functional equation.
Companion continued fraction	Constructed with exact bracketing and all-orders recessive asymptotics; global priority not asserted.
Convolution and inverse expansions	Deductions with explicit formulas and integer-rounding bounds; the inverse expansion is an instance of the repository’s inversion apparatus (Section 11).
Lucas congruences	Not resolved or used here.

A targeted search did not locate an earlier proof of this particular constant formula. This is not an exhaustive priority search. In particular, recent work of Cohen and Zudilin on continued fractions for Chowla–Selberg quotients [14] shows that the broader area is active. We do not claim a first continued fraction for a gamma quotient, an identification of the companion limit with such a quotient, or an irrationality measure. Manuscript 56’s own, independent audit is printed in Section 14.1.

1.2 Relation to ProveIt

The repository’s Analysis/Transseries project separates asymptotic scales, flatness, power–logarithmic expressions, and inversion from the Fabius-function development [15]. This report follows that separation: an analytic existence proof precedes exact coefficient recursions and inverse expansions. No unverified repository theorem is needed. The inspected snapshot was a5ad4ac2e2d4e642a6c015836a70492d40fcbc71; inspection was not an exhaustive review of archived reports.

[Merge note, batch 73O2: The Lambert- W inverse of Section 11 is the instance $a = \log 27$, $\beta = -3/2$ of Theorem t2:thm:balanced-inverse of the repository’s volume *Combinatorial Transseries and Their Inverses* [16], which neither manuscript cites; it is printed as an application, with no novelty claimed for it. Manuscript 58’s rounding proposition is the sequence-specific part.]

1.3 Two sources, one report

This report merges two manuscripts that reached ProveIt on the same day, October 1, 2026, in batch 73O2 of its incoming-reports intake, and that were written independently: manuscript 58 (*An exact gamma constant for an Apéry-like sequence*) and manuscript 56 (*Exact asymptotics of OEIS A279619*). Manuscript 58 inspected the repository at a commit made before manuscript 56 arrived, and manuscript 56 cites neither manuscript 58 nor any repository revision. Both prove the same leading constant, the same boundary value $A(\rho)$, and the same rational corrections (they agree as rationals through order 10, the last order manuscript 56 prints; manuscript 58 records them to order 12). They do so by *different* proofs. Two independent derivations of the same constant arriving on the same day is mild corroboration, not a substitute for review.

Manuscript 58 is the base text: Sections 1–5 and 7–12 are its text, with its labels prefixed, its hard-coded section numbers replaced by references, and manuscript 56’s additions inserted as marked passages. Manuscript 56 contributes the second proof of the constant (Section 6), the sum at $1/125$ and the critical-tail, ratio and root corollaries (Section 2), the integer form of the Frobenius recurrence and the coefficients $c_6, \dots, c_{10}$ (Section 7), its inverse remark (Section 11), its numerical tables (Section 12), ten research problems (Section 13.13), its novelty audit (Section 14.1), two derivations (Appendix B) and its OEIS proposal (Appendix C). Results proved in both manuscripts are printed once and marked “proved in both sources” (for Theorem 2.1, now “proved in all three sources”). Section 15 records the provenance.

[Write note, 2026-10-02, batch 77P5: A third manuscript, manuscript 32 of batch 77P5 (*The exact asymptotic constant for A279619 and its fixed weight family*, dated October 2, 2026), reached ProveIt a day later. It names no repository revision and cites neither this report nor manuscripts 56 and 58, and it proves Theorem 2.1 a third time, by a third route: the singular coefficient directly at the Fricke fixed point, and the eta value by an explicit Epstein-zeta calculation. It contains no theorem that is not already in this report. Its route is printed in Section 16, placed after the provenance section so that no existing section, theorem or equation number changes, with its own notation table (Table 7); its explicit convolution generator and its all-weight rounding brackets are Remarks 10.2 and 11.3.] [Write note, 2026-10-04, batch 97G: A fourth manuscript, manuscript 07 of batch 97 (*A Sharp Positivity Threshold for OEIS A279619*, dated October 4, 2026), was written with this report in hand (it inspected revision a21208b3f) and answers research question R6 of

Section 13.13: the sequence is strictly log-convex at every index, and its Hankel matrix (a_{i+j}) is strictly totally positive of order five but not totally nonnegative of order six, with an exact truncated-moment obstruction. It is printed as Section 17, after Section 16 so that no existing number changes, with its own notation table (Table 8). Manuscript 07 calls the sequence c_n and its corrections a_j , this report's letters swapped; it is rewritten in this report's notation. This write adds Subsection 17.10: the generating function is singular at $z = -1$, so no tail of (a_n) is a Stieltjes moment sequence.]

1.4 Notation of the two manuscripts

The two manuscripts use several letters for different objects. This report uses manuscript 58's notation throughout; manuscript 56's material has been rewritten in it. The dictionary, with the tempting false readings, is Table 1.

Table 1: Notation of manuscripts 58 and 56.

Object	This report (= 58)	Manuscript 56	Caution
the sequence A279619	a_n	c_n	here c_j are <i>corrections</i>
its generating function	$A(z)$	$F(x)$	here F is ${}_2F_1(\frac{1}{12}, \frac{5}{12}; 1; \cdot)$
gamma product	$\mathcal{G}_7$	$\mathcal{G}_7$	same
leading constant	C	C_c	—
boundary value $A(\rho)$	$A_\star$	$F(1/27), A(0)$	56's $A(s)$ is not the GF
corrections	c_j	α_j	equal as rationals
analytic part at ρ	$U(s)$	$A(s)$	—
singular coefficient	B	B_0	—
singular Frobenius series	$V(s) = \sum v_j s^j$	$B(s) = \sum b_m s^m$	here b_n is the <i>companion</i>
square sequence (A183204)	$t_n = a_n^{(2)}, A(z)^2$	$t_n, G(x) = F(x)^2$	G is not $\mathcal{G}_7$
its leading constant	$C_t = 2CA_\star$	C_t	—
log 27	λ	L	here L is the companion limit
entropy function	$\text{Ent}(u)$	$H(u)$	here H is Clausen's ${}_3F_2$
Laplace variable, weight	$\xi = k/n, \kappa(\xi)$	$a, g(a)$	a is not a term
scaled term	$R_n = a_n n^{3/2}/27^n$	Q_n	here $Q_n, Q(m)$ differ
critical partial sum, shift	$\Sigma_N, \nu = N + 1$	S_N, q	here S_N sums $F(q_0)$
continuous inverse	$x_0(Y)$	$n_0(y)$	—
formal series at infinity	$S_{27}(t)$	$\mathcal{A}(t)$	—
Weber invariant power	$u = f_2(\tau_0)^{24}$	u	here u_n is a solution
level-7 Hauptmodul	$w_7(\tau)$	$w_7(\tau)$	here w is the rational parameter
modular j -invariant	$j(\tau)$	$j(\tau)$	here $j(m)$ is a pullback

No normalization changes: $C = C_c, B = B_0$ (both with the singular Frobenius series normalized to 1 at $s = 0$) and $\lambda = L_{56}$. Manuscript 56's index range " $M \geq 1$, error $O(n^{-M})$, sum to $M - 1$ " is manuscript 58's " $M \geq 0$, error $O(n^{-M-1})$, sum to M " shifted by one.

2 Principal results

Theorem 2.1 (Dominant asymptotics and the exact constant; proved in all three sources). *Put*

$$C = \frac{\sqrt{3\pi}}{\mathcal{G}_7}, \quad A_\star = \frac{3\mathcal{G}_7}{8\pi^2}. \quad (3)$$

Then $A(\rho) = A_\star$. For every fixed $M \geq 0$,

$$a_n = C \frac{27^n}{n^{3/2}} \left(\sum_{j=0}^M \frac{c_j}{n^j} + O_M(n^{-M-1}) \right), \quad (4)$$

where the c_j are uniquely determined rational numbers with $c_0 = 1$. In particular,

$$\begin{aligned} c_1 &= -\frac{215}{1008}, & c_2 &= -\frac{1265}{290304}, & c_3 &= \frac{4683055}{877879296}, \\ c_4 &= \frac{924908071}{3539609321472}, & c_5 &= -\frac{6088275239005}{3567926196043776}. \end{aligned} \quad (5)$$

An exact triangular algorithm computes every coefficient.

The theorem also evaluates a convergent boundary sum:

$$\sum_{n=0}^{\infty} \frac{a_n}{27^n} = \frac{3\Gamma(1/7)\Gamma(2/7)\Gamma(4/7)}{8\pi^2}, \quad CA_\star = \frac{3\sqrt{3}}{8\pi^{3/2}}. \quad (6)$$

The gamma product in the boundary value cancels against its reciprocal in the coefficient amplitude. Numerically $C = 0.0955223052681267146513079107870296256727946666510071798669 \dots$ and $A_\star = 1.221131097467566669284224585685993462375827056379 \dots$; the two manuscripts' decimal values agree in every printed digit.

[Merge note, batch 73O2: Manuscript 58 proves Theorem 2.1 in Sections 3–7; manuscript 56 proves it in Section 6 and in its part of Section 7. Manuscript 56's statement is the same theorem with $\alpha_j = c_j$ and the shifted index range of Section 1.4; it lists the same five coefficients.] [Write note, 2026-10-02, batch 77P5: Manuscript 32 proves the theorem a third time (Section 16), with $p_m = c_m$ in its notation; it prints $c_1, \dots, c_6$, equal as rationals to (5) and (76).] [Write note, 2026-10-04, batch 97G: Manuscript 07 proves the expansion part of the theorem (existence of $C > 0$ and of the full expansion with rational c_j) once more, by a recurrence-only stable-ratio argument without singularity analysis (Theorem 17.11, printed as a marked second route, Remark 17.8). It prints $c_1, \dots, c_4$, equal to (5), and takes the value of C from this theorem.]

Theorem 2.2 (Exact critical and subcritical sums; manuscript 56). *The generating function has the exact values*

$$A(1/27) = \frac{3\mathcal{G}_7}{8\pi^2}, \quad A(1/125) = \sum_{n=0}^{\infty} \frac{a_n}{125^n} = \frac{5\mathcal{G}_7}{16\pi^2}. \quad (7)$$

Numerically, $A(1/125) = 1.017609247889638891070187154738327885313189213649 \dots$

The first value is (6). The second is proved in Section 6.6 from the first and a functional equation; it appears only in manuscript 56.

What is resolved (manuscript 56). The decimal constant in the OEIS asymptotic conjecture is exactly the gamma quotient in (3). More is true: the complete expansion exists to arbitrary algebraic order, all its normalized coefficients are rational, the singular connection coefficient is explicit, and the critical value of the generating function is a closed gamma product.

2.1 Immediate corollaries

The all-orders form gives several useful consequences without further modular input. The three corollaries below are manuscript 56's.

Corollary 2.3 (Ratio asymptotics and eventual log-convexity). *As $n \rightarrow \infty$,*

$$\frac{a_{n+1}}{a_n} = 27 \left(1 - \frac{3}{2n} + \frac{2105}{1008n^2} - \frac{18815}{7056n^3} + \frac{3425255}{1053696n^4} + O(n^{-5}) \right). \quad (8)$$

Consequently $a_{n+1}/a_n \rightarrow 27$, and (a_n) is strictly log-convex for all sufficiently large n .

Proof. Insert (4) into the quotient. The difference of successive ratios is

$$\frac{a_{n+2}}{a_{n+1}} - \frac{a_{n+1}}{a_n} = \frac{81}{2n^2} + O(n^{-3}) > 0$$

for all sufficiently large n . Increasing ratios are equivalent to strict log-convexity. □

[Write note, 2026-10-04, batch 97G: The qualifier “for all sufficiently large n ” is no longer needed: Proposition 17.5 (manuscript 07) proves a_{n+1}/a_n strictly increasing for every $n \geq 0$, that is $a_{n-1}a_{n+1} > a_n^2$ for all $n \geq 1$, by elementary ratio barriers, and Theorem 17.1 strengthens this to strict total positivity of order five of (a_{i+j}) . The coefficients of (8) are $\omega_j/27$, $j \leq 4$, of Subsection 17.5, which continues them to order eleven, with a remainder proof (Proposition 17.10).] Manuscript 58 proves the first three terms of (8) and eventual log-convexity through second differences of $\log a_n$ (Section 7.4); the two increments are consistent, $81/(2n^2) \div 27 = 3/(2n^2)$.

Corollary 2.4 (Root and logarithmic scales). *With $\lambda = \log 27$,*

$$\log a_n = n\lambda - \frac{3}{2} \log n + \log C - \frac{215}{1008n} + O(n^{-2}), \quad (9)$$

$$a_n^{1/n} = 27 \exp \left(\frac{\log C - \frac{3}{2} \log n}{n} + O(n^{-2}) \right). \quad (10)$$

The first line is the case $M = 1$ of manuscript 58's logarithmic expansion (77).

Corollary 2.5 (Critical-tail asymptotics). *Let $\Sigma_N = \sum_{n=0}^N a_n/27^n$ and $\nu = N + 1$. Then*

$$A(1/27) - \Sigma_N = C \left(2\nu^{-1/2} + \frac{541}{1512} \nu^{-3/2} + \frac{2411}{145152} \nu^{-5/2} + O(\nu^{-7/2}) \right). \quad (11)$$

In particular, convergence at the critical point is only of order $N^{-1/2}$.

Proof. Sum (4) termwise and use the standard Hurwitz-zeta expansion for $\zeta(s, N + 1)$. □

[Merge note, batch 7302: The intake re-derived 541/1512 and 2411/145152 by the Euler–Maclaurin formula, and checked that the partial sum through $N = 5000$ plus the three-term tail matches $3\mathcal{G}_7/(8\pi^2)$ to 4.9×10^{-16} , consistent with the stated $O(v^{-7/2})$.]

Theorem 2.6 (An exact recessive companion; manuscript 58). *Let $b_0 = 0$, $b_1 = 1$, and impose (1) on b_n for $n \geq 1$. Then the limit*

$$L = \lim_{n \rightarrow \infty} \frac{b_n}{a_n} = 0.46721176888437312394211005141309126172410854995971 \dots \quad (12)$$

exists and satisfies

$$\frac{b_{2m}}{a_{2m}} < L < \frac{b_{2m+1}}{a_{2m+1}}, \quad m \geq 0. \quad (13)$$

For $e_n = b_n - La_n$ and every fixed $M \geq 0$,

$$e_n = (-1)^{n+1} \frac{\sqrt{3}}{56\pi C} n^{-3/2} \left(\sum_{j=0}^M \frac{d_j}{n^j} + O_M(n^{-M-1}) \right), \quad (14)$$

where $d_0 = 1$ and

$$d_1 = -\frac{57}{112}, \quad d_2 = \frac{655}{3584}, \quad d_3 = -\frac{13365}{401408}, \quad d_4 = -\frac{3627123}{179830784}. \quad (15)$$

The d_j are rational and computable to arbitrary order. Consequently,

$$L - \frac{b_n}{a_n} \sim (-1)^n \frac{\sqrt{3}}{56\pi C^2} 27^{-n}. \quad (16)$$

The two scales are $27^n n^{-3/2}$ and $(-1)^n n^{-3/2}$. The second is exponentially small *relative* to the first, not in absolute magnitude. We do not assign an unproved canonical subdominant coefficient to the asymptotic expansion of a_n itself.

3 From the recurrence to a controlled singularity

3.1 Positivity, growth, and the differential equation

The recurrence immediately gives $a_n > 0$. Induction also gives $a_n \leq 27^n$, since

$$26n^2 + 13n + 2 + \frac{3(3n-1)(3n-2)}{27} = 27n^2 + 12n + \frac{20}{9} < 27(n+1)^2.$$

Thus A is analytic in $|z| < \rho$. Multiplying the recurrence by z^n and summing, including $a_1 = 2a_0$, gives

$$zD(z)A'' + (1 - 39z - 54z^2)A' - (2 + 6z)A = 0, \quad D(z) = (1 - 27z)(1 + z). \quad (17)$$

The coefficient of z^n on the left is the recurrence residual, so the analytic solution normalized by $A(0) = 1$ is unique. Outside $0, \rho, -1$, the equation is ordinary; analytic continuation along paths avoiding these points follows from linear differential-equation theory. (Manuscript 56 derives the same equation, written with $zD(z) = z(1 - 26z - 27z^2)$, in Appendix B. Its positive singularity ρ is separated from the other nonzero finite singularity -1 by a factor of 27 in modulus.)

3.2 The local exponents

In $s = 1 - 27z$, the equation becomes

$$s(1-s)(28-s)y'' + (14-43s+2s^2)y' - \frac{20-2s}{9}y = 0. \quad (18)$$

The indicial equation is $14\alpha(2\alpha-1) = 0$. Its exponents $0, 1/2$ have nonintegral difference, hence Frobenius theory gives a convergent local expansion

$$A(z) = U(s) + B\sqrt{s}V(s), \quad V(0) = 1, \quad (19)$$

where U, V are analytic at 0. Branches are positive for $0 < z < \rho$. Below we show $B < 0$, so the singularity is not removable.

The self-adjoint form provides a useful sign check:

$$(z\sqrt{D(z)}A'(z))' = \frac{2+6z}{\sqrt{D(z)}}A(z), \quad 0 < z < \rho. \quad (20)$$

Frobenius theory makes $A(\rho)$ finite. Positivity of the right side implies that the limiting flux is positive, whereas (19) identifies it as $-\frac{1}{2}\sqrt{28/27}B$.

There is no other singularity on $|z| = \rho$. The chosen solution extends to a dented disk of radius greater than ρ , cut to the right from ρ . This supplies the global analytic condition for singularity analysis, not just a formal local expansion [12, Chapter VI].

4 A rational certificate for the generating function

Set

$$w = \sqrt{\frac{1-27z}{1+z}}, \quad z = \frac{1-w^2}{27+w^2}. \quad (21)$$

The points $0, \rho$ correspond to $w = 1, 0$. Define

$$\begin{aligned} T(w) &= 15w^2 + 96w + 85, \\ P(w) &= \left(\frac{7(w^2 + 27)}{T(w)} \right)^{1/4}, \\ J(w) &= \frac{1728(1-w)^7(1+w)}{(w^2 + 27)T(w)^3}, \quad F(x) = {}_2F_1\left(\begin{matrix} 1/12, 5/12 \\ 1 \end{matrix}; x\right). \end{aligned} \quad (22)$$

Fourth roots are positive on $[0, 1]$.

Proposition 4.1. *As analytic germs near $w = 1$, and throughout $0 \leq w \leq 1$,*

$$A\left(\frac{1-w^2}{27+w^2}\right) = P(w)F(J(w)). \quad (23)$$

Proof. The equation for $y(w) = A(z(w))$ is $y'' + py' + qy = 0$, where

$$p = \frac{2w}{(w-1)(w+1)}, \quad q = -\frac{16(w^2-15)}{(w-1)(w+1)(w^2+27)^2}. \quad (24)$$

Let $h = P'/P = \frac{1}{4}(2w/(w^2 + 27) - T'/T)$. The hypergeometric equation [13, §15.10] is

$$x(1-x)F'' + (1 - \frac{3}{2}x)F' - \frac{5}{144}F = 0. \quad (25)$$

Substitution of $y = PF(J)$, followed by eliminating F'' , reduces the claim to two rational identities:

$$\begin{aligned} J'' + (2h + p)J' + (J')^2 \frac{3J/2 - 1}{J(1-J)} &= 0, \\ h' + h^2 + ph + q + \frac{5}{144} \frac{(J')^2}{J(1-J)} &= 0. \end{aligned} \quad (26)$$

Clearing denominators and expanding proves both; the attached code checks them identically in $\mathbb{Q}(w)$. This is an exact certificate rather than a fit to finitely many coefficients. At $w = 1$, $P = 1$, $J = 0$, and $z'(1) \neq 0$. The right side is analytic there and normalized to 1. Uniqueness at $z = 0$ proves the germ identity. For $0 \leq w < 1$,

$$\frac{J'}{J} = -\frac{7}{1-w} + \frac{1}{1+w} - \frac{2w}{w^2 + 27} - 3\frac{30w + 96}{T(w)} < 0.$$

Thus $0 \leq J(w) \leq J(0) = 64/85^3 < 1$, and every expression continues regularly along the interval, proving the full claim. $\square$

Differentiation at a singular z -value has now become ordinary differentiation at the regular parameter value $w = 0$.

5 Evaluating the connection constant (first proof)

5.1 An explicit hypergeometric expression

Put

$$q_0 = \frac{64}{85^3}, \quad P_0 = \left(\frac{189}{85}\right)^{1/4}. \quad (27)$$

As $s \rightarrow 0$,

$$w = \sqrt{\frac{27s}{28-s}} = \sqrt{\frac{27}{28}}\sqrt{s}(1 + O(s)).$$

At $w = 0$ we have $P'/P_0 = -24/85$ and $J'/q_0 = -798/85$. Hence

$$B = -\sqrt{\frac{27}{28}} \frac{P_0}{85} (24F(q_0) + 798q_0F'(q_0)) < 0. \quad (28)$$

Positivity follows from the positive hypergeometric coefficients. Coefficient transfer for $\sqrt{1-27z}$ and $\Gamma(-1/2) = -2\sqrt{\pi}$ gives

$$C = -\frac{B}{2\sqrt{\pi}} = \sqrt{\frac{27}{28}} \frac{P_0}{170\sqrt{\pi}} (24F(q_0) + 798q_0F'(q_0)). \quad (29)$$

This already gives an explicit positive constant. The following special-value reductions explain its unexpectedly simple form.

5.2 Ramanujan's identity removes the derivative

The relevant Clausen identity is

$$F(q)^2 = H(q) := \sum_{n \geq 0} \frac{(1/6)_n (1/2)_n (5/6)_n}{(n!)^3} q^n. \quad (30)$$

It follows directly from differential equations: the symmetric square of (25) is

$$[\vartheta^3 - q(\vartheta + 1/6)(\vartheta + 1/2)(\vartheta + 5/6)]H = 0, \quad \vartheta = q \frac{d}{dq},$$

whose analytic solution with $H(0) = 1$ is unique. The rational coefficients of this symmetric-square identity are checked in the code.

The classical level-seven Ramanujan identity, reproduced and obtained by the construction in [6, Eq. (98)], is

$$\sum_{n \geq 0} (133n + 8) \frac{(1/6)_n (1/2)_n (5/6)_n}{(n!)^3} q_0^n = \frac{85\sqrt{255}}{54\pi}. \quad (31)$$

The source uses the coefficient $(6n)! / ((3n)!(n!)^3)$, which is 1728^n times the coefficient here, and $1728/255^3 = 64/85^3$. This normalization check is essential. Equation (31) is a cited established identity, not an equality inferred from numerical recognition. Using (30), it becomes

$$8F(q_0)^2 + 266q_0F(q_0)F'(q_0) = \frac{85\sqrt{255}}{54\pi}.$$

Multiply by $3/F(q_0)$ and substitute into (29). Since $A_\star = P_0F(q_0)$ by Proposition 4.1, this yields

$$C = \frac{3\sqrt{3}}{8\pi^{3/2}P_0F(q_0)} = \frac{3\sqrt{3}}{8\pi^{3/2}A_\star}. \quad (32)$$

5.3 An elliptic period supplies the gamma product

Let

$$m_7 = \frac{8 - 3\sqrt{7}}{16}, \quad k_7 = \sqrt{m_7}, \quad K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - k^2 \sin^2 \theta}}. \quad (33)$$

Here K takes the *modulus* k , not the parameter $m = k^2$. The classical singular-period evaluation is

$$K(k_7) = \frac{\mathcal{G}_7}{4\pi 7^{1/4}}. \quad (34)$$

This is an established complex-multiplication special value [7, 8], not a new identity claimed here. Appendix A explicitly reduces the beta-function form in Zucker's table to this normalization.

The required hypergeometric transformation is

$$F\left(\frac{27m^2(1-m)^2}{4(1-m+m^2)^3}\right) = (1-m+m^2)^{1/4} {}_2F_1\left(\begin{matrix} 1/2, 1/2 \\ 1 \end{matrix}; m\right). \quad (35)$$

For a direct verification put $Q(m) = 1 - m + m^2$ and $j(m) = 27m^2(1-m)^2/(4Q(m)^3)$. Then $Q^{-1/4}F(j)$ satisfies

$$m(1-m)y'' + (1-2m)y' - \frac{1}{4}y = 0.$$

Substituting and clearing denominators proves two rational identities, exactly as in (26). Both solutions are analytic at 0 and equal to 1, proving the transformation near 0. Continuation on $[0, m_7]$ proves the value used here. The program verifies both identities.

At $m = m_7$,

$$m_7(1 - m_7) = 1/256, \quad Q(m_7) = 255/256, \quad j(m_7) = q_0.$$

Euler's integral gives ${}_2F_1(1/2, 1/2; 1; m_7) = 2K(k_7)/\pi$. Therefore

$$\begin{aligned} A_\star &= P_0 F(q_0) = \left(\frac{189}{85} \frac{255}{256} \right)^{1/4} \frac{2K(k_7)}{\pi} \\ &= \frac{3 \cdot 7^{1/4}}{4} \frac{2}{\pi} \frac{\mathcal{G}_7}{4\pi 7^{1/4}} = \frac{3\mathcal{G}_7}{8\pi^2}. \end{aligned}$$

Combining this with (32) proves $C = \sqrt{3\pi}/\mathcal{G}_7$. Since the coefficients are positive, the finite limit at ρ also gives the boundary sum (6), by monotone convergence.

Remark 5.1 (Dependencies of the exact evaluation). The two non-elementary special values are (31) and (34); both are classical inputs. The recurrence manipulations, rational pullbacks, connection-coefficient extraction, and asymptotic deductions are proved here. We do not purport to give a new derivation of Chowla–Selberg theory or of Ramanujan's general $1/\pi$ formulas.

6 Second proof: the square sequence and a CM evaluation (manuscript 56)

This section is manuscript 56's proof of $A_\star = 3\mathcal{G}_7/(8\pi^2)$ and $C = \sqrt{3\pi}/\mathcal{G}_7$, rewritten in this report's notation (Section 1.4). It is independent of Sections 4 and 5: it uses neither the ${}_2F_1$ pullback, nor Ramanujan's series, nor the elliptic period. Its classical inputs are O'Brien's modular parametrization, the Chowla–Selberg formula at discriminant -7 , and Hirschhorn's asymptotic for A183204, whose leading constant it re-derives.

The proof combines four structures already present around this level-seven sequence: the square identity $A(z)^2 = \sum t_n z^n$ with the Apéry-like sequence A183204, a one-dimensional Laplace analysis of the binomial sum for t_n , O'Brien's modular parametrization, and a Chowla–Selberg evaluation of a Dedekind eta value of discriminant -7 . They come from two apparently different descriptions of the same object: a square root of an Apéry-like generating function and a modular parametrization on $X_0(7)$.

6.1 The square identity

Let

$$A(z)^2 = \sum_{n \geq 0} t_n z^n, \quad t_n = a_n^{(2)}. \quad (36)$$

O'Brien's level-seven construction, equivalently the relation recorded in OEIS, identifies (t_n) with A183204, one of the sporadic Apéry-like sequences [9]. (Manuscript 58 checks the identification through the symmetric-square equation (93).) Its coefficients have the finite binomial sum

$$t_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{2k}{n} \binom{n+k}{k}, \quad (37)$$

where terms with $2k < n$ vanish. The initial values are

$$1, 4, 48, 760, 13840, 273504, \dots$$

[Merge note, batch 7302: Manuscript 58 checks the convolution against the different binomial form $\sum_k \binom{n}{k}^2 \binom{n+k}{n} \binom{2n-k}{n}$; the intake checked that both forms equal $[z^n]A(z)^2$ for $n \leq 40$.] Hirschhorn proved a complete asymptotic expansion for this sequence [4]. We rederive its leading constant in Section 6.3, because that constant is the bridge from the known square to the previously unknown square root.

6.2 The modular parametrization

Let η denote Dedekind's eta function [13, Ch. 23],

$$\eta(\tau) = e^{\pi i \tau / 12} \prod_{m \geq 1} (1 - e^{2\pi i m \tau}), \quad \Im \tau > 0. \quad (38)$$

O'Brien's parametrization may be written

$$w_7(\tau) = \left(\frac{\eta(7\tau)}{\eta(\tau)} \right)^4, \quad (39)$$

$$X_7(\tau) = \frac{w_7(\tau)}{1 + 13w_7(\tau) + 49w_7(\tau)^2}, \quad (40)$$

$$X_7(\tau) = \left(\frac{\eta(\tau)\eta(7\tau)}{z_7(\tau)} \right)^3, \quad A(X_7(\tau)) = z_7(\tau). \quad (41)$$

The last identity is initially a q -series identity near the cusp and then extends by analytic continuation. It is the crucial reason the critical value of A can be reduced to a CM eta product.

6.3 The square sequence by Laplace's method

We now obtain the leading asymptotic of t_n directly from (37). This gives a self-contained derivation of the one published constant from which the new constant will be extracted.

For $\xi \in (1/2, 1)$ and $k = \xi n$, Stirling's formula gives, uniformly on closed subintervals of $(1/2, 1)$,

$$\binom{n}{k}^2 \binom{2k}{n} \binom{n+k}{k} = n^{-2} \kappa(\xi) e^{n\Phi(\xi)} (1 + O(n^{-1})), \quad (42)$$

where, with $\text{Ent}(u) = -u \log u - (1-u) \log(1-u)$,

$$\Phi(\xi) = 2 \text{Ent}(\xi) + 2\xi \text{Ent}\left(\frac{1}{2\xi}\right) + (1+\xi) \text{Ent}\left(\frac{\xi}{1+\xi}\right), \quad (43)$$

$$\kappa(\xi) = \frac{1}{2\pi^2 \xi(1-\xi)} \sqrt{\frac{1+\xi}{2(2\xi-1)}}. \quad (44)$$

A convenient derivative is

$$\Phi'(\xi) = 2 \log(1-\xi) + \log\left(4 + \frac{4}{\xi}\right) - 2 \log(2\xi-1). \quad (45)$$

Furthermore,

$$\Phi''(\xi) = -\frac{2}{1-\xi} - \frac{1}{\xi(1+\xi)} - \frac{4}{2\xi-1} < 0,$$

so the critical point is the unique global maximum. Direct substitution gives

$$\xi_0 = \frac{4}{5}, \quad e^{\Phi(\xi_0)} = 27, \quad \Phi''(\xi_0) = -\frac{625}{36}, \quad \kappa(\xi_0) = \frac{25\sqrt{6}}{16\pi^2}. \quad (46)$$

The endpoint ranges are exponentially smaller than the saddle range. Replacing the sum by a Riemann sum and applying the Gaussian form of Laplace's method therefore yields

$$\begin{aligned} t_n &\sim 27^n n^{-3/2} \kappa(\xi_0) \sqrt{\frac{2\pi}{-\Phi''(\xi_0)}} \\ &= C_t 27^n n^{-3/2}, \quad C_t = \frac{3\sqrt{3}}{4\pi^{3/2}} = \frac{1}{4} \left(\frac{3}{\pi}\right)^{3/2}. \end{aligned} \quad (47)$$

This agrees with Hirschhorn's theorem, whose further terms begin

$$t_n \sim C_t 27^n n^{-3/2} \left(1 - \frac{65}{144n} + \frac{3865}{41472n^2} + \frac{111727}{17915904n^3} + \dots\right). \quad (48)$$

The first correction $-65/144$ is also the case $m = 2$ of manuscript 58's convolution theorem, (92); the two manuscripts agree.

Remark 6.1 (Why the square alone does not determine the answer). From $A^2 = \sum t_n z^n$, the asymptotic constant of t_n is not simply the square of the constant of a_n . Both series converge at $z = \rho$, and their dominant singularity is additive rather than multiplicative: $A = A(\rho) + B\sqrt{1 - z/\rho} + \dots$. Thus the singular coefficient of A^2 is $2A(\rho)B$. The missing datum is the exact boundary value $A(\rho)$, which is evaluated next.

6.4 The Fricke fixed point

Take

$$\tau_* = \frac{i}{\sqrt{7}}. \quad (49)$$

Dedekind's inversion law [11, Ch. 3]

$$\eta(-1/\tau) = \sqrt{-i\tau} \eta(\tau) \quad (50)$$

shows that

$$\eta(i\sqrt{7}) = 7^{-1/4} \eta(i/\sqrt{7}). \quad (51)$$

It follows from (39) that $w_7(\tau_*) = 1/7$, hence

$$X_7(\tau_*) = \frac{1/7}{1 + 13/7 + 1} = \frac{1}{27} = \rho. \quad (52)$$

All relevant quantities are positive on the imaginary axis. Therefore (41) gives

$$A(\rho) = z_7(\tau_*) = 3\eta(i/\sqrt{7})\eta(i\sqrt{7}). \quad (53)$$

The remaining task is an exact evaluation of this product.

6.5 Chowla–Selberg at discriminant -7

Put

$$\tau_0 = \frac{1 + i\sqrt{7}}{2}. \quad (54)$$

The class number of discriminant -7 is one [10]. Specializing the Chowla–Selberg formula in the form proved by van der Poorten and Williams [5] gives

$$|\eta(\tau_0)| = (14\pi)^{-1/4} \left(\frac{\Gamma(1/7)\Gamma(2/7)\Gamma(4/7)}{\Gamma(3/7)\Gamma(5/7)\Gamma(6/7)} \right)^{1/4}. \quad (55)$$

Reflection [13, Ch. 5] and the classical trigonometric product

$$\sin \frac{\pi}{7} \sin \frac{2\pi}{7} \sin \frac{3\pi}{7} = \frac{\sqrt{7}}{8} \quad (56)$$

imply

$$\mathcal{G}_7 \Gamma(3/7)\Gamma(5/7)\Gamma(6/7) = \frac{8\pi^3}{\sqrt{7}}.$$

Consequently (55) simplifies to

$$|\eta(\tau_0)| = \frac{\sqrt{\mathcal{G}_7}}{2 \cdot 7^{1/8} \pi}. \quad (57)$$

We still need to move from τ_0 to $i\sqrt{7}$. Introduce the Weber function

$$\mathfrak{f}_2(\tau) = \sqrt{2} \frac{\eta(2\tau)}{\eta(\tau)}, \quad j(\tau) = \frac{(\mathfrak{f}_2(\tau)^{24} + 16)^3}{\mathfrak{f}_2(\tau)^{24}}. \quad (58)$$

The Hilbert class polynomial of discriminant -7 is $X + 3375$ [10], so $j(\tau_0) = -3375$. Setting $u = \mathfrak{f}_2(\tau_0)^{24}$ gives

$$(u + 16)^3 + 3375u = (u + 1)(u^2 + 47u + 4096) = 0. \quad (59)$$

Here the nome is $e^{2\pi i \tau_0} = -e^{-\pi\sqrt{7}}$, and the product definition of η shows that u is a negative real number. The two roots of the quadratic factor in (59) are nonreal. Hence $u = -1$ and $|\mathfrak{f}_2(\tau_0)| = 1$. Since $2\tau_0 = 1 + i\sqrt{7}$ and $|\eta(\tau + 1)| = |\eta(\tau)|$, we obtain

$$\eta(i\sqrt{7}) = \frac{|\eta(\tau_0)|}{\sqrt{2}} = \frac{\sqrt{\mathcal{G}_7}}{2\sqrt{2} \cdot 7^{1/8} \pi}. \quad (60)$$

Combining (51) and (60),

$$\eta(i/\sqrt{7})\eta(i\sqrt{7}) = 7^{1/4} \eta(i\sqrt{7})^2 = \frac{\mathcal{G}_7}{8\pi^2}. \quad (61)$$

Finally, (53) becomes

$$\boxed{A(1/27) = \frac{3\mathcal{G}_7}{8\pi^2}}. \quad (62)$$

This is the second proof of $A(\rho) = A_\star$ in Theorem 2.1.

6.6 A second exact series from the functional equation

The square-root generating function satisfies

$$\frac{1}{1+4y}A\left(\frac{y}{(1+4y)^3}\right) = \frac{1}{1+2y}A\left(\frac{y^2}{(1+2y)^3}\right). \quad (63)$$

This follows either from O'Brien's modular equation or by taking the branch equal to 1 at the origin in the corresponding identity for A^2 . At $y = 1/8$, the two arguments are $1/27$ and $1/125$, and (63) reads

$$\frac{2}{3}A(1/27) = \frac{4}{5}A(1/125).$$

Together with (62), this proves the second identity of Theorem 2.2.

[Merge note, batch 7302: Provenance of (63). Manuscript 56 derives it from O'Brien's modular equation (or from the identity for A^2) and uses it only here. Manuscript 58 mentions, without using it, a functional equation added to the OEIS entry in 2026 (August 2026, in its source audit) as a possible input for the Lucas congruences (Section 13.6); the dossier of the intake identifies it with (63). The intake checked (63) as an identity of power series in y through y^{24} , and the value $A(1/125) = 5\mathcal{G}_7/(16\pi^2)$ by direct summation, to 3×10^{-61} . The second identity of Theorem 2.2 therefore rests on the functional equation, taken as an input from O'Brien's modular equation (the finite check is not a proof), as a power-series identity continued along $0 \leq y \leq 1/8$, where both arguments increase within $[0, \rho]$ and reach $1/27$ and $1/125$ at $y = 1/8$.]

6.7 The singular connection coefficient

Because the exponents at $s = 0$ are 0 and $1/2$, the local form of A is (19), $A(z) = U(s) + B\sqrt{s}V(s)$, with $U(0) = A(\rho)$ and $V(0) = 1$. The coefficient B is nonzero: otherwise A , and hence A^2 , would be analytic at $1/27$, contradicting (47).

Squaring (19),

$$A(z)^2 = A(\rho)^2 + 2A(\rho)Bs^{1/2} + O(s). \quad (64)$$

The transfer formula (73) at exponent $1/2$ [12, Ch. VI],

$$[z^n](1 - 27z)^{1/2} \sim -\frac{27^n}{2\sqrt{\pi}n^{3/2}}, \quad (65)$$

then shows that the constant in (47) satisfies

$$C_t = -\frac{2A(\rho)B}{2\sqrt{\pi}} = -\frac{A(\rho)B}{\sqrt{\pi}}. \quad (66)$$

(The transfer from the local expansion of A^2 to t_n uses the same dented-disk continuation as in Section 3; see the merge note in Section 7.2.) Using (47) and (62),

$$B = -\frac{2\pi\sqrt{3}}{\mathcal{G}_7}. \quad (67)$$

Applying (65) directly to A gives

$$C = -\frac{B}{2\sqrt{\pi}} = \frac{\sqrt{3}\pi}{\mathcal{G}_7},$$

which is the second proof of the leading assertion of Theorem 2.1. In manuscript 58's terms, (66) is $C_t = 2CA_\star$, the case $m = 2$ of (90); manuscript 56 obtains the product $CA_\star$ from the known square, manuscript 58 from Ramanujan's series (32).

Exact local connection formula (manuscript 56). At the dominant singularity,

$$A(z) = \frac{3\mathcal{G}_7}{8\pi^2} - \frac{2\pi\sqrt{3}}{\mathcal{G}_7} \sqrt{1-27z} + O(1-27z). \quad (68)$$

The first constant is modular/CM; the second is obtained by matching the known square A^2 to the first. Formula (3) is precisely the transfer of the second constant.

Manuscript 58 obtains the same singular coefficient in hypergeometric form, (28), and the same value through $B = -2\sqrt{\pi}C$.

[Write note, 2026-10-02, batch 77P5: Manuscript 32 obtains B a third way, without the square sequence and without a hypergeometric identity: by differentiating the Fricke involution of z_7 at τ_* (Proposition 16.2, $BA_\star = -3\sqrt{3}/(4\pi)$). It also re-derives (60) from an Epstein zeta function instead of citing the Chowla–Selberg formula (Theorem 16.6), and checks it by the route of this section (Remark 16.7).]

7 All orders: existence, remainders, and coefficients

7.1 The convergent local expansion

Write $V(s) = \sum_{j \geq 0} v_j s^j$, $v_0 = 1$, $v_{-1} = 0$. Substitution of $s^{1/2}V(s)$ into (18) gives

$$\begin{aligned} 28j(j + \tfrac{1}{2})v_j &= \left(29(j - \tfrac{1}{2})^2 + 14(j - \tfrac{1}{2}) + \tfrac{20}{9}\right)v_{j-1} \\ &\quad - \left((j - \tfrac{3}{2})^2 + (j - \tfrac{3}{2}) + \tfrac{2}{9}\right)v_{j-2}, \quad j \geq 1. \end{aligned} \quad (69)$$

Thus all v_j are rational, with $v_1 = 593/1512$. Frobenius theory makes this a *convergent local* power series, not merely a formal solution at infinity.

Manuscript 56 writes the same recurrence with integer coefficients,

$$504(m+1)(2m+3)v_{m+1} - (1044m^2 + 1548m + 593)v_m + (36m^2 - 1)v_{m-1} = 0, \quad (70)$$

and lists

$$\begin{aligned} v_1 &= \frac{593}{1512}, & v_2 &= \frac{7493}{31104}, & v_3 &= \frac{400316059}{2304433152}, \\ v_4 &= \frac{5678514224315}{41811635109888}, & v_5 &= \frac{25841240646919967}{231803705049219072}. \end{aligned} \quad (71)$$

(The two recurrences give the same v_j ; the intake checked $j \leq 9$ exactly.) Thus a more detailed version of (68) is

$$A(z) = U(1-27z) - \frac{2\pi\sqrt{3}}{\mathcal{G}_7} (1-27z)^{1/2} \left(1 + \frac{593}{1512}(1-27z) + \frac{7493}{31104}(1-27z)^2 + \dots\right). \quad (72)$$

7.2 Transfer to coefficients

For nonintegral α , the exact coefficient formula is

$$[z^n](1-27z)^\alpha = 27^n \frac{\Gamma(n-\alpha)}{\Gamma(-\alpha)\Gamma(n+1)}. \quad (73)$$

Apply it to $B \sum_{j=0}^M v_j (1 - 27z)^{j+1/2}$ and expand each gamma ratio [13, §5.11]. Singularity analysis in the dented disk from Section 3 transfers the remainder. The locally analytic part contributes no algebraic terms from this singularity. Equivalently one may subtract arbitrarily many Taylor terms of U before applying the transfer theorem. The first omitted nonintegral power is $(1 - 27z)^{M+3/2}$, and its coefficient is $O(27^n n^{-M-5/2})$. This proves (4) for every fixed M .

[Merge note, batch 7302: Manuscript 56's version of this step reads: "The gamma quotient has a complete expansion in descending powers of n . Combining the Frobenius series and (73) therefore gives the expansion. The analytic branch contributes only an exponentially smaller term after the local singular part is removed, because the next finite singularity is at $z = -1$." It does not state the continuation domain. The hypothesis singularity analysis needs, continuation to a dented disk of radius greater than ρ with no other singularity on $|z| = \rho$, is manuscript 58's (Section 3); this report relies on it for both proofs.]

The first correction is transparent: the gamma ratio for $\alpha = 1/2$ contributes $3/(8n)$; the $s^{3/2}$ term contributes $-3v_1/(2n)$, relative to the leading term. Hence

$$c_1 = \frac{3}{8} - \frac{3}{2} \frac{593}{1512} = -\frac{215}{1008}.$$

No numerical fitting of the coefficients is needed.

7.3 A triangular recurrence at infinity

For efficient computation insert $u_n = r^n n^{-3/2} S(1/n)$ in the recurrence, with $r \in \{27, -1\}$. The corresponding formal operator is

$$\begin{aligned} \mathcal{E}_r[S](t) &= r(1+t)^{1/2} S\left(\frac{t}{1+t}\right) - (26 + 13t + 2t^2)S(t) \\ &\quad - \frac{27 - 27t + 6t^2}{r} (1-t)^{-3/2} S\left(\frac{t}{1-t}\right). \end{aligned} \tag{74}$$

The characteristic equation is $r^2 - 26r - 27 = 0$, with roots $27, -1$.

Proposition 7.1 (Exact coefficient algorithm). *For either root there is a unique formal series $S_r(t) = 1 + \sum_{j \geq 1} \gamma_j t^j$ with $\mathcal{E}_r[S_r] = 0$. Writing $S_{m-1} = \sum_{j=0}^{m-1} \gamma_j t^j$, one has*

$$\gamma_m = \frac{[t^{m+1}]\mathcal{E}_r[S_{m-1}](t)}{m(r + 27/r)}, \quad m \geq 1. \tag{75}$$

All coefficients are rational; the denominators in the recursion are $28m$ and $-28m$, respectively.

Proof. A newly inserted $\gamma_m t^m$ contributes zero in order m , by the characteristic equation. Its coefficient in order $m+1$ is

$$r\left(\frac{1}{2} - m\right) - 13 - \frac{27}{r}\left(\frac{1}{2} + m\right) = -m(r + 27/r).$$

This is nonzero, giving the assertion inductively. At the first two orders the normalization is compatible with both characteristic roots. $\square$

For $r = 27$, the analytic existence proof above identifies this formal series with the asymptotic series of a_n/C . For $r = -1$, Section 9 will construct a genuine solution realizing the series. Formal coefficient generation alone is not used to assert existence of an asymptotic expansion.

Manuscript 56 derives the case $r = 27$ independently: with $S_{27} = \mathcal{A}$ in its notation, $\mathcal{E}_{27}[S] = 0$ is its formal identity, since $(27 - 27t + 6t^2)/27 = 1 - t + \frac{2}{9}t^2$ (Appendix B). It records the next five coefficients beyond (5):

$$\begin{aligned} c_6 &= \frac{5073267927236405}{3082688233381822464}, \\ c_7 &= \frac{141980078251330365605}{21751448174742139305984}, \\ c_8 &= \frac{1180675715260847223544061}{175403678081120611363454976}, \\ c_9 &= -\frac{541114776511779121907204365}{227323166793132312327037648896}, \\ c_{10} &= -\frac{10501970724197421668190580597385}{458283504254954741651307900174336}. \end{aligned} \quad (76)$$

These agree with manuscript 58's recorded coefficients, which continue to c_{12} in `data/58-gamma-verification.json`. The existence of the local convergent Frobenius series, together with standard singularity analysis, upgrades the formal computation to the error statement in Theorem 2.1 for every fixed order.

7.4 Logarithms, ratios, and convexity

Let $\log S_{27}(t) = \sum_{j \geq 1} \ell_j t^j$. Then

$$\log a_n = n \log 27 - \frac{3}{2} \log n + \log C + \sum_{j=1}^M \frac{\ell_j}{n^j} + O_M(n^{-M-1}), \quad (77)$$

with

$$\ell_1 = -\frac{215}{1008}, \quad \ell_2 = -\frac{85}{3136}, \quad \ell_3 = \frac{99905}{85349376}, \quad \ell_4 = \frac{1657}{2458624}. \quad (78)$$

In particular,

$$\frac{a_{n+1}}{a_n} = 27 \left(1 - \frac{3}{2n} + \frac{2105}{1008n^2} + O(n^{-3}) \right). \quad (79)$$

(Corollary 2.3, from manuscript 56, adds the terms in n^{-3} and n^{-4} .) The sequence is strictly increasing, since the coefficient of a_n in (1) already exceeds 1 for $n \geq 1$, and $a_1 > a_0$. Taking sufficiently many terms before forming second differences gives

$$\log a_{n+1} - 2 \log a_n + \log a_{n-1} = \frac{3}{2n^2} + O(n^{-3}) > 0$$

for sufficiently large n . Thus it is eventually strictly log-convex. No explicit smallest threshold is asserted. [Write note, 2026-10-04, batch 97G: The smallest threshold is $n = 1$: the sequence is strictly log-convex at every index (Proposition 17.5, manuscript 07), and more strongly (a_{i+j}) is strictly totally positive of order five (Theorem 17.1).]

8 A continued fraction and exact rational bracketing

8.1 The discrete Wronskian

Let b_n be as in Theorem 2.6 and set $W_n = a_n b_{n+1} - a_{n+1} b_n$. Eliminating the two forward terms gives

$$W_n = -\frac{3(3n-1)(3n-2)}{(n+1)^2} W_{n-1}, \quad W_0 = 1.$$

Grouping factors by residue modulo 3 yields the exact formula

$$W_n = (-1)^n \frac{(3n)!}{n!(n+1)!^2}. \quad (80)$$

Consequently

$$\delta_n := \frac{b_{n+1}}{a_{n+1}} - \frac{b_n}{a_n} = (-1)^n \frac{(3n)!}{n!(n+1)!^2 a_n a_{n+1}}. \quad (81)$$

Its magnitude decreases strictly because

$$\left| \frac{\delta_{n+1}}{\delta_n} \right| = \frac{3(3n+2)(3n+1)a_n}{(n+2)^2 a_{n+2}} < 1.$$

The numerator is one positive summand in the recurrence for the denominator. The ratio tends to $1/27$ by Theorem 2.1. Thus $\sum \delta_n$ converges absolutely, and its alternating tails prove the existence of L and the brackets (13). Even approximants increase and odd approximants decrease; consecutive ones enclose L with the exact width

$$\left| \frac{b_{n+1}}{a_{n+1}} - \frac{b_n}{a_n} \right| = \frac{(3n)!}{n!(n+1)!^2 a_n a_{n+1}}. \quad (82)$$

No unknown constants enter this stopping criterion.

8.2 Polynomial continued-fraction coefficients

The rescaled sequences $\mathcal{H}_n = (n!)^2 a_n$ and $\mathcal{K}_n = (n!)^2 b_n$ satisfy

$$u_{n+1} = P_n u_n + Q_n u_{n-1}, \quad P_n = 26n^2 + 13n + 2, \quad Q_n = 3n^2(3n-1)(3n-2), \quad (83)$$

with initial pairs $(1, 2)$ and $(0, 1)$. Their integrality follows directly from this recurrence, independently of integrality of the unscaled sequence. The continuant recurrence gives

$$L = \frac{1}{2 + \frac{6}{41 + \frac{240}{132 + \frac{1512}{275 + \ddots}}}}. \quad (84)$$

The convergent with n denominator levels is $\mathcal{K}_n/\mathcal{H}_n = b_n/a_n$.

Remark 8.1 (The initial relation matters). The companion satisfies the recurrence only for $n \geq 1$; $b_1 = 2b_0$ is false. Its generating function $\mathcal{B}(z) = \sum b_n z^n$ therefore satisfies (17) with right side 1, not 0. The same holds for $\mathcal{B} - LA$. These must not be called second analytic solutions of the homogeneous equation at 0. The two-dimensional solution space here consists of sequences satisfying the recurrence for $n \geq 1$.

9 The recessive scale and its full expansion

Stirling's formula in (80) gives

$$W_n = (-1)^n \frac{\sqrt{3}}{2\pi} \frac{27^n}{n^3} (1 + O(n^{-1})). \quad (85)$$

Together with the dominant asymptotic this yields

$$\delta_n = (-1)^n \frac{\sqrt{3}}{54\pi C^2} 27^{-n} (1 + O(n^{-1})). \quad (86)$$

The exact tail identities are

$$L - \frac{b_n}{a_n} = \sum_{j \geq 0} \delta_{n+j}, \quad e_n = -a_n \sum_{j \geq 0} \delta_{n+j}. \quad (87)$$

Since $\sum_{j \geq 0} (-1/27)^j = 27/28$, the leading terms of (16) and (14) follow. The recessive amplitude can also be written

$$D_{\min} = \frac{\sqrt{3}}{56\pi C} = \frac{\mathcal{G}_7}{56\pi^{3/2}}. \quad (88)$$

9.1 A remainder proof for the geometric tail

The full Stirling expansion and Theorem 2.1 imply, for each fixed M ,

$$\delta_n = (-1)^n K_0 27^{-n} \left(\sum_{k=0}^M \eta_k n^{-k} + O_M(n^{-M-1}) \right), \quad K_0 = \frac{\sqrt{3}}{54\pi C^2}, \quad \eta_0 = 1,$$

where all η_k are rational. The error bound is valid for all sufficiently large integer arguments, hence also at $n + j$ uniformly in $j \geq 0$, with a bound by a constant times $27^{-j} n^{-M-1}$ after factoring out 27^{-n} . Expand $(n + j)^{-k}$ for $0 \leq j \leq n/2$ by Taylor's formula, retaining its bounded remainder. The range $j > n/2$ is exponentially small, even with any fixed polynomial weight. Every coefficient involves moments

$$\sum_{j \geq 0} j^h (-1/27)^j,$$

which are rational: apply $(q d/dq)^h$ to $1/(1-q)$ and set $q = -1/27$. Thus the tail in (87) has an expansion to every fixed order with a controlled error and rational normalized coefficients. Multiplication by the expansion of a_n proves existence of (14). Since e_n satisfies the recurrence, Proposition 7.1 with $r = -1$ proves that its coefficients are precisely the d_j already given. This completes the proof of Theorem 2.6.

9.2 What the two-scale expansion does and does not say

Because $W_n \neq 0$, a_n and e_n form an exact basis of the recurrence solutions for $n \geq 1$. Every solution has a unique form $u_n = \alpha a_n + \beta e_n$. In this specified basis its two asymptotic scales are

$$\begin{aligned} u_n \sim & \alpha C 27^n n^{-3/2} \sum_{j \geq 0} c_j n^{-j} \\ & + \beta (-1)^{n+1} D_{\min} n^{-3/2} \sum_{j \geq 0} d_j n^{-j}. \end{aligned} \quad (89)$$

Each exact basis sequence has its own fixed-order expansion. A finite truncation of the dominant expansion generally has an error far larger than the entire recessive term. Equation (89) is therefore not a claim that two finite sums resolve exponentially small effects in a generic solution. A canonical exponentially improved expansion of a_n alone would require additional summation or Stokes data; that question is left open.

10 Convolution powers and A183204

For a fixed positive integer m , let

$$a_n^{(m)} = [z^n] A(z)^m = \sum_{n_1 + \dots + n_m = n} a_{n_1} \cdots a_{n_m}.$$

The exact boundary value controls this family as well as the original sequence.

Theorem 10.1 (Convolution asymptotics; manuscript 58). *For every fixed $m \geq 1$,*

$$a_n^{(m)} = mA_\star^{m-1} C \frac{27^n}{n^{3/2}} \left(1 + \frac{c_1^{(m)}}{n} + O_m(n^{-2}) \right), \quad (90)$$

where

$$c_1^{(m)} = -\frac{215}{1008} - \frac{5}{21}(m-1) - \pi \frac{C^2}{A_\star^2}(m-1)(m-2). \quad (91)$$

There is a computable expansion to every fixed order, and all its normalized coefficients lie in $\mathbb{Q}(\pi C^2/A_\star^2)$.

Proof. The analytic part of (19) satisfies

$$U(s) = A_\star \left(1 + \frac{10}{63}s + O(s^2) \right),$$

as the constant coefficient of (18) shows. Its subsequent normalized coefficients are rational. The singular part is $B\sqrt{s}(1 + v_1 s + \dots)$, with $B = -2\sqrt{\pi}C$ and $v_1 = 593/1512$. In the m th power the coefficient of $\sqrt{s}$ is $mA_\star^{m-1}B$. Relative to that coefficient, the coefficient of $s^{3/2}$ is

$$v_1 + (m-1)\frac{10}{63} + \frac{(m-1)(m-2)}{6} \frac{B^2}{A_\star^2}.$$

Transfer as in Section 7 gives (91). At higher orders only odd powers of B multiply half-integral powers of s . After division by the leading singular coefficient, the remaining expressions are polynomials in $B^2/A_\star^2 = 4\pi C^2/A_\star^2$ with rational coefficients. The convergent Puiseux expansion and the same dented domain provide the fixed-order remainder estimates. $\square$

Remark 10.2 (An explicit generator for every order; manuscript 32). [Write note, 2026-10-02, batch 77P5: Manuscript 32 proves Theorem 10.1 again (its Theorem 7.1, with weight $k = m$) and makes the “computable expansion to every fixed order” explicit. Notation is that of Table 7.] Put

$$\mathfrak{q} = \frac{B^2}{A_\star^2} = \frac{4\pi C^2}{A_\star^2} = \frac{256\pi^6}{3\mathcal{G}_7^4} = 0.0768944281311685853765\dots,$$

the quantity $B^2/A_\star^2$ of the proof above (it is *not* the hypergeometric argument $q_0 = 64/85^3$ of Section 5; manuscript 32’s replay output calls it $\mathfrak{q}\mathbf{0}$). With $U(s) = A_\star \widehat{U}(s)$, $\widehat{U}(0) = V(0) = 1$, the coefficient of $s^{j+1/2}$ in $A(z)^m = (A_\star \widehat{U}(s) + B\sqrt{s}V(s))^m$ is $B_j^{(m)} = mA_\star^{m-1}B\mathcal{R}_{m,j}$, where the binomial theorem gives the finite sum

$$\mathcal{R}_{m,j} = \sum_{h=0}^{\min(j, \lfloor (m-1)/2 \rfloor)} \frac{1}{m} \binom{m}{2h+1} \mathfrak{q}^h [s^{j-h}] \widehat{U}(s)^{m-2h-1} V(s)^{2h+1}, \quad \mathcal{R}_{m,0} = 1.$$

Transfer as in Section 7.2 then gives, for every fixed M ,

$$a_n^{(m)} = 27^n \sum_{j=0}^M B_j^{(m)} \frac{\Gamma(n-j-\frac{1}{2})}{\Gamma(-j-\frac{1}{2})\Gamma(n+1)} + O_{m,M}(27^n n^{-M-5/2}),$$

and, with the gamma-ratio coefficients h_i of (16.14),

$$c_i^{(m)} = \sum_{j=0}^i \mathcal{R}_{m,j} (-1)^j \left(\frac{3}{2}\right)_j h_{i-j}(-j-\frac{1}{2}, 1).$$

All operations are finite and all exponents nonnegative integers, so $c_i^{(m)}$ lies in the polynomial ring $\mathbb{Q}[\mathbf{q}]$, a sharper form of the field statement of Theorem 10.1. For $i = 1$, $\mathcal{R}_{m,1} = \frac{593}{1512} + \frac{10}{63}(m-1) + \frac{(m-1)(m-2)}{6}\mathbf{q}$ and $h_1(-\frac{1}{2}, 1) = \frac{3}{8}$, so $c_1^{(m)} = \frac{3}{8} - \frac{3}{2}\mathcal{R}_{m,1}$, which is (91) because $\pi C^2/A_\star^2 = \mathbf{q}/4 = 64\pi^6/(3\mathcal{G}_7^4)$; for $m = 3$ it is $-0.72793134104971127682\dots$. For $m \leq 2$ only $h = 0$ occurs, so every $c_i^{(1)}$ and $c_i^{(2)}$ is rational; for $m \geq 3$ the first correction is a polynomial of degree one in $\mathbf{q}$. For $m = 2$ the leading constant and $c_1^{(2)} = -65/144$ are Hirschhorn's (92); manuscript 32 recovers them as a consistency check on prior work, not as a new result, and it checks the binomial sum (37) only over a finite range (its replay: $n \leq 40$). It asserts no OEIS identification for $m \geq 3$.

Manuscript 32 also records that each $a_n^{(m)}$ is strictly increasing in n : raising the index of the first factor maps the terms of degree n injectively to strictly larger terms of degree $n+1$, and the remaining terms of degree $n+1$ are positive. Its replay checks the local expansion of A^m by direct series multiplication for $m = 1, \dots, 6$ and the convolutions for $m = 2, 3$ through $n = 400$. These are finite checks.

For $m = 2$ this is A183204, and the first two terms simplify to

$$[z^n]A(z)^2 = \frac{3\sqrt{3}}{4\pi^{3/2}} \frac{27^n}{n^{3/2}} \left(1 - \frac{65}{144n} + O(n^{-2})\right). \quad (92)$$

The leading formula is already recorded in its OEIS entry [2]; it is recovered here, not claimed as new. The cancellation of gamma factors at this convolution power is a useful check on the evaluation of C . (Both terms are Hirschhorn's [4], as quoted by manuscript 56 in (48), which runs on to the n^{-3} term. In the second proof of Section 6, the logic is reversed: the leading constant of (92) is the input from which C is extracted.)

For an algebraic check of the generating-function identification, the symmetric-square equation of (17) is

$$\begin{aligned} (-z^2 + 26z^3 + 27z^4)Y''' + (-3z + 117z^2 + 162z^3)Y'' \\ + (-1 + 86z + 186z^2)Y' + (4 + 24z)Y = 0. \end{aligned} \quad (93)$$

This is the equation recorded for A183204, and its normalized analytic solution is unique. The program verifies this symmetric-square identity and checks the convolution against

$$\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{n} \binom{2n-k}{n}$$

for $0 \leq n \leq 40$. The finite check supplements the differential-equation argument; it does not replace it.

11 Inversion and the integer-rounding boundary

[Merge note, batch 7302: Both manuscripts invert the asymptotic by Lambert's W_{-1} . The core and its triangular corrections are the instance $a = \lambda = \log 27$, $\beta_{\text{CTI}} = -3/2$, $q_j^{\text{CTI}} = \ell_j$ of Theorem

`t2:thm:balanced-inverse` of the repository's volume *Combinatorial Transseries and Their Inverses* [16]: the coefficients q_1, q_2 of (97) below are that theorem's d_1, d_2 (checked by hand at intake). Neither manuscript cites it, and this report claims no novelty for the inverse expansion. The rounding proposition, Proposition 11.2, is the sequence-specific part; its conclusion has the form of the separation condition (part 2) of Theorem `p0:thm:staircase` in the repository's canonical volume *Transseries and Inversion* [17].]

Put $\lambda = \log 27$ and $\beta = 3/2$. The leading smooth model is $Y = Ce^{\lambda x}x^{-\beta}$. Its large positive solution is

$$x_0(Y) = -\frac{\beta}{\lambda} W_{-1} \left(-\frac{\lambda}{\beta} (C/Y)^{1/\beta} \right), \quad (94)$$

where W_{-1} is the real negative branch of Lambert's function. The principal branch selects the small solution and is inappropriate for inversion of the large- n sequence [13, §4.13]. (Manuscript 56 derives the same formula, raising the envelope to the power $-2/3$.)

11.1 A triangular inverse expansion

Seek

$$x_M(Y) = x_0 + \sum_{j=1}^M \frac{q_j}{x_0^j}. \quad (95)$$

Writing $\delta = x - x_0$, the formal equation determining the coefficients is

$$\lambda \delta - \beta \log(1 + \delta/x_0) + \sum_{k \geq 1} \ell_k (x_0 + \delta)^{-k} = 0. \quad (96)$$

The newly introduced coefficient q_j has multiplier λ , so this is triangular at every order. The first coefficients are

$$\begin{aligned} q_1 &= -\frac{\ell_1}{\lambda} = \frac{215}{1008\lambda}, \\ q_2 &= \frac{\beta q_1 - \ell_2}{\lambda} = \frac{215}{672\lambda^2} + \frac{85}{3136\lambda}, \\ q_3 &= \frac{\beta q_2 + \ell_1 q_1 - \ell_3}{\lambda}. \end{aligned} \quad (97)$$

All q_j belong to $\mathbb{Q}(\lambda)$.

An expansion involving only logarithms follows by putting $X = \log(Y/C)$:

$$\begin{aligned} x(Y) &= \frac{1}{\lambda} \left\{ X + \beta \log \frac{X}{\lambda} + \frac{\beta^2 \log(X/\lambda) - \lambda \ell_1}{X} \right\} \\ &\quad + O\left(\frac{(\log X)^2}{X^2}\right). \end{aligned} \quad (98)$$

Here $x(Y)$ denotes the asymptotic inverse scale, not a previously specified analytic interpolation of the integer sequence. Formula (94) is often more accurate numerically because it retains the entire leading model instead of truncating its logarithmic inversion.

Remark 11.1 (Manuscript 56's form of the inverse). Expanding W_{-1} at 0^- , manuscript 56 records

$$x_0(Y) = \frac{1}{\lambda} \left(\log Y + \frac{3}{2} \log \log Y - \frac{3}{2} \log \lambda - \log C \right) + O\left(\frac{\log \log Y}{\log Y}\right), \quad (99)$$

consistent with the first two terms of (98). It notes that the rational correction series can be inserted and reverted recursively to produce an arbitrarily long expansion in $1/\log Y$ with nested powers of $\log \log Y$, and that for the discrete inverse $N(Y)$ below, formula (94), “rounded and corrected by one or two exact recurrence evaluations, gives an efficient certified locator”. [Merge note, batch 73O2: The last statement is a practical recipe, not a theorem: a *certified* locator needs explicit constants K_M, Y_M in Proposition 11.2, which neither manuscript supplies (manuscript 58 lists them as a research direction, Section 13.1). Checking a candidate n by exact evaluation of $a_{n-1} < Y \leq a_n$ is of course rigorous.]

11.2 A rigorous discrete statement

Define

$$N(Y) = \min\{n \geq 0 : a_n \geq Y\}.$$

The sequence is strictly increasing and unbounded, so this is well-defined for $Y > 1$. An inverse approximation cannot uniformly remove the ambiguity when its value is exceptionally close to an integer.

Proposition 11.2 (Rounding brackets). *For every fixed $M \geq 0$ there exist $K_M > 0$ and Y_M such that, for $Y \geq Y_M$,*

$$\lceil x_M - K_M x_M^{-M-1} \rceil \leq N(Y) \leq \lceil x_M + K_M x_M^{-M-1} \rceil. \quad (100)$$

When the two ceilings coincide, they determine the discrete inverse exactly. For $M = 0$, x_M means x_0 .

Proof. Use the smooth finite model

$$g_M(x) = \log C + \lambda x - \beta \log x + \sum_{k=1}^M \ell_k x^{-k}.$$

The all-orders theorem gives $\log a_n = g_M(n) + O_M(n^{-M-1})$. Also $g'_M(x) = \lambda + O(x^{-1})$, so it is bounded away from zero for large x . Let t_M be the large solution of $g_M(t_M) = \log Y$. Formal substitution of (95) into g_M , with ordinary finite Taylor remainders, gives $g_M(x_M) - \log Y = O_M(x_M^{-M-1})$. The mean value theorem therefore implies $t_M - x_M = O_M(x_M^{-M-1})$.

For integers near t_M , comparison of $g_M(n)$ with $g_M(t_M)$, using its derivative lower bound and the error in $\log a_n$, shows that integers sufficiently below t_M cannot reach Y , whereas integers sufficiently above it do reach Y . Enlarge the constant to absorb $t_M - x_M$, and use monotonicity of a_n for all remaining integers. This gives the two ceilings in (100). $\square$

The proposition asserts the existence of fixed-order constants, not explicit numerical values of K_M or Y_M . Computing certified, practical values is one of the research directions below. No claim that $N(Y) = \lceil x_M(Y) \rceil$ holds uniformly is made.

Remark 11.3 (Every fixed weight, and an implicit-function proof; manuscript 32). [Write note, 2026-10-02, batch 77P5: Manuscript 32 states the rounding brackets for every fixed convolution weight m (its Theorems 8.1 and 8.2, weight $k = m$), with notation as in Table 7. For $m = 1$ they are Proposition 11.2. The Lambert- W_{-1} core and its triangular corrections are, as for $m = 1$, instances of the repository’s inversion apparatus: `t2:thm:balanced-inverse` of [16], and `p0:thm:lambert-core`, `p0:thm:perturbed-inversion` and `p0:thm:staircase` of [17]. Neither manuscript 32 nor this report claims novelty for them.] Let $N^{(m)}(Y) = \min\{n \geq 0 : a_n^{(m)} \geq Y\}$, well defined for every real $Y > 0$ because $a_n^{(m)}$ is strictly increasing and unbounded (Remark 10.2). With $C^{(m)} = m A_{\star}^{m-1} C$ and $S_M^{(m)}(u) = \sum_{j=0}^M c_j^{(m)} u^j$, define the smooth model

$$\mathcal{F}_M^{(m)}(v) = C^{(m)} e^{\lambda v} v^{-\beta} S_M^{(m)}(1/v), \quad \frac{d}{dv} \log \mathcal{F}_M^{(m)}(v) = \lambda + O_{m,M}(1/v),$$

and let $\tilde{x}_M^{(m)}(Y)$ be its unique large solution of $\mathcal{F}_M^{(m)}(v) = Y$. Then there are $K_M^{(m)} > 0$ and $Y_M^{(m)}$ such that, for $Y \geq Y_M^{(m)}$ and $\tilde{x} = \tilde{x}_M^{(m)}(Y)$,

$$\lceil \tilde{x} - K_M^{(m)} \tilde{x}^{-M-1} \rceil \leq N^{(m)}(Y) \leq \lceil \tilde{x} + K_M^{(m)} \tilde{x}^{-M-1} \rceil.$$

The proof is that of Proposition 11.2: the forward expansion gives $|\log a_n^{(m)} - \log \mathcal{F}_M^{(m)}(n)| \leq Dn^{-M-1}$; the logarithmic derivative is at least $\lambda/2$ beyond a fixed point; for $\epsilon = K\tilde{x}^{-M-1}$ the mean value theorem gives $a_{n_+}^{(m)} \geq Y$ at $n_+ = \lceil \tilde{x} + \epsilon \rceil$ and $a_{n_-}^{(m)} < Y$ at $n_- = \lceil \tilde{x} - \epsilon \rceil - 1$, and monotonicity gives $n_- + 1 \leq N^{(m)}(Y) \leq n_+$.

For the centre, let $x_0^{(m)}(Y)$ be (94) with C replaced by $C^{(m)}$, let $\ell_j^{(m)} = [u^j] \log \sum_{i \geq 0} c_i^{(m)} u^i$ (a formal logarithm; convergence of the coefficient series is not assumed), and define $q_j^{(m)}$ by $\delta(v) = \sum_{j \geq 1} q_j^{(m)} v^j$ and

$$\lambda \delta - \beta \log(1 + v\delta) + \sum_{j \geq 1} \ell_j^{(m)} \left(\frac{v}{1 + v\delta} \right)^j = 0,$$

triangular because the new $q_j^{(m)}$ enters with multiplier λ . Then $\tilde{x}_M^{(m)}(Y) = x_0^{(m)} + \sum_{j=1}^M q_j^{(m)} (x_0^{(m)})^{-j} + O_{m,M}((x_0^{(m)})^{-M-1})$. Manuscript 32 proves this by the *analytic* implicit-function theorem rather than by formal substitution: with $\tilde{x} = x_0^{(m)} + \delta$ and $v = 1/x_0^{(m)}$, the exact equation $\lambda \delta - \beta \log(1 + v\delta) + \log S_M^{(m)}(v/(1 + v\delta)) = 0$ is analytic near $(0, 0)$ with δ -derivative $\lambda \neq 0$, so it has a convergent solution $\delta(v)$, which is the large inverse for small $v > 0$; its first M Taylor coefficients are the $q_j^{(m)}$, and raising M leaves the determined coefficients unchanged. Explicitly $q_1^{(m)} = -\ell_1^{(m)}/\lambda$, $q_2^{(m)} = -\beta \ell_1^{(m)}/\lambda^2 - \ell_2^{(m)}/\lambda$ and $q_3^{(m)} = -\beta^2 \ell_1^{(m)}/\lambda^3 - (\beta \ell_2^{(m)} + (\ell_1^{(m)})^2)/\lambda^2 - \ell_3^{(m)}/\lambda$; for $m = 1$ these are q_1, q_2, q_3 of (97) (checked by hand at this write). The replay records the universal coefficients through order 6. Combining the two statements, the centre $\tilde{x}$ of the brackets may be replaced by the truncated expansion after enlarging the constant; for $m = 1$ this is Proposition 11.2 with centre x_M .

As in Proposition 11.2, $K_M^{(m)}$ and $Y_M^{(m)}$ are existential. Once evaluated effectively, the brackets decide $N^{(m)}(Y)$ whenever the two ceilings agree; without such values they supply no practical rounding certificate, and no unqualified exact ceiling at or near a threshold is asserted. The replay's inverse residuals are taken at $Y = a_n^{(m)}$, exactly at discrete thresholds, and are not a ceiling rule.

12 Reproducible computations and certified constants

[Merge note, batch 7302: File names in this section are the shipped ones. Manuscript 58 named its README, SOURCE_AUDIT.md and data/verification.json; they are this report's README.md, 58-gamma-SOURCE_AUDIT.md and data/58-gamma-verification.json.]

Build and run instructions are in the accompanying README.md; manuscript 58's source audit is in 58-gamma-SOURCE_AUDIT.md. The PDF can be built without running Python, since its numerical tables are part of the TeX source. Manuscript 58's program (code/58-gamma-verify.py) uses exact integers and rational functions for its algebraic checks. Arbitrary-precision floating-point arithmetic is used separately for numerical diagnostics. It requires only Python, SymPy, and mpmath, and makes no network requests.

12.1 Certificates and reproducible output

The following assertions were executed successfully: eight rational identities for the two pullbacks and two symmetric squares; coefficient recurrences through order 12 for both characteristic roots; exact recurrence

evaluation through $n = 1000$; 100 discrete Wronskian identities; and 41 comparisons with the binomial sum for A183204. The division in the a_n recurrence is exact at every tested index. This last computation is a finite integrality check, not an independent proof of integrality for all indices.

The constants are

$$\begin{aligned} C &= 0.09552230526812671465130791078702962567279466665100\dots, \\ A_\star &= 1.22113109746756666928422458568599346237582705637935\dots, \\ L &= 0.46721176888437312394211005141309126172410854995971\dots, \\ D_{\min} &= 0.10306659600794607491874204424062776206902297866964\dots \end{aligned}$$

The first displayed value is not inferred by integer-relation searching: its gamma formula is proved in Section 5 (and again in Section 6), and the following interval method supplies independent certified decimal evaluation.

12.2 A rational enclosure for the asymptotic constant

Write $F(q_0) = \sum_{j \geq 0} h_j$. Its positive terms satisfy

$$\frac{h_{j+1}}{h_j} = q_0 \frac{(12j+1)(12j+5)}{144(j+1)^2} < q_0 < 1.$$

Hence, for $S_N = \sum_{j=0}^{N-1} h_j$,

$$S_N \leq F(q_0) \leq S_N + \frac{h_N}{1 - q_0}. \quad (101)$$

Every quantity in this enclosure is rational. The implementation bounds π using Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$$

and exact alternating-series bounds. Square and fourth roots are bracketed by rational numbers using integer root extraction on a fixed decimal grid. Combining positive upper and lower bounds in (32) gives a rational enclosure for C of width less than 10^{-120} . Outward rounding produces the following interval of width 10^{-100} ; each row continues on its second line.

$$\begin{aligned} \text{Lower: } & \mathbf{0.09552230526812671465130791078702962567279466665100} \\ & \mathbf{71798669948234917659264524531457675812093972181446} \\ \text{Upper: } & \mathbf{0.09552230526812671465130791078702962567279466665100} \\ & \mathbf{71798669948234917659264524531457675812093972181447} \end{aligned}$$

For L the enclosure uses no transcendental functions at all: $b_{90}/a_{90} < L < b_{91}/a_{91}$, with exact rational endpoints whose difference is less than 10^{-125} . The full endpoints and their 100-place decimal enclosures are stored in `data/58-gamma-verification.json`. The JSON also records every coefficient through order 12 and the zero values of all eight rational certificates.

12.3 Numerical diagnostics

For Table 2, the relative error is

$$\frac{C 27^n n^{-3/2} \sum_{j=0}^M c_j n^{-j}}{a_n} - 1.$$

Table 2: Dominant expansion: signed relative errors (manuscript 58).

n	$M = 0$	$M = 3$	$M = 6$	$M = 12$
20	1.079×10^{-2}	-1.143×10^{-9}	-5.413×10^{-12}	2.818×10^{-19}
100	2.138×10^{-3}	-2.449×10^{-12}	-6.609×10^{-17}	6.083×10^{-28}
1000	2.133×10^{-4}	-2.597×10^{-16}	-6.536×10^{-24}	6.727×10^{-41}

The values are rounded floating-point diagnostics, not new rigorous upper bounds on the asymptotic remainder. All were computed with 190 decimal digits of working precision.

The normalized recessive error in Table 3 is $\sum_{j=0}^M d_j n^{-j} - e_n / [(-1)^{n+1} D_{\min} n^{-3/2}]$. The value of L used in this diagnostic is the midpoint of the exact r_{150}, r_{151} enclosure, where $r_n = b_n/a_n$. Its error remains negligible after multiplication by a_n at the displayed indices.

Table 3: Recessive expansion: signed normalized errors (manuscript 58).

n	$M = 0$	$M = 3$	$M = 6$
10	4.910×10^{-2}	1.721×10^{-6}	-1.532×10^{-9}
30	1.676×10^{-2}	2.361×10^{-8}	-6.532×10^{-13}
75	6.753×10^{-3}	6.240×10^{-10}	-1.036×10^{-15}

Table 4 tests inversion at $Y = a_n$. It reports the signed continuous error $x_M(a_n) - n$, not an integer rounding error. The uncorrected Lambert- W scale is already accurate; the algebraic corrections produce the predicted further improvement.

Table 4: Inverse expansion at exact sequence values (manuscript 58).

n	$x_0(a_n) - n$	$x_1(a_n) - n$	$x_3(a_n) - n$
20	-3.332×10^{-3}	-9.581×10^{-5}	-6.315×10^{-9}
100	-6.509×10^{-4}	-3.780×10^{-6}	-1.064×10^{-11}
1000	-6.475×10^{-5}	-3.769×10^{-8}	-1.075×10^{-15}

12.4 Manuscript 56's tables and checks

All numerical checks of manuscript 56 were generated from exact integers using (1); no OEIS term table was used after the initial comparison. Define the scaled quantity

$$R_n = \frac{a_n n^{3/2}}{27^n}. \quad (102)$$

Table 5 compares R_n with the exact constant. The last column subtracts the approximation through c_3/n^3 and divides the residual by C .

The residual column is the $M = 3$ column of Table 2 with the opposite sign convention: manuscript 56 tabulates $R_n/C - \sum_{j \leq 3} c_j n^{-j}$, manuscript 58 tabulates $C \sum_{j \leq 3} c_j n^{-j}/R_n - 1$, which is minus the former times $C/R_n = 1 + O(n^{-1})$. At $n = 100$ and 1000 the two agree up to sign to the printed digits (2.4441×10^{-12} against -2.449×10^{-12} , and 2.5960×10^{-16} against -2.597×10^{-16}).

The critical series converges slowly, exactly as Corollary 2.5 predicts. Table 6 gives the exact remainder computed from (62) and its ratio to the leading estimate $2C/\sqrt{N+1}$.

Table 5: Convergence to the exact asymptotic constant (manuscript 56).

n	R_n	$R_n/C - 1$	residual after c_3
10	0.0934812234164389792	$-2.1367594 \times 10^{-2}$	1.1427×10^{-8}
50	0.0951146568282489133	$-4.2675733 \times 10^{-3}$	3.6462×10^{-11}
100	0.0953185211418410836	$-2.1333669 \times 10^{-3}$	2.4441×10^{-12}
500	0.0954815550048036949	$-4.2660469 \times 10^{-4}$	4.1263×10^{-15}
1000	0.0955019305511748892	$-2.1329800 \times 10^{-4}$	2.5960×10^{-16}
5000	0.0955182303912366763	$-4.2658904 \times 10^{-5}$	4.1754×10^{-19}

Table 6: Critical-series tail at $z = 1/27$ (manuscript 56).

N	$A(1/27) - \sum_{n=0}^N a_n/27^n$	ratio to $2C/\sqrt{N+1}$
10	0.058541895880356	1.016314999
50	0.026845518682688	1.003510906
100	0.019043336251750	1.001772100
500	0.008538291869020	1.000357123
1000	0.006039421866043	1.000178732
5000	0.002701605297719	1.000035774

Manuscript 56's script (`code/56-exact-verify_a279619.py`; its recorded output is `data/56-exact-verification.txt`) performs the following independent checks:

1. generates $a_0, \dots, a_N$ as exact Python integers and verifies divisibility by $(n+1)^2$ at every recurrence step;
2. derives the c_j by exact symbolic expansion of the formal identity $\mathcal{E}_{27}[S] = 0$ of (74);
3. derives the v_j by the exact recurrence (70);
4. evaluates the gamma products and both special sums at high precision;
5. prints Tables 5 and 6;
6. checks the algebraic identities for $A(1/27)$, B , C , and the square-sequence constant C_t .

Items 4 and 6 are high-precision numerical consistency checks of identities proved in Section 6; they are not proofs.

13 Further research questions and topics

The questions below are concrete extensions of the proved results. They are not assertions that every item is globally open: before a priority claim, the relevant modular-form and continued-fraction literature should be checked in greater depth. Sections 13.1–13.12 are manuscript 58's twelve; Section 13.13 lists manuscript 56's ten and records where they meet manuscript 58's.

13.1 Explicit remainder constants

Theorems 2.1 and 2.6 give remainders at every fixed order but not practical values of the constants. Construct computable K_M, N_M such that

$$\left| \frac{a_n n^{3/2}}{C 27^n} - \sum_{j=0}^M c_j n^{-j} \right| \leq K_M n^{-M-1} \quad (n \geq N_M).$$

One route is to bound the analytic Frobenius functions on a specified complex neighborhood and make the transfer contour explicit. A second is to construct upper and lower recurrence solutions from truncated series plus a controlled correction. The latter may give smaller thresholds and could turn the inverse brackets into a practical certified algorithm.

13.2 Late-order growth and optimal truncation

Determine the large- j behavior of c_j and d_j . The proof of fixed-order asymptotics does not imply convergence of the series in $1/n$. A useful target is a sharp factorial-over-power law, with its constant and phase, or a proof of a specified Gevrey bound. This would identify an optimal truncation index and quantify what accuracy is available before the recessive scale becomes relevant.

13.3 Canonical exponentially improved asymptotics

The exact basis a_n, e_n removes ambiguity from (89), but it does not specify a canonical summation of the dominant formal series. Develop a Borel or contour normalization and compute the associated connection data between the singularities ρ and -1 . The desired result is an error theorem resolving the alternating contribution after optimal truncation, not merely a formal addition of a second exponential scale.

13.4 Recognizing the continued-fraction limit

Find an exact special-value expression for L . The inhomogeneous operator in the remark following (84) provides a starting point: variation of parameters expresses its solutions through integrals involving A and algebraic functions. The condition that the $\sqrt{1-27z}$ coefficient vanish selects $\mathcal{B} - LA$ and hence L . A successful reduction might involve an elliptic integral, an iterated modular integral, or a special L -value. No such identification is proved in this report.

13.5 Arithmetic strength of the approximants

Does the continued fraction prove irrationality of L , or a useful irrationality measure? The exponential error 27^{-n} alone is not enough: the natural continuant denominators contain factorial growth. The first step is to analyze the reduced denominators and common factors of $\mathcal{H}_n$ and $\mathcal{K}_n$. Any arithmetic conclusion must compare the error with the actual denominator after cancellation. The general results in [14] are relevant context, but do not automatically apply to the specific limit constructed here.

13.6 The arithmetic conjecture in the entry

The OEIS entry also displays Lucas congruences for primes in the specified residue classes modulo 7 [1]. The present asymptotic argument neither proves nor disproves them. A separate project should first audit

subsequent work on the Chan–Cooper–Sica conjecture and then exploit the modular parametrization or the functional equation added to OEIS in 2026 (printed as (63)). This is a different target from the asymptotic conjecture resolved here; analytic evidence should not be substituted for an arithmetic proof.

13.7 Coefficient fields for convolution powers

Theorem 10.1 bounds the coefficient field by $\mathbb{Q}(\pi C^2/A_\star^2)$. Determine exactly which convolution powers and which correction orders eliminate the period ratio. The square is special at its first correction, and its third-order differential equation may expose cancellations more systematically. A symbolic classification would clarify when a convolution inherits purely rational normalized corrections and when new period combinations necessarily appear. [Write note, 2026-10-02, batch 77P5: Partly answered by manuscript 32 (Remark 10.2): every normalized coefficient $c_i^{(m)}$ is an explicit polynomial in $q = B^2/A_\star^2 = 4\pi C^2/A_\star^2$ with rational coefficients; for $m = 1$ and $m = 2$ every correction is rational, at every order; for $m \geq 3$ the first correction is a polynomial of degree one in q , so it is rational only if q is, and the rationality of $q = 256\pi^6/(3\mathcal{G}_7^4)$ is not known. Still open: for $m \geq 3$, whether any higher correction loses its dependence on q , and the arithmetic nature of q .]

13.8 Noninteger powers and growing convolution order

Extend the convolution theorem to a specified branch of $A(z)^u$ for noninteger real or complex u . Zero avoidance and possible additional singularities must be addressed rather than assumed. A separate uniform problem lets the integer $m = m(n)$ grow. The fixed- m proof is then insufficient because powers of the analytic boundary part and the singular part can compete. Determine the applicable uniform ranges and the transition scales from the actual local expansion.

13.9 Positive integral representations

Find a direct integral or moment representation for a_n that makes positivity and possible stronger inequalities transparent. A compact real support reflecting both characteristic roots would be especially informative, but such a representation and positivity of its density are not established here. Potential consequences include explicit log-convexity thresholds, Hankel determinant signs, and alternative error bounds for coefficient asymptotics. [Write note, 2026-10-04, batch 97G: Partly answered by manuscript 07 (Section 17). The log-convexity threshold is $n = 1$ (Proposition 17.5); the Hankel matrix (a_{i+j}) is TP₅ but not TN₆ (Theorem 17.1), so (a_n) has no positive representing measure on $[0, \infty)$, while $a_0, \dots, a_{10}$ do have one and a_{11} falls short by an exact amount (Theorem 17.2). This write adds that no tail $(a_{N+n})_{n \geq 0}$ has one either (Theorem 17.17), because A is singular at $z = -1$ (Proposition 17.16), and that a positive measure on $\mathbb{R}$ with moments a_n , if one exists, is unique and supported in $[-1, 27]$, a compact real support reflecting both characteristic roots, as anticipated above (Proposition 17.19). Its existence, and a direct integral formula, remain open (Subsection 17.13, item 1).]

13.10 Automated connection-constant recognition

For families of quadratic-coefficient recurrences with two distinct characteristic roots, combine three exact procedures: derive the Fuchsian equation, certify a rational hypergeometric pullback, and test whether a singular endpoint is a known complex-multiplication value. The intended output should include a symbolic

certificate and a branch check, not just a high-precision guess. A279619 is a compact test case because the rational parameter turns the dominant endpoint into $w = 0$.

13.11 A direct modular proof of the product identity

Can the product $CA_\star = 3\sqrt{3}/(8\pi^{3/2})$ be obtained directly from a single level-seven modular relation? The proof here combines a Ramanujan identity with a period evaluation. A direct connection or Wronskian argument on the modular side may unify the two normalizations and generalize to other Apéry-like equations. Such a proof could also explain which sequences exhibit reciprocal boundary and singular periods. [*Merge note, batch 7302*: Manuscript 56’s proof (Section 6) obtains $CA_\star$ as half of Hirschhorn’s constant C_l of the square, and $A_\star$ from a single CM eta product. It is modular on the boundary value but still takes the singular coefficient from the separately proved asymptotic of A183204, so the question as posed, a single modular relation giving the product, remains open.] [*Write note, 2026-10-02, batch 77P5*: Answered in substance for A279619 by manuscript 32 (Proposition 16.2). Differentiating the Fricke involution $z_7(-1/(7\tau)) = -i\sqrt{7}\tau z_7(\tau)$ at its fixed point τ_* gives $BA_\star = -3\sqrt{3}/(4\pi)$, that is $CA_\star = 3\sqrt{3}/(8\pi^{3/2})$; the only other input is O’Brien’s identity $dw_7/d\tau = 2\pi i w_7 z_7^2$, which converts the τ -derivative into a derivative in $\sqrt{1-27z}$. No hypergeometric identity, no period evaluation and no input from A183204 is used. The mechanism is not new: it is the level-seven case of Guillera and Zudilin’s modular limit formula [18, eq. (27)], which is not specific to level seven. What remains open is the last part of the question: which sequences exhibit reciprocal boundary and singular periods, and why.]

13.12 Machine-checked certificates and analytic infrastructure

A staged Lean development can begin with the recurrence, positivity, integer continuants, and the exact Wronskian. These statements need only algebra and induction. The rational identities in (26) and the triangular coefficient algorithm form a second self-contained layer. A later analytic layer must connect convergent local solutions to coefficient asymptotics, including branch choices and remainder estimates. The complex-multiplication identities should be stated as explicit dependencies until they too are formalized. This separation prevents an exact symbolic certificate from being mistaken for a proof of analytic continuation or of a special value.

13.13 Manuscript 56’s questions

Manuscript 56 wrote: the exact constant closes one visible gap, but it exposes several deeper ones. Its list is ordered roughly by mathematical importance and by the likelihood that the existing level-seven structure can be used. Each item is followed, where it applies, by its relation to the items above.

- R1. Prove the Lucas-congruence conjecture for A279619.** The OEIS entry conjectures Lucas congruences for primes $p \equiv 0, 1, 2, 4 \pmod{7}$. A promising route is to combine a diagonal or constant-term representation with the Cartier operator and Dwork-style unit-root arguments. The exceptional residue classes should be explained geometrically rather than checked case by case. (Same target as Section 13.6; different suggested route.)
- R2. Find a direct binomial or multisum formula for a_n .** The square coefficients t_n have the compact sum (37), while the square-root coefficients are presently most natural through their recurrence or modular expansion. A positive multisum would likely clarify integrality, congruences, and combinatorial meaning simultaneously.

- R3. Determine supercongruences and the underlying motive.** A183204 belongs to the sporadic Apéry-like family [9]. The square-root sequence may encode a lower-weight or half-integral structure. Identify the relevant crystalline Frobenius polynomial and test whether a_{mpr} satisfies congruences stronger than Lucas congruence.
- R4. Classify all CM evaluations of $A(X_7(\tau))$.** For imaginary quadratic τ , Chowla–Selberg expresses eta values in gamma products. Determine which CM points give algebraic $X_7(\tau)$, then tabulate exact series evaluations $\sum a_n X_7(\tau)^n$. The values at $1/27$ and $1/125$ should be the first entries of a larger level-seven atlas.
- R5. Compute exact connection matrices.** Equation (17) has singular points $0, 1/27, -1$, and ∞ . The present article computes one connection coefficient between 0 and $1/27$. Determine the full 2×2 connection matrices between Frobenius bases at all singular points, preferably in closed gamma form. (Manuscript 58’s companion adds the exact recessive amplitude $D_{\min}$ of (88), a connection datum of the recurrence at the root -1 ; the full matrices remain open.) [Write note, 2026-10-04, batch 97G: Advanced at this write (Proposition 17.16): continued along $(-1, 0]$, $A = U_- + B_- \sqrt{1+z} V_-$ near $z = -1$, with an explicit hypergeometric expression for B_- and $-0.2075 < B_- < -0.1754$, so A is genuinely singular at -1 . The closed forms $B_- = -2\pi/\mathcal{G}_7$ and $U_-(0) = \mathcal{G}_7/(4\pi^2)$ are observed numerically, not proved (Subsection 17.13, item 11). This is one connection datum between 0 and -1 , bearing also on R7 and on the question on exponentially improved asymptotics in Section 13; the full matrices remain open.]
- R6. Establish global log-convexity or stronger total positivity.** We proved eventual log-convexity. Exact computation suggests stronger positivity phenomena. Seek a Stieltjes-moment representation, a continued fraction, or a production matrix proving log-convexity for all n and perhaps total positivity of the Hankel matrix (a_{i+j}) . (Overlaps “Positive integral representations” above.) [Write note, 2026-10-04, batch 97G: Answered by manuscript 07 (Section 17): log-convexity holds at every index (Proposition 17.5); the Hankel matrix (a_{i+j}) is strictly totally positive of order exactly five (Theorem 17.1); and there is no Stieltjes-moment representation, of the sequence (Theorems 17.1 and 17.2) or, by this write, of any of its tails (Theorem 17.17). The continued-fraction and production-matrix routes were not pursued. Whether a positive Hamburger measure exists remains open (Subsection 17.13, item 1).]
- R7. Develop rigorous optimal truncation and exponentially small terms.** The algebraic expansion is controlled by the singularity at $1/27$. After optimal truncation, the next singularity at -1 should generate an exponentially small, alternating contribution. Determine the associated transseries and Stokes constants. (Same target as manuscript 58’s late-order and exponentially improved items above. Manuscript 58’s Theorem 2.6 constructs the alternating recessive solution of the recurrence exactly; what remains open is the canonical summation and the Stokes data of a_n itself.)
- R8. Invert the full expansion.** Extend (94) to a systematic inverse transseries for the integer threshold $N(Y)$, including rational corrections and a proof of a uniform rounding rule. (Re-scoped: the systematic inverse with rational corrections is (95)–(97), an instance of `t2:thm:balanced-inverse`, and the rounding brackets are Proposition 11.2. A uniform rounding rule is not available: Proposition 11.2 decides $N(Y)$ only when the two ceilings coincide, and manuscript 58 makes no claim that $N(Y) = \lceil x_M(Y) \rceil$ uniformly. What remains open is explicit K_M, Y_M , Section 13.1.)
- R9. Formalize the algebraic core.** A realistic Lean project would formalize the recurrence-to-ODE step, the Frobenius coefficient recurrence, the formal identity $\mathcal{E}_{27}[S] = 0$, and finite-order transfer algebra. The Dedekind eta and Chowla–Selberg inputs would initially be isolated as analytic theorems, making the boundary between formalized algebra and classical complex analysis explicit. (Same programme as Section 13.12.)

R10. Audit neighboring level-seven OEIS entries. The same thesis introduced the quartic-recurrence sequence A279613 and related modular functions. Its oscillatory asymptotic is still listed conjecturally. The exact-CM/connection strategy used here suggests a systematic attack on its amplitude and phase constants, although the multiple dominant singularities make the problem substantially harder.

14 Scope, audits, and conclusion

14.1 Manuscript 56's scope and novelty audit

The mathematical proof of Section 6 uses published structural identities, but the closed form (3) is obtained there by combining them in a way not stated in the inspected sources. The audit performed for manuscript 56 covered:

- the current OEIS pages for A279619 and A183204;
- O'Brien's 2016 thesis, including its recurrence, modular parametrization, square relation, and numerical asymptotic work;
- Hirschhorn's proof of the A183204 asymptotic;
- van der Poorten and Williams' explicit eta-value formula;
- targeted web and repository searches for A279619 together with the gamma product or the displayed decimal constant;
- the existing OEIS-asymptotics reports in the ProveIt repository.

No prior exact identification of the A279619 constant was found in that search: the constant appears not to have been recorded in the OEIS entry, O'Brien's thesis, Hirschhorn's paper, or the targeted searches. This is evidence of novelty, not a substitute for a full bibliographic review or peer review, and not a claim that no equivalent formula can exist elsewhere. All theorem statements are supported by the derivations given and by the reproducible checks shipped with the source. Manuscript 58's audit (Section 1 and 58-gamma-SOURCE-AUDIT.md) reached the same conclusion independently.

14.2 Integration roadmap and conclusion

A natural repository layout separates four components: sequence recurrences and exact certificates; local analytic representation; asymptotic coefficient calculus; and inverse expansions. The first and third components can be tested independently of modular-form machinery. The second supplies the analytic existence and connection constants that formal manipulations alone cannot establish. The fourth should retain an explicit distinction between a continuous inverse scale and an integer threshold function.

No existing Lean declaration is assumed to certify the special-function steps of this report. A useful first integration milestone is a machine-checked discrete Wronskian with rational bracketing of L . The next is a generic formal-series lemma giving (75); both are substantially smaller than a full formalization of the gamma constant. The supplied Python verification programs are a reproducible algebraic audit and a source of test data, not a replacement for that formalization.

The principal outcome is a precise resolution of the asymptotic statement displayed for A279619:

$$a_n \sim \frac{\sqrt{3\pi}}{\Gamma(1/7)\Gamma(2/7)\Gamma(4/7)} \frac{27^n}{n^{3/2}}.$$

The same argument supplies the boundary sum, every algebraic correction with a fixed-order error bound, and a useful convolution calculus. The independent companion construction reveals an alternating recessive solution with an exact gamma-product amplitude. Inversion then yields a Lambert- W scale and rigorous rounding brackets. The remaining questions concern sharper effective bounds, canonical exponentially improved expansions, arithmetic properties, and broader families, rather than the existence or value of the leading asymptotic constant proved here.

15 Provenance of this report

- **Two manuscripts**, both of batch 73O2 of ProveIt’s incoming-reports intake, both dated October 1, 2026:
 - manuscript 58, *An exact gamma constant for an Apéry-like sequence* (author line “Research report prepared for Vladimir Reshetnikov”; 22-page PDF), archive A279619_Gamma_Asymptotics.zip, arrived in commit aa43cc555; pin a5ad4ac2e2d4e642a6c015836a70492d40fcb71 (the repository snapshot it inspected). It is the base text: everything not marked otherwise.
 - manuscript 56, *Exact asymptotics of OEIS A279619: a level-7 CM evaluation, the missing connection constant, and a full coefficient expansion* (author line “A mathematical analysis prepared for Vladimir Reshetnikov”; 15-page PDF), archive A279619_exact_asymptotics.zip, arrived in commit 58dbe2cd3; no pin (it cites no repository revision). It contributes the material listed in Section 1.3.
- **Placement:** commit 6e193dd4f staged manuscript 58’s article, README and source audit and both manuscripts’ code and data (prefixes 58-gamma- and 56-exact-); manuscript 56’s article and README and both PDFs were not staged and survive in their arrival commits.
- **Independence.** Manuscript 58’s pin (12:35 PDT) is earlier than manuscript 56’s arrival (12:56 PDT); manuscript 56 cites no revision. Neither saw the other. Their texts share formulas, the opening terms and the O’Brien bibliography entry, but no prose.
- **Where the merge chose.** (i) Manuscript 58’s analytic justification (the dented disk) is used for both proofs; manuscript 56’s transfer step is kept verbatim in a merge note (Section 7.2). (ii) Results proved in both are printed once in manuscript 58’s wording (the main theorem, $A(\rho)$, the differential equation, the Frobenius recurrence, the formal identity at infinity, the Lambert- W core, the A183204 constant), with manuscript 56’s distinct proofs and forms kept. (iii) Manuscript 56’s notation was rewritten in manuscript 58’s (Section 1.4). (iv) Both abstracts were replaced by one; both title pages by one. (v) Manuscript 56’s reproducibility appendix, which lists its delivery file names, is replaced by the report README. (vi) The two OEIS proposals are both printed (Appendix C); manuscript 56’s gains the credit to O’Brien that manuscript 58’s already had. (vii) The two bibliographies are merged; five entries that manuscript 56 listed but never cited (Cooper, Cox, Apostol, DLMF, Flajolet–Sedgewick) are now cited where they are used.
- [Write note, 2026-10-02, batch 77P5: **A third manuscript**, added in batch 77P5: manuscript 32 of batch 77, *The exact asymptotic constant for A279619 and its fixed weight family* (dated October 2, 2026;

empty author line; 15-page PDF), archive `level7-asymptotics-replay.zip`, arrived in commit `096ee7b87` and placed in commit `4f11bc9c0`. It has no pin: it names no repository revision, and it cites neither this report nor manuscripts 56 and 58. It shares 11 of its 445 long source lines with manuscript 58's source and 19 with manuscript 56's (formulas and bibliography), and is not a version of either. The placement staged only its replay program (`code/77-32-level7-replay.py`) and that program's reference output (`data/77-32-level7-replay_results.json`). Its article, PDF, README, build script, manifest (8 of 8 entries verified at placement and retired) and manifest checker were not staged, and its `requirements.txt` is byte-identical to `data/58-gamma-requirements.txt`; all survive in the arrival commit. It contributes the third route of Section 16 and Remarks 10.2 and 11.3. **Where this addition chose.** (i) Its material is a new last section rather than a section after Section 6, so that no existing number changes; its equations are numbered within that section. (ii) Results it shares with this report (the main theorem, $c_1, \dots, c_6$, the differential equation, the triangular algorithm, the convolution theorem, the A183204 case, the Lambert- W core, the rounding brackets for $m = 1$) are printed once, by reference; its different proofs are kept. (iii) Its notation is rewritten in this report's (Table 7). (iv) Two sentences of its Section 1.1 that present O'Brien's Question 10.2 as answered there and the fixed-weight generators as its contribution are not carried; they are replaced by the scope statement of Section 16. (v) Its abstract and title page are replaced by this report's; its reproducibility section by the report README.]

- [Write note, 2026-10-04, batch 97G: **A fourth manuscript**, added in batch 97: manuscript 07 of batch 97, *A Sharp Positivity Threshold for OEIS A279619: strict total positivity through order five, a twelfth-moment obstruction, and all-orders Hankel asymptotics* (dated October 4, 2026; author line "Research report prepared for Vladimir Reshetnikov"; 23-page PDF), archive `A279619_Hankel_Positivity.zip`, arrived in commit `e88ed8bf6` and placed in commit `a4542b8a3`. Its pin is `a21208b3ff14a07a4c8318dbf916d543acbef043`, the main-tree revision it inspected; this report's directory did not change between the pin and the placement. The placement staged its checker, its two generators and its Makefile (`code/97-07-hankel-*`), its certificates, records, logs, requirements and draft OEIS comment (`data/97-07-hankel-*`) and its source audit (`97-07-hankel-SOURCE_AUDIT.md`); its article, README, PDF and checksum manifest (18 of 18 entries verified at placement and retired) were not staged and survive in the arrival commit. It contributes Section 17 apart from Subsection 17.10, and the notes marked "batch 97G". **Where this addition chose.** (i) A new last section, with equations numbered within it, so that no existing number changes. (ii) Its notation is rewritten in this report's (Table 8); in particular its sequence c_n and corrections a_j are this report's a_n and c_j . (iii) Results it shares with the report are printed once, by reference (the differential equation, $c_1, \dots, c_4$, the ratio coefficients through order four); its recurrence-only proof of the expansion is kept as a marked second route, and the value of C is the report's. (iv) Its inverse for D_r is printed with a note relating it to Section 11 and to `t2:thm:balanced-inverse`. (v) Its citation of this report became internal references, its OEIS reference is [1], and its three literature references are added. (vi) Its two appendices became Subsections 17.14 and 17.11; its title page, abstract and reproducibility instructions are replaced by this report's and by the README. (vii) One unproved sentence (perturbation stability) is moved to the further questions (item 7). (viii) Added at this write: Subsection 17.10, item 11 of Subsection 17.13, and this write's additions to items 1, 2, 3, 7 and 12 there; no claim of manuscript 07 was found wrong.]

16 A third derivation of the constant (manuscript 32, batch 77P5)

[Write note, 2026-10-02, batch 77P5: This section is manuscript 32 of batch 77P5 rewritten in this report's notation. It is a third derivation of the constant first proved in Sections 5 and 6 (manuscripts 58 and 56). Manuscript 32 was written without knowledge of either; it presents its proof as answering O'Brien's Question 10.2 [3], which asks for the constant in familiar terms, and its fixed-weight generators as part of its contribution. Relative to this report both are earlier: Theorem 2.1 already answers Question 10.2, and Theorem 10.1 already gives the fixed-weight family to every order. The family theorem is Theorem 10.1; the explicit generator $\mathcal{R}_{m,j}$ of Remark 10.2 is the addition. Equations in this section are numbered within it.]

What is new to the report is the route. (i) The singular coefficient B is computed directly at the Fricke fixed point $\tau_* = i/\sqrt{7}$ from the Fricke involution of the weight-one theta series (Proposition 16.2); the first proof uses a hypergeometric pullback and Ramanujan's series, the second the square sequence A183204. (ii) The value $\eta(i\sqrt{7})^2$ is derived from an unnormalized Epstein zeta function, with an explicit conductor-two factor (Theorem 16.6); the second proof cites the Chowla–Selberg formula. (iii) The Fricke computation is identified as the level-seven case of Guillera and Zudilin's modular limit formula. (iv) In Remarks 10.2 and 11.3: an explicit generator for the convolution coefficients, rounding brackets for every fixed weight, and an implicit-function proof of the inverse expansion. The numerical values in manuscript 32 are checks on the exact expressions, not inputs to their proofs.

16.1 Notation

Manuscript 32 reuses many letters that this report already gives other meanings. Table 7 maps its notation to the one used here; the last column names the tempting false reading. No normalization changes: its B , C , λ , β , ℓ_j and ρ are this report's, and its odd Frobenius series is V with $V(0) = 1$.

Table 7: Notation of manuscript 32 and of this section.

Object	Here	Manuscript 32	Caution
generating-function variable	z	x	—
local variable $1 - 27z$	s	t	—
local coordinate $\sqrt{1 - 27z}$	$\sqrt{s}$	s	32's s is a square root
Fricke fixed point $i/\sqrt{7}$	τ_*	τ_0	here $\tau_0 = (1 + i\sqrt{7})/2$
the point $(1 + i\sqrt{7})/2$	τ_0	α	here α is an exponent
weight-one theta series	$z_7(\tau)$	$\Theta(\tau)$	—
Hauptmodul and its pullback	w_7, X_7	w, X	here w is 58's parameter
norm form $j_1^2 + j_1 j_2 + 2j_2^2$	$Q_7(j)$	$Q(m, n)$	here $Q(m) = 1 - m + m^2$
its Gram matrix	$\mathbf{M}$	H	here H is Clausen's ${}_3F_2$
boundary value $A(\rho) = z_7(\tau_*)$	$A_\star$	Θ_0, A_0	—
coefficients of a_n	c_j	p_m	—
formal series at infinity	$S_{27}(u)$	$P(u)$	32 also has $P(x) = (1 + x)(1 - 27x)$
$(1 - 27z)(1 + z)$	$D(z)$	$P(x)$	—
analytic Frobenius series	$\widehat{U} = U/A_\star$	$E(t)$	here E_y is Epstein's series
singular Frobenius series	V	$O(t)$	—
gamma-ratio coefficients	$h_i(a, b)$	$g_m(a, b)$	here g_M is a smooth model
convolution weight	m	k	—
$[z^n]A(z)^m$	$a_n^{(m)}$	$a_{k,n}$	—
its constant $mA_\star^{m-1}C$	$C^{(m)}$	C_k	—

Table 7, continued

Object	Here	Manuscript 32	Caution
its corrections	$c_i^{(m)}$	$p_{k,m}$	—
$B^2/A_\star^2 = 256\pi^6/(3\mathcal{G}_7^4)$	$\mathfrak{q}$	q_* (replay: $\mathfrak{q}\mathfrak{0}$)	here $q_0 = 64/85^3$
generator	$\mathcal{R}_{m,j}$	$R_{k,j}$	here R_n is a scaled term
smooth model of $a_n^{(m)}$	$\mathcal{F}_M^{(m)}$	$F_{k,M}$	here F is a ${}_2F_1$
its truncated series	$S_M^{(m)}$	$P_{k,M}$	—
its large inverse	$\tilde{x}_M^{(m)}(Y)$	$\nu_{k,M}(Y)$	here $\nu = N + 1$ (tail)
inverse coefficients	$q_j^{(m)}$	d_j	here d_j are recessive
integer threshold	$N^{(m)}(Y)$	$N_k(Y)$	$N^{(1)} = N$
Epstein argument	σ	z	here z is the GF variable
conductor-two factor	$\varphi_2(\sigma)$	$f(z)$	—
character $(\cdot/7)$ and its L -function	$\chi_{-7}, L(\sigma, \chi_{-7})$	$\chi, L(z, \chi)$	here L alone is the companion limit
$\Gamma(3/7)\Gamma(5/7)\Gamma(6/7)$	$\mathcal{G}_7^-$	$\mathcal{G}_-$	—
Ramanujan’s class invariant	G_7	G_7	not the gamma product $\mathcal{G}_7$

16.2 The Fricke fixed point and the local coefficient

Write $Q_7(j) = j_1^2 + j_1j_2 + 2j_2^2$ for $j = (j_1, j_2) \in \mathbb{Z}^2$, and

$$z_7(\tau) = \sum_{j \in \mathbb{Z}^2} e^{2\pi i \tau Q_7(j)}, \quad \Im \tau > 0. \quad (16.1)$$

It has weight one and the quadratic character of discriminant -7 . With w_7 and X_7 as in (39)–(41), the modular inputs are O’Brien’s published identities [3, Theorems 3.7, 3.8, 3.10, 4.6 and 6.1]:

$$A(X_7(\tau)) = z_7(\tau), \quad \frac{dw_7}{d\tau} = 2\pi i w_7(\tau) z_7(\tau)^2, \quad X_7(\tau) = \left(\frac{\eta(\tau)\eta(7\tau)}{z_7(\tau)} \right)^3. \quad (16.2)$$

(The first and third are (41), which identifies z_7 with this theta series.) They identify germs at the cusp, where $X_7 = e^{2\pi i \tau} + O(e^{4\pi i \tau})$; the germs are then continued along an explicit path. A global inverse of X_7 is neither needed nor asserted.

The branch. By (51), $\eta(\tau_*) = 7^{1/4}\eta(i\sqrt{7})$, $w_7(\tau_*) = 1/7$ and $X_7(\tau_*) = \rho$; all eta and theta values below are positive. For $\tau = iy$ with $y \geq 1/\sqrt{7}$, (16.2) gives $\frac{d}{dy}w_7(iy) = -2\pi w_7(iy)z_7(iy)^2 < 0$, so w_7 increases from 0 to $1/7$ as y decreases from ∞ to $1/\sqrt{7}$; and

$$\frac{dX_7}{dw_7} = \frac{1 - 49w_7^2}{(1 + 13w_7 + 49w_7^2)^2} > 0 \quad (0 < w_7 < 1/7).$$

So X_7 increases from 0 to ρ along the path, its inverse is regular in the interior, and (16.2) identifies the branch reached from the generating function at zero:

$$\lim_{z \uparrow \rho} A(z) = z_7(\tau_*) = A_\star. \quad (16.3)$$

This makes explicit the branch step behind (53).

Proposition 16.1 (Fricke involution). $z_7(-1/(7\tau)) = -i\sqrt{7}\tau z_7(\tau)$.

Proof. The Gram matrix of $2Q_7$ is $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$, with $\det \mathbf{M} = 7$ and $7\mathbf{M}^{-1} = \begin{pmatrix} 4 & -1 \\ -1 & 2 \end{pmatrix}$, which is integrally equivalent to $\mathbf{M}$ (exchange the coordinates and change the sign of one). The two-dimensional Poisson formula for $\sum_{v \in \mathbb{Z}^2} \exp(\pi i \zeta v^T \mathbf{M} v)$ transforms it, at argument $-1/\zeta$, into the sum with matrix $\mathbf{M}^{-1}$ times the factor $(-i\zeta)/\sqrt{\det \mathbf{M}}$. Taking $\zeta = 7\tau$ and using the integral equivalence gives the identity exactly; no unspecified multiplier remains. $\square$

Proposition 16.2 (The local coefficient at the Fricke point). *Let $\sqrt{s} = \sqrt{1 - 27X_7}$, positive on the real path. Then*

$$\sqrt{s} = \frac{1 - 7w_7}{\sqrt{1 + 13w_7 + 49w_7^2}}, \quad \left. \frac{d\sqrt{s}}{d\tau} \right|_{\tau_*} = -2\pi i A_\star^2 \sqrt{\frac{7}{27}} \neq 0, \quad (16.4)$$

and the singular coefficient of (19) is

$$B = \left. \frac{dz_7}{d\sqrt{s}} \right|_{\sqrt{s}=0} = -\frac{3\sqrt{3}}{4\pi A_\star} < 0. \quad (16.5)$$

Equivalently $CA_\star = 3\sqrt{3}/(8\pi^{3/2})$, which is (32) of the first proof and (66) of the second.

Proof. The algebraic identity for $\sqrt{s}$ follows on squaring, since $1 + 13w + 49w^2 - 27w = (1 - 7w)^2$; both square roots are positive for $0 < z < \rho$. Differentiating Proposition 16.1 at τ_* , where $d(-1/(7\tau))/d\tau = -1$ and $-i\sqrt{7}\tau_* = 1$, gives $-z'_7(\tau_*) = -i\sqrt{7}A_\star + z'_7(\tau_*)$, so $z'_7(\tau_*) = \frac{i\sqrt{7}}{2}A_\star$. By (16.2), $w'_7(\tau_*) = \frac{2\pi i}{7}A_\star^2$; since $1 - 7w_7$ vanishes at τ_* , $d\sqrt{s}/dw_7 = -7\sqrt{7/27}$ there, which gives (16.4). The inverse-function theorem lets z_7 be expanded analytically in $\sqrt{s}$, and $B = z'_7(\tau_*)/(d\sqrt{s}/d\tau)(\tau_*)$ is (16.5). Thus the branch of (16.3) satisfies the convergent local relation $A(z) = A_\star + B(1 - 27z)^{1/2} + O(1 - 27z)$. This fixes the sign and the magnitude of the singular term without an asymptotic formula for A^2 and without a numerical fit. The product form follows from $C = -B/(2\sqrt{\pi})$, (29). $\square$

Remark 16.3 (Guillera–Zudilin). For $Z = A^2$ the local expansion gives

$$\lim_{z \uparrow \rho} \sqrt{D(z)} z Z'(z) = -A_\star B \sqrt{\frac{28}{27}} = \frac{\sqrt{7}}{2\pi}, \quad (16.6)$$

which is exactly the level-seven case of Guillera and Zudilin’s equation (27), consistent with their Example 5 and equation (23) [18]. The general connection mechanism and this weight-two limit are prior results; the Fricke computation above is an explicit instance of that mechanism, and the remaining task is the weight-one normalization. In this report’s terms the left side is $2A_\star$ times the limiting flux of (20), which Section 3 identifies as $-\frac{1}{2}\sqrt{28/27}B$.

[Write note, 2026-10-02, batch 77P5: Reciprocal note. The report cooper-level-15-moving-zeros in this directory (batch 77, manuscript 33) applies the same Fricke-point connection mechanism (its theorem “A specified real connection”) to Cooper’s sporadic families of levels 11, 14, 15 and 24, proving the ten constants of Cooper’s Table 8, and credits the same Guillera–Zudilin equation (27). The two reports share no theorem, and neither manuscript cites the other; that report already points here.]

16.3 The CM period from an Epstein zeta function

It remains to evaluate $A_\star$. The result is classical in CM theory and is not claimed as new; the purpose is a self-contained derivation in a fixed normalization, because a normalized instead of an unnormalized Epstein zeta function would introduce a spurious power of 7.

Proposition 16.4 (Kronecker limit, derivative form). *For $y > 0$ and $\Re \sigma > 1$ let $E_y(\sigma) = \sum_{(j_1, j_2) \neq (0,0)} (j_1^2 + y^2 j_2^2)^{-\sigma}$. Then E_y continues meromorphically to a neighbourhood of $\sigma = 0$, and*

$$E'_y(0) = -2 \log(2\pi) - 4 \log \eta(iy). \quad (16.7)$$

Proof. Separate $j_2 = 0$, apply the Mellin identity to the remaining terms, and use $\sum_{j \in \mathbb{Z}} e^{-j^2 t} = \sqrt{\pi/t} \sum_{r \in \mathbb{Z}} e^{-\pi^2 r^2/t}$; absolute convergence for $\Re \sigma > 1$ permits the interchanges. The zero frequency gives a gamma integral, and the nonzero frequencies are evaluated by $\int_0^\infty t^{\nu-1} e^{-at-b/t} dt = 2(b/a)^{\nu/2} K_\nu(2\sqrt{ab})$ ($a, b > 0$). The result,

$$\begin{aligned} E_y(\sigma) &= 2\zeta(2\sigma) + \frac{2\sqrt{\pi} \Gamma(\sigma - \frac{1}{2})}{\Gamma(\sigma)} y^{1-2\sigma} \zeta(2\sigma - 1) \\ &\quad + \frac{8\pi^\sigma}{\Gamma(\sigma)} y^{1/2-\sigma} \sum_{n,r \geq 1} n^{1/2-\sigma} r^{\sigma-1/2} K_{\sigma-1/2}(2\pi ynr), \end{aligned} \quad (16.8)$$

gives the continuation; the double Bessel series and its derivatives in the order converge locally uniformly by exponential decay, so termwise differentiation is legitimate. At $\sigma = 0$ the first term contributes $4\zeta'(0) = -2 \log(2\pi)$; since $1/\Gamma(\sigma) = \sigma + O(\sigma^2)$, $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$ and $\zeta(-1) = -\frac{1}{12}$, the second contributes $\pi y/3$; and $K_{-1/2}(u) = \sqrt{\pi/(2u)} e^{-u}$ makes the third contribute $4 \sum_{n,r \geq 1} e^{-2\pi ynr}/r = -4 \sum_{n \geq 1} \log(1 - e^{-2\pi yn})$. With the product for η the three combine to (16.7). $\square$

In the determinant-one normalization $\tilde{E}_y(\sigma) = y^\sigma E_y(\sigma)$ one has instead $\tilde{E}'_y(0) = -2 \log(2\pi) - \log y - 4 \log \eta(iy)$, because $E_y(0) = -1$; the extra $-\log y$ disappears when the normalization is undone. This agrees with the classical limit-formula literature [21]; the derivation makes the convention independent of a citation.

Proposition 16.5 (The conductor-two factor). *Let $\chi_{-7}(a) = (a/7)$ and $\mathcal{O} = \mathbb{Z}[\tau_0]$, $\tau_0 = (1 + i\sqrt{7})/2$. For $\Re \sigma > 1$,*

$$\sum_{j \neq 0} Q_7(j)^{-\sigma} = 2\zeta(\sigma) L(\sigma, \chi_{-7}), \quad E_{\sqrt{7}}(\sigma) = 2\varphi_2(\sigma) \zeta(\sigma) L(\sigma, \chi_{-7}), \quad (16.9)$$

with $\varphi_2(\sigma) = 1 - 2^{1-\sigma} + 2^{1-2\sigma}$, and both identities continue meromorphically.

Proof. $\mathcal{O}$ is the maximal order of discriminant -7 . Minkowski's bound $2\sqrt{7}/\pi < 2$ puts an integral ideal of norm one in every class, so the class number is one, and the only units are ± 1 . Counting a generator and its negative for each nonzero principal ideal gives the first identity (equivalently, compare Euler factors at split, inert and ramified primes). The form $j_1^2 + 7j_2^2$ belongs to the suborder $\mathbb{Z}[i\sqrt{7}] = \mathbb{Z} + 2\mathbb{Z}\tau_0$. Modulo 2 the minimal polynomial $T^2 - T + 2$ becomes $T(T-1)$, so $\mathcal{O}/2\mathcal{O} \simeq \mathbb{F}_2 \times \mathbb{F}_2$, and the suborder consists of the elements with equal residues in the two factors. A nonzero such element has residues $(1, 1)$, hence odd norm, or $(0, 0)$, hence lies in $2\mathcal{O}$. The prime 2 splits into two primes of norm 2; excluding both multiplies the ideal sum by $(1 - 2^{-\sigma})^2$ and counts exactly the odd-norm elements, while multiplication by 2 multiplies norms by 4 and counts the other class. The two sets are disjoint, which gives $\varphi_2 = (1 - 2^{-\sigma})^2 + 4^{-\sigma}$. The identity holds first for $\Re \sigma > 1$ and then by continuation. No CM formula for a nonfundamental discriminant is assumed. $\square$

Theorem 16.6 (The eta value; classical).

$$\eta(i\sqrt{7})^2 = \frac{\mathcal{G}_7}{8 \cdot 7^{1/4}\pi^2}, \quad A_\star = 3\eta(\tau_\star)\eta(i\sqrt{7}) = 3 \cdot 7^{1/4}\eta(i\sqrt{7})^2 = \frac{3\mathcal{G}_7}{8\pi^2}, \quad B = -\frac{2\pi\sqrt{3}}{\mathcal{G}_7}. \quad (16.10)$$

Proof. From $L(\sigma, \chi_{-7}) = 7^{-\sigma} \sum_{a=1}^6 \chi_{-7}(a) \zeta(\sigma, a/7)$, $\zeta(0, u) = \frac{1}{2} - u$ and $\zeta'(0, u) = \log \Gamma(u) - \frac{1}{2} \log(2\pi)$ (the character is +1 at 1, 2, 4 and -1 at 3, 5, 6),

$$L(0, \chi_{-7}) = 1, \quad L'(0, \chi_{-7}) = -\log 7 + \log(\mathcal{G}_7/\mathcal{G}_7^-).$$

With $\varphi_2(0) = 1$, $\varphi_2'(0) = -2 \log 2$, $\zeta(0) = -\frac{1}{2}$ and $\zeta'(0) = -\frac{1}{2} \log(2\pi)$, differentiating (16.9) gives $E'_{\sqrt{7}}(0) = \log(14/\pi) - \log(\mathcal{G}_7/\mathcal{G}_7^-)$. Comparing with (16.7) at $y = \sqrt{7}$ and using Gauss's multiplication formula $\mathcal{G}_7 \mathcal{G}_7^- = \prod_{a=1}^6 \Gamma(a/7) = (2\pi)^3/\sqrt{7}$ gives $\eta(i\sqrt{7})^4 = \mathcal{G}_7^2/(64\sqrt{7}\pi^4)$, and positivity fixes the square root. At $\tau_\star$ the third identity of (16.2) and positivity give $A_\star = 3\eta(\tau_\star)\eta(i\sqrt{7})$, and $\eta(\tau_\star) = 7^{1/4}\eta(i\sqrt{7})$. Finally Proposition 16.2 gives B . $\square$

The first value is (60) squared, the second is (62), and the third is (67).

Remark 16.7 (A classical cross-check, and its relation to the second proof). Manuscript 32 compares the value with the fundamental-discriminant formula. Zagier gives [20, p. 95] $7^{1/4}|\eta(\tau_0)|^2 = \mathcal{G}_7/(4\pi^2)$. In Rebák's notation the class invariant is $\mathcal{G}_n = 2^{-1/4}r^{-1/24} \prod_{j \geq 0} (1 + r^{2j+1})$ with $r = e^{-\pi\sqrt{n}}$, and the classical evaluation $\mathcal{G}_7 = 2^{1/4}$ is recalled in the proof of his Theorem 4.1 [22]. Dividing the eta products with $r = e^{-\pi\sqrt{7}}$ gives $|\eta(\tau_0)|/|\eta(i\sqrt{7})| = r^{-1/24} \prod_{j \geq 0} (1 + r^{2j+1}) = \sqrt{2}$, which again gives (16.10). In substance this is the route of Section 6: Zagier's value is (57) (there from van der Poorten and Williams's form of Chowla–Selberg), and the ratio $\sqrt{2}$ is (60) (there from Weber's function with $u = -1$). Neither classical evaluation is claimed as new. The cross-check separates the discriminant -7 period at τ_0 from the conductor-two point $i\sqrt{7}$.

16.4 Continuation, transfer and the third proof

Manuscript 32 justifies the continuation without the dented disk of Section 3. The equation (17) has finite singular points 0, ρ and -1 only. The germ at zero is analytic, and every other point of $|z| < \rho$ is ordinary, so A is single-valued and analytic there; this also justifies continuing (16.2) along the path above. For $\rho < R < 1$ the slit disk $\{|z| < R\} \setminus [\rho, R)$ contains no singular point other than 0; loops in it minus $\{0\}$ have monodromy generated by a loop about zero, which fixes the analytic germ. So the continued solution is single-valued, its singularity at zero is removable, and it supplies the complex Δ -domain at ρ that singularity analysis needs, not merely a real-axis limit. In $s = 1 - 27z$ the equation is (18), with exponents 0 and $\frac{1}{2}$, so $A = A_\star \widehat{U}(s) + B\sqrt{s} V(s)$ with no logarithmic term; the constants are fixed by (16.3) and Theorem 16.6, and $B \neq 0$ makes ρ a genuine singularity, the only one on $|z| = \rho$.

For the transfer, expand $\widehat{U}$ through degree $M + 1$ and V through degree M . The subtracted integral-power part is a polynomial in z ; it is not necessary to assert that the full analytic branch $U(1 - 27z)$ extends to $z = 0$. The remainder is analytic in the slit domain and of local size $O(s^{M+3/2})$, so its coefficients are $O_M(27^n n^{-M-5/2})$ by a Hankel-type contour near ρ and an outer contour beyond ρ [12, Chapter VI]. With (73) this gives

$$a_n = 27^n B \sum_{j=0}^M v_j \frac{\Gamma(n-j-\frac{1}{2})}{\Gamma(-j-\frac{1}{2})\Gamma(n+1)} + O_M(27^n n^{-M-5/2}), \quad (16.11)$$

whose leading term is $-B/(2\sqrt{\pi}) = \sqrt{3}\pi/\mathcal{G}_7 = C$. The leading bound makes $\sum a_n \rho^n$ convergent, and Abel's theorem with (16.3) identifies its sum as $A_\star$. This completes the third proof of Theorem 2.1.

16.5 Generators for every fixed order

Both Frobenius series come from one recurrence. For $r \in \{0, \frac{1}{2}\}$ put $f_{r,-1} = 0$, $f_{r,0} = 1$; then for $j \geq 1$ and $v = r + j$,

$$f_{r,j} = \frac{(29(v-1)^2 + 14(v-1) + \frac{20}{9})f_{r,j-1} - (v - \frac{4}{3})(v - \frac{5}{3})f_{r,j-2}}{14v(2v-1)}, \quad (16.12)$$

with no vanishing denominator. For $r = \frac{1}{2}$ this is (69) ($f_{1/2,j} = v_j$). For $r = 0$ it gives the normalized analytic part, which the proof of Theorem 10.1 uses only through its first coefficient:

$$\widehat{U}(s) = \frac{U(s)}{A_\star} = 1 + \frac{10}{63}s + \frac{986}{11907}s^2 + \frac{89984}{1607445}s^3 + O(s^4). \quad (16.13)$$

Both converge near zero by Frobenius theory, not merely as formal series.

Let B_k be the Bernoulli polynomials and define rational polynomials $h_i(a, b)$ by

$$\sum_{i \geq 0} h_i(a, b)u^i = \exp\left(\sum_{r \geq 1} \frac{(-1)^{r+1}(B_{r+1}(a) - B_{r+1}(b))}{r(r+1)}u^r\right), \quad (16.14)$$

which is the fixed-order expansion $\Gamma(n+a)/\Gamma(n+b) \sim n^{a-b} \sum_i h_i(a, b)n^{-i}$ obtained by subtracting two Stirling series for $\log \Gamma$ [13, §5.11]. Since $\Gamma(-\frac{1}{2})/\Gamma(-j - \frac{1}{2}) = (-1)^j(\frac{3}{2})_j$, (16.11) gives

$$c_i = \sum_{j=0}^i v_j (-1)^j (\frac{3}{2})_j h_{i-j}(-j - \frac{1}{2}, 1), \quad (16.15)$$

each coefficient by finitely many rational operations. This is the “Bernoulli-polynomial gamma-ratio” route by which the batch-73O2 intake checked $c_0, \dots, c_8$. Manuscript 32’s second, independent generator is the formal equation $\mathcal{E}_{27}[S_{27}] = 0$ of (74) with the triangular step (75) (denominator $28m$), proved in both earlier manuscripts and printed once there; it gives uniqueness and rationality and checks (16.15). The formal equation identifies the coefficients of an actual expansion only because the expansion has already been established; a formal ansatz alone determines neither the connection constant nor the remainder. Manuscript 32 prints $c_1, \dots, c_6$, equal to (5) and (76).

16.6 Scope, prior work and the replay

Manuscript 32’s statement of prior work and scope, kept here:

- O’Brien’s thesis establishes the recurrence and the modular parametrization; its Conjecture 8.1 gives the corrections c_1, c_2, c_3 , its Section 8.4.3 estimates the leading constant numerically, and its Question 10.2 asks for an exact expression [3]. The OEIS entry inspected on October 2, 2026 still labelled the leading equivalent conjectural [1].
- The weight-two sequence A183204, its constant and its corrections through the sixth order are Hirschhorn’s [4]. Cooper’s 2025 article [19], inspected in its author preprint of May 8, 2024, recovers this known weight-two asymptotic in its Section 10; its Section 11 discusses the weight-one sequence in connection with congruences and cites O’Brien, but does not supply the weight-one connection constant.
- The general modular radial-connection mechanism predates the manuscript (Remark 16.3); its role here is to connect the weight-one amplitude to the CM value $A_\star$.

- The source search behind these comparisons is bounded and does not establish exhaustive priority. The modular identities, the classical CM values, Poisson summation, Frobenius theory, singularity transfer and implicit inversion are established tools.
- All asymptotic statements have a fixed weight m and a fixed truncation order, and the implied constants may depend on both. No claim is made about a growing weight, a growing truncation order, uniformity in either, effective inverse constants or cutoffs, exponentially improved expansions, optimal truncation, or the oscillatory companion A279613.

The replay program `code/77-32-level7-replay.py` uses exact rational arithmetic for the Frobenius, recurrence, family and inverse generators and high-precision floating point for the numerical checks; its reference output is `data/77-32-level7-replay_results.json` (order 6, $n \leq 1600$, 100 digits). It compares the two rational generators, checks the displayed coefficients, generates exact a_n from (1) with integrality and monotonicity checked through $n = 1600$, checks the weight-two binomial sum for $n \leq 40$, evaluates the eta, theta, Fricke and Epstein identities to relative error below 10^{-80} at 100 digits, and compares forward and inverse residuals for $m = 1, 2, 3$; for instance $n(a_n/(C27^n n^{-3/2}) - 1) \rightarrow -215/1008$, and multiplying by n^{M+1} after subtracting M corrections exposes the next coefficient. These are finite symbolic identities and numerical consistency checks. They do not replace the analytic proof, establish integrality for all indices, prove interval bounds, or turn the existential inverse brackets into a certified rounding algorithm. The intake reran it on a copy; the output agreed with the reference in every entry except the elapsed time and the Python version (README).

17 Hankel positivity: a sharp order-five threshold and the twelfth-moment obstruction (manuscript 07, batch 97)

[Write note, 2026-10-04, batch 97G: This section is manuscript 07 of batch 97 (*A Sharp Positivity Threshold for OEIS A279619: strict total positivity through order five, a twelfth-moment obstruction, and all-orders Hankel asymptotics*, dated October 4, 2026), rewritten in this report's notation. Unlike the three earlier sources, it was written with this report in hand: it inspected the repository at revision a21208b3f, at which this report was the text of Sections 1–16, and it addresses research question R6 of Section 13.13 (global log-convexity or stronger total positivity, perhaps through a Stieltjes-moment representation) and part of Section 13's question on positive integral representations. Its gamma constant is this report's Theorem 2.1, credited as prior work and not re-derived; none of its new sign or moment results uses it. Subsection 17.10 and Remark 17.8 were added at this write and are marked there. Equations in this section are numbered within it.]

What the section adds to the report is the following. (i) Global strict log-convexity, $a_{n-1}a_{n+1} > a_n^2$ for every $n \geq 1$, by an elementary ratio argument (Proposition 17.5); the report had only the eventual statement of Corollary 2.3 and Section 7.4. (ii) The Hankel matrix $(a_{i+j})_{i,j \geq 0}$ is strictly totally positive of order five but not totally nonnegative of order six (Theorem 17.1), the positive half by 21 explicit rational polynomials with 1,090 positive coefficients and an exact, standard-library checker. (iii) $a_0, \dots, a_{10}$ are the moments of a positive measure on $[0, \infty)$ supported at zero and five positive points, but a_{11} is not: the least admissible eleventh moment is explicit and attained by a unique measure (Theorem 17.2). (iv) A fixed-order asymptotic theorem, to every order, for shifted Hankel determinants of any sequence with a full power-law expansion, an explicit coefficient generator, and its A279619 instance (Theorems 17.12 and 17.3); each fixed order of total positivity is restored by a sufficiently far shift (Corollary 17.14). (v) A recurrence-only route to the expansion part of Theorem 2.1 (Subsection 17.6) and a Lambert- W inverse for the Hankel determinants (Subsection 17.9). (vi) Added at this write: the generating function is singular at $z = -1$, with an explicit negative connection coefficient (Proposition 17.16); hence no tail $(a_{N+n})_{n \geq 0}$ is a Stieltjes moment sequence

(Theorem 17.17), which answers manuscript 07's third question, and a positive Hamburger measure, if one exists, lives on $[-1, 27]$ and charges every neighbourhood of -1 (Proposition 17.19).

17.1 Notation

Manuscript 07 calls the sequence c_n and its asymptotic corrections a_j : exactly this report's letters, swapped. This section uses the report's letters throughout, so a_n is A279619 and c_j are the corrections of (4). Its other symbols that collide with the report's are renamed as in Table 8, whose last column names the tempting false reading. No normalization changes: its C is the report's C , its corrections are the report's c_j , and its generating function is A . Its own internal overloads (E for three families, Δ for a finite difference and a Vandermonde product, w for a vector and for weights, M for a matrix and for an order) are separated as well.

Table 8: Notation of manuscript 07 and of this section.

Object	Here	Manuscript 07	Caution
the sequence A279619	a_n	c_n	here c_j are corrections
its asymptotic corrections	c_j	a_j	here a_n is the sequence
generating function	$A(z)$	$F(z)$	here F is ${}_2F_1(\frac{1}{12}, \frac{5}{12}; 1; \cdot)$
recurrence coefficients	$P_n, \tilde{Q}_n$	$p(n), q(n)$	P_n as in (83); there $Q_n = n^2 \tilde{Q}_n$
backward ratio a_n/a_{n-1}	q_n	r_n	here r is a characteristic root
forward ratio a_{n+1}/a_n	q_{n+1}	R_n	here $R_n = a_n n^{3/2}/27^n$
ratio map	f_n	$f_n; F_n = f_{n+1}$	—
elementary barriers	lo_n, hi_n	$\ell_n^{\text{basic}}, u_n^{\text{basic}}$	here ℓ_j are log coefficients
ratio coefficients and series	$\omega_j, \Omega(t)$	$b_j, B(t)$	here b_n is the companion, B the singular coefficient
barrier half-width	K	K	here $K(k), K_M, K_0$
barrier polynomials	$\Pi_-(n), \Pi_+(n)$	$\ell(n), h(n)$	—
ratio enclosure	$\Pi_{\pm}(n)/n^{11}$	$L(n), U(n)$	here L is the companion limit, U the analytic part
barrier certificates	Ξ_0, Ξ_-, Ξ_+	E_0, E_-, E_+	here $\mathcal{E}_r$ is a formal operator
normalized ratios	$\psi_k(n, y)$	$x_k(n, R)$	here $x_0(Y), x_M$ are inverses
normalized determinant	$\widehat{D}_r(n, y)$	$d_r(n, R)$	here d_j are recessive coefficients
positive denominators	$\text{Den}_r(n), \text{Den}_r^*(n)$	$E_r(n), d(n)$	here $D(z) = (1 - 27z)(1 + z)$
certificate polynomial	$Q_r(n, u)$	$Q_r(n, u)$	here $Q(m), Q_7, Q_n$
its power and Bernstein coefficients	$Q_{r,j}(n), Q_{r,k}^B(n)$	$q_{r,j}(n), B_{r,k}(n)$	here q_j are inverse coefficients
shifted Hankel matrix	$\mathbf{H}^{(N)}$	$H^{(N)}$	here H is Clausen's ${}_3F_2$
moment witness, its monic form	$\mathcal{W}(x), \mathcal{W}_1(x)$	$Q(x), P(x)$	here $P(w), P_n$
moment functional	$\mathfrak{M}$	$\mathcal{L}$	here L is the companion limit
least admissible eleventh moment	$a_{11}^{\min}$	$m_{11}^{\min}$	here m is a convolution weight
quadrature lemma: moments, matrices	$\mathbf{m}_k, \mathbf{G}_i, \mathbf{J}$	s_k, M, B, T, J	here $s = 1 - 27z$
nodes and weights there	x_p, ϖ_p	λ_ν, w_ν	here $\lambda = \log 27, \nu = N + 1$
stable-error lemma: error, rate, constant, exponent	$\epsilon_n, \theta, K', \gamma$	e_n, θ, C_0, p	here $e_n = b_n - La_n$
logarithmic scale	$\mathfrak{z}_n$	z_n	here z is the GF variable
formal correction series	$S_{27}(t)$	$\mathcal{A}(t)$	as in (74)
general sequence, base, corrections	$\mathbf{s}_n, \Lambda, \mathbf{a}_j$	s_n, λ, a_j	here $\lambda = \log 27$

Table 8, continued

Object	Here	Manuscript 07	Caution
power exponent (general)	α	α	same
$\prod_{j < r} j!(\alpha)_j$	$\mathcal{V}_r(\alpha)$	$h_r(\alpha)$	here h_i are gamma-ratio coefficients
Hankel exponent $r\alpha + r(r-1)$	$\alpha_r (= r^2 + \frac{r}{2})$	κ_r, κ	here $\kappa(\xi)$ is a Laplace weight
Hankel leading constant	C_r^H	A_r, A	here $A(z), A_\star$
Hankel corrections	$\mathfrak{b}_{r,j}$	$\beta_{r,j}$	here $\beta = 3/2$ in Section 11
normalized entries, differenced entries	$\mathcal{Z}_k(t), \mathcal{X}_s(t)$	$E_k(t), \mathcal{K}_s(t)$	here $\mathcal{K}_n = (n!)^2 b_n$
leading matrices, index matrix	$\mathbf{P}, \mathbf{P}_1, \mathbf{N}$	G, G_1, D	here $\mathcal{G}_7$ is the gamma product
the trace $\text{tr}(\mathbf{P}^{-1}\mathbf{P}_1)$	$\mathcal{T}$	T	here $T(w) = 15w^2 + 96w + 85$
row and column offsets	g_i, g'_j	u_i, v_j	here u_n solutions, v_j Frobenius coefficients
Vandermonde product	$\text{Vand}(g)$	$\Delta(u)$	here Δ is a forward difference
least shift giving TP_r	N_r^{TP}	N_r	here $N(Y), N_M$
inverse of D_r : rate, core, model	$r\lambda, x_0^{[r]}, \mathcal{F}_M^{[r]}$	L_r, x_0, F_M	here $x_0(Y)$ inverts a_n
its log corrections	$\ell_1^{[r]}, \ell_2^{[r]}$	β_1, γ_2	here γ_j in (75)
its discrete inverse, constants	$N^{[r]}(Y), n_*^{[r]}, K_M^{[r]}$	$T_r(Y), n_*, K_{r,M}$	here $N(Y), K_M$ are for a_n

Symbols introduced at this write (Subsection 17.10): $s_- = 1 + z$, the local variable at $z = -1$, and the Puiseux data U_-, B_-, V_- there, the analogues of s, U, B, V at ρ in (19); $\check{w} = 1/w$, the reciprocal of the parameter w of (21) (the letter u is kept for the Bernstein variable of Subsection 17.5); S_μ , the Stieltjes transform of a measure μ ; and $r^*(N)$, the order of first failure of total nonnegativity. Throughout, a real matrix is *strictly totally positive of order r* , written TP_r , if every minor of size at most r is strictly positive, and *totally nonnegative of order r* , TN_r , if these minors are nonnegative; row and column indices of a minor are listed increasingly, and “strictly” is said explicitly because naming conventions in the literature differ. For $r \geq 1, n \geq 0$,

$$D_r(n) = \det[a_{n+i+j}]_{0 \leq i, j < r}, \quad D_0(n) = 1, \quad \mathbf{H}^{(N)} = (a_{N+i+j})_{i, j \geq 0}. \quad (17.1)$$

The $D_r(n)$ are the contiguous minors of $\mathbf{H}^{(0)}$. (There is no other D_r in the report; $D(z)$ and $D_{\min}$ keep their meanings.) A Hankel minor here is not an iteration of the nonlinear log-convexity operator; those are related but different objects.

17.2 The statements

Write the recurrence (1) as

$$P_n = 26n^2 + 13n + 2, \quad \tilde{Q}_n = 27n^2 - 27n + 6, \quad (n+1)^2 a_{n+1} = P_n a_n + \tilde{Q}_n a_{n-1}. \quad (17.2)$$

Only $a_0 = 1, a_1 = 2$ and (17.2) are used for the global positivity and moment results; in particular they depend on no modular-form identity and on no assumed asymptotic formula. Since $P_n, \tilde{Q}_n > 0$ for $n \geq 1$, every a_n is positive. Manuscript 07 writes the differential equation (17) as $z(1 - 26z - 27z^2)A'' + (1 - 39z - 54z^2)A' - 2(1 + 3z)A = 0$, manuscript 56’s form; it is background for the analytic questions, and the sign proofs below work directly with the recurrence.

Theorem 17.1 (Sharp global order; manuscript 07). *The sequence is strictly log-convex at every index: $a_n a_{n+2} > a_{n+1}^2$ for all $n \geq 0$. More strongly, $\mathbf{H}^{(0)}$ is TP_5 , whereas*

$$D_6(1) = -26875777009408537346562624 < 0. \quad (17.3)$$

Consequently five is the largest order of global total nonnegativity or strict total positivity, and (a_n) is not a Stieltjes moment sequence.

The positive assertion covers *all* minors of sizes one through five, with arbitrary gaps in their row and column indices, not merely the first few principal determinants; the reduction from contiguous minors to all minors is Subsection 17.5. The negative minor uses rows $0, \dots, 5$ and columns $1, \dots, 6$ of $\mathbf{H}^{(0)}$.

Theorem 17.2 (Exact truncation boundary; manuscript 07). *There is a positive measure supported at zero and five distinct positive points whose moments of degrees $0, \dots, 10$ are $a_0, \dots, a_{10}$. Among all positive measures on $[0, \infty)$ with these eleven moments, the least possible moment of degree eleven is*

$$a_{11}^{\min} = \frac{87170364947144797711187234}{6107572673541}, \quad (17.4)$$

attained by a unique such six-point measure. The actual value $a_{11} = 14272471122560$ falls short by exactly

$$a_{11}^{\min} - a_{11} = \frac{210335094301031002274}{6107572673541} = 34438410.4035 \dots \quad (17.5)$$

Thus eleven is the greatest length of an initial segment of (a_n) that a positive measure on $[0, \infty)$ represents.

The distinction between a Stieltjes measure, on $[0, \infty)$, and a Hamburger measure, on all of $\mathbb{R}$, is essential: manuscript 07 rules out the former and not the latter, and a positive Hamburger representation remains open (Subsection 17.13; Proposition 17.19 restricts where its support could lie).

Theorem 17.3 (Fixed-order Hankel asymptotics; manuscript 07). *For every fixed $r \geq 1$ and every fixed $M \geq 0$,*

$$D_r(n) = C_r^H 27^{rn} n^{-\alpha_r} \left(\sum_{j=0}^M \frac{\mathbf{b}_{r,j}}{n^j} + O_{r,M}(n^{-M-1}) \right), \quad (17.6)$$

where

$$\alpha_r = r^2 + \frac{r}{2}, \quad C_r^H = C^r 27^{r(r-1)} \prod_{j=0}^{r-1} j! \left(\frac{3}{2}\right)_j, \quad \mathbf{b}_{r,0} = 1. \quad (17.7)$$

All $\mathbf{b}_{r,j}$ are rational, and

$$\mathbf{b}_{r,1} = -r \left(r + \frac{1}{2} \right) \left(r - 1 + \frac{215}{1512} \right). \quad (17.8)$$

For every fixed r there is N_r^{TP} such that $\mathbf{H}^{(N)}$ is TP_r for all $N \geq N_r^{\text{TP}}$.

The general determinant mechanism is a consequence of classical finite-difference and Vandermonde algebra, developed in Subsection 17.7 in a usable all-orders form; the sequence-specific contribution is its combination with the sharp global order, the exact truncated-moment obstruction, and certificates covering every index. Manuscript 07 does not claim that asymptotic positivity methods for polynomial-recursive sequences are new: see Hou–Zhang [24] and Mao–Pei [25], whose iterated log-convexity must not be identified with TP_r .

17.3 Global strict log-convexity

Put $\varrho_n = a_n/a_{n-1}$ for $n \geq 1$. The recurrence becomes

$$\varrho_{n+1} = f_n(\varrho_n), \quad f_n(x) = \frac{P_n + \tilde{Q}_n/x}{(n+1)^2}. \quad (17.9)$$

For $x > 0$, f_n is strictly decreasing: a lower bound for an input produces an upper bound for the next ratio.

Lemma 17.4 (Elementary ratio barriers). *For every $n \geq 3$,*

$$\text{lo}_n := \frac{27(2n-3)}{2n} < \varrho_n < \text{hi}_n := \frac{54(n-1)}{2n+1}. \quad (17.10)$$

Proof. At $n = 3$, $\varrho_3 = 168/11$. Direct rational simplification gives

$$f_n(\text{hi}_n) - \text{lo}_{n+1} = \frac{283n - 277}{18(n-1)(n+1)^2} > 0, \quad (17.11)$$

$$\text{hi}_{n+1} - f_n(\text{lo}_n) = \frac{136n^2 - 417n + 162}{9(n+1)^2(2n-3)(2n+3)} > 0; \quad (17.12)$$

with $n = 3 + t$ the second numerator is $136t^2 + 399t + 135 > 0$. Since f_n is decreasing, the two identities carry the bounds from n to $n + 1$. $\square$

Proposition 17.5 (Global strict log-convexity; manuscript 07). *The ratios ϱ_n are strictly increasing, stay below 27 and converge to 27; so (a_n) is strictly log-convex at every index. For $n \geq 3$,*

$$\varrho_{n+1} - \varrho_n > \frac{81}{2(n+1)(2n+1)}. \quad (17.13)$$

Moreover $a_n^{1/n}$ is strictly increasing for $n \geq 1$, with limit 27, and $a_n/27^n$ is strictly decreasing.

Proof. The barrier intervals are separated: $\text{lo}_{n+1} - \text{hi}_n = 81/(2(n+1)(2n+1)) > 0$. Thus $\varrho_{n+1} > \varrho_n$ for $n \geq 3$; the first comparisons follow from $\varrho_1 = 2$, $\varrho_2 = 11$, $\varrho_3 = 168/11$. Both barriers tend to 27, and the upper ones are below 27. Increasing ratios are exactly strict log-convexity. The root assertion follows by taking geometric means of the increasing positive ratios, and the last from $a_{n+1}/a_n < 27$. $\square$

This argument settles the global log-convexity question without computer assistance, and supplies the positive lower bounds and the limiting ratio used in Subsection 17.6. The order-five result needs much tighter ratio control, not a stronger assumption on the sequence. (The increment (17.13) is consistent with the asymptotic increment $81/(2n^2)$ in the proof of Corollary 2.3.)

17.4 The exact Stieltjes obstruction and its optimal repair

A negative determinant is not a numerical instability. Exact recurrence evaluation and integer determinant arithmetic give

r	$D_r(0)$	$D_r(1)$
1	1	2
2	18	188
3	14132	777000
4	504704088	132810611472
5	839827568468592	780401205934374816
6	66730435339909773988512	-26875777009408537346562624

No floating-point determinant is used; the checker regenerates these entries from (17.2). If $a_n = \int_0^\infty x^n d\mu(x)$ for a positive measure, then for every polynomial $v(x) = \sum_{j=0}^5 v_j x^j$,

$$\sum_{i,j=0}^5 v_i v_j a_{i+j+1} = \int_0^\infty x v(x)^2 d\mu(x) \geq 0,$$

so $(a_{i+j+1})_{0 \leq i,j \leq 5}$ would be positive semidefinite, contradicting (17.3). Its inertia is exactly (5, 1): its first five leading principal minors are positive and the sixth negative, so an LDL^T decomposition has five positive pivots and one negative pivot.

A single polynomial witness. The obstruction can be displayed without a determinant. Define

$$\begin{aligned} \mathcal{W}(x) = & 6107572673541x^5 - 427862903728833x^4 \\ & + 10913975768286866x^3 - 122491788886528941x^2 \\ & + 570493806130562193x - 782476408841608767. \end{aligned} \quad (17.14)$$

For the linear functional $\mathfrak{M}[x^j] = a_j$, direct integer arithmetic gives

$$\mathfrak{M}[x \mathcal{W}(x)^2] = -1284636874239646271231341030632234 \quad (17.15)$$

and

$$\mathfrak{M}[x^{j+1} \mathcal{W}(x)] = 0 \quad (0 \leq j \leq 4). \quad (17.16)$$

These finite identities involve only $a_1, \dots, a_{11}$, and the checker verifies each. Since $x \mathcal{W}(x)^2 \geq 0$ on $[0, \infty)$, (17.15) certifies that no positive measure on $[0, \infty)$ has these moments.

A finite spectral construction. The positive truncated representation is given too, so that the obstruction is not overstated.

Lemma 17.6 (A zero-plus-positive-nodes quadrature; classical, proof as in manuscript 07). *Let $m_0, \dots, m_{2m}$ be real numbers such that*

$$\mathbf{G}_0 = (m_{i+j})_{0 \leq i,j \leq m} \succ 0, \quad \mathbf{G}_1 = (m_{i+j+1})_{0 \leq i,j < m} \succ 0$$

(positive definite). Then a positive measure supported at zero and m distinct positive points has the moments $m_0, \dots, m_{2m}$.

Proof. Let $\mathbf{G}_2 = (m_{i+j+2})_{0 \leq i,j < m}$, a positive-definite principal submatrix of $\mathbf{G}_0$. Give the coefficient vectors of polynomials of degree at most $m-1$ the inner product $\langle u, v \rangle_{\mathbf{G}_1} = u^T \mathbf{G}_1 v$, and put $\mathbf{J} = \mathbf{G}_1^{-1} \mathbf{G}_2$. Then $\mathbf{J}$ is self-adjoint and positive definite for this inner product, and $\mathbf{J}e_k = e_{k+1}$ for $k < m-1$, because the k th column of $\mathbf{G}_2$ is the $(k+1)$ st column of $\mathbf{G}_1$. So e_0 is cyclic: $e_0, \mathbf{J}e_0, \dots, \mathbf{J}^{m-1}e_0$ are the standard basis vectors. Consequently $\mathbf{J}$ has m distinct positive eigenvalues $x_1, \dots, x_m$, and the spectral weights $\varpi_p = \|\text{proj}_{x_p} e_0\|_{\mathbf{G}_1}^2$ are all positive. For $0 \leq k \leq 2m-2$, split $k = i+j$ with $i, j \leq m-1$: $\sum_{p=1}^m \varpi_p x_p^k = \langle \mathbf{J}^i e_0, \mathbf{J}^j e_0 \rangle_{\mathbf{G}_1} = m_{k+1}$. For $k = 2m-1$ use $\langle \mathbf{J}^{m-1} e_0, \mathbf{J} \mathbf{J}^{m-1} e_0 \rangle_{\mathbf{G}_1} = (\mathbf{G}_2)_{m-1, m-1} = m_{2m}$. Thus $\sum_p \varpi_p \delta_{x_p}$ has moments $m_1, \dots, m_{2m}$ in degrees $0, \dots, 2m-1$. Put $\mathbf{v} = (m_1, \dots, m_m)^T = \mathbf{G}_1 e_0$. The total mass of the proposed positive nodes is $\sum_p \varpi_p / x_p = \langle e_0, \mathbf{J}^{-1} e_0 \rangle_{\mathbf{G}_1} = \mathbf{v}^T \mathbf{G}_2^{-1} \mathbf{v}$, and the Schur complement in $\mathbf{G}_0 = \begin{bmatrix} m_0 & \mathbf{v}^T \\ \mathbf{v} & \mathbf{G}_2 \end{bmatrix}$ is positive. Therefore

$$\mu = (m_0 - \mathbf{v}^T \mathbf{G}_2^{-1} \mathbf{v}) \delta_0 + \sum_{p=1}^m \frac{\varpi_p}{x_p} \delta_{x_p}$$

is a positive measure of the required kind. □

Apply the lemma with $m = 5$ and $m_j = a_j$: the determinant table and Sylvester's criterion show that both required matrices are positive definite, which proves representability through degree ten. The proof uses the eigenvalues themselves; numerically the five positive nodes are about 2.28210, 7.53071, 14.22768, 20.69126 and 25.32274.

The least possible twelfth term. Now let $\mathbf{G}_1 = (a_{i+j+1})_{0 \leq i, j < 5}$ and $\mathbf{g} = (a_6, a_7, a_8, a_9, a_{10})^\top$. A putative eleventh moment a'_{11} must make $\begin{bmatrix} \mathbf{G}_1 & \mathbf{g} \\ \mathbf{g}^\top & a'_{11} \end{bmatrix}$ positive semidefinite; since $\mathbf{G}_1 \succ 0$, this is equivalent to

$$a'_{11} \geq \mathbf{g}^\top \mathbf{G}_1^{-1} \mathbf{g}. \quad (17.17)$$

The polynomial $\mathcal{W}_1 = \mathcal{W}/6107572673541$ is the monic polynomial $x^5 - \sum_{j=0}^4 (\mathbf{G}_1^{-1} \mathbf{g})_j x^j$. So, for any positive measure with the prescribed moments through degree ten and finite next moment a'_{11} , $\int x \mathcal{W}_1(x)^2 d\mu(x) = a'_{11} - \mathbf{g}^\top \mathbf{G}_1^{-1} \mathbf{g}$. The six-point measure constructed above has its five positive nodes at the zeros of $\mathcal{W}_1$: the spectral quadrature makes the moments of the x -weighted measure agree through degree nine, and (17.16) then forces $\mathcal{W}_1$ to vanish at each node, by the invertibility of a Vandermonde matrix. Hence equality holds in (17.17). Conversely, equality forces support on $\{0\} \cup \{\mathcal{W}_1 = 0\}$, and the masses are then fixed by the first six moments; this is uniqueness at the minimum. Finally

$$a_{11} - \mathbf{g}^\top \mathbf{G}_1^{-1} \mathbf{g} = \frac{D_6(1)}{D_5(1)},$$

which gives (17.4) and (17.5). Shorter initial segments inherit the same representing measure, and every longer one contains the impossible first twelve terms. This proves Theorem 17.2.

17.5 Exact certification of strict total positivity through order five

Reduction to contiguous determinants. The strict form of Fekete's criterion says that a finite matrix whose contiguous minors are all positive is totally positive; its finite-order version, that positivity of all contiguous minors of sizes at most r implies TP_r , is Fallat–Johnson–Sokal [23, Lemma 2.4]. The reduction can be seen in two stages. Restrict first to a consecutive block of at most r rows and apply the full criterion to that rectangular matrix; this permits arbitrary column gaps. Then fix at most r columns and apply the full criterion to the tall submatrix, whose contiguous minors were just shown positive; this permits arbitrary row gaps. Applying this to finite rectangles of an infinite matrix gives the infinite version. For $\mathbf{H}^{(0)}$ it therefore suffices to prove

$$D_r(n) > 0 \quad (n \geq 0, 1 \leq r \leq 5). \quad (17.18)$$

The case $r = 1$ is the positivity of a_n .

A tighter ratio interval. This subsection works with the forward ratio $\varrho_{n+1} = a_{n+1}/a_n$ (manuscript 07's R_n), which satisfies $\varrho_{n+2} = f_{n+1}(\varrho_{n+1})$. The following rational numbers specify a polynomial; they are not an assumption that a divergent or unproved approximation bounds the ratios.

j	ω_j	j	ω_j
0	27	6	65617849635/550731776
1	-81/2	7	-1040841454725/7710244864
2	6315/112	8	520118139886395/3454189699072
3	-56445/784	9	-8041730705126175/48358655787008
4	30827295/351232	10	988630328840718285/5416169448144896
5	-508917075/4917248		

($\omega_0, \dots, \omega_4$ are 27 times the coefficients of (8).) Define

$$K = \frac{14943122191587178247}{37913186137014272}, \quad (17.19)$$

$$\Pi_-(n) = \sum_{j=0}^{10} \omega_j n^{11-j} - K, \quad \Pi_+(n) = \Pi_-(n) + 2K. \quad (17.20)$$

Lemma 17.7 (Certified forward-ratio enclosure; manuscript 07). *For every integer $n \geq 5$,*

$$0 < \frac{\Pi_-(n)}{n^{11}} < \varrho_{n+1} < \frac{\Pi_+(n)}{n^{11}}. \quad (17.21)$$

Proof. Consider the three rational polynomials

$$\Xi_0(n) = \Pi_-(n), \quad (17.22)$$

$$\Xi_-(n) = [P_{n+1}\Pi_+(n) + \tilde{Q}_{n+1}n^{11}](n+1)^{11} - (n+2)^2\Pi_+(n)\Pi_-(n+1), \quad (17.23)$$

$$\Xi_+(n) = (n+2)^2\Pi_-(n)\Pi_+(n+1) - [P_{n+1}\Pi_-(n) + \tilde{Q}_{n+1}n^{11}](n+1)^{11}. \quad (17.24)$$

Every coefficient of each of $\Xi_0(5+x)$, $\Xi_-(5+x)$, $\Xi_+(5+x)$ is strictly positive; their degrees are 11, 13, 13. The exact coefficient lists are `data/97-07-hankel-certificates.json`, checked as described below. It follows that $\Pi_-(n) > 0$. Since $\Pi_+(n) > \Pi_-(n) > 0$, the remaining signs give

$$\begin{aligned} f_{n+1}\left(\frac{\Pi_+(n)}{n^{11}}\right) - \frac{\Pi_-(n+1)}{(n+1)^{11}} &= \frac{\Xi_-(n)}{(n+2)^2\Pi_+(n)(n+1)^{11}} > 0, \\ \frac{\Pi_+(n+1)}{(n+1)^{11}} - f_{n+1}\left(\frac{\Pi_-(n)}{n^{11}}\right) &= \frac{\Xi_+(n)}{(n+2)^2\Pi_-(n)(n+1)^{11}} > 0. \end{aligned}$$

At $n = 5$, $\varrho_6 = 2628/127$, with exact positive margins

$$\varrho_6 - \frac{\Pi_-(5)}{5^{11}} = \frac{82060445111646705659}{14694136472780800000000000}, \quad \frac{\Pi_+(5)}{5^{11}} - \varrho_6 = \frac{1241292957438397992097}{117553091782246400000000000}.$$

Since f_{n+1} is decreasing on positive arguments, these inequalities and the base case prove the enclosure by induction. $\square$

Determinants as polynomials in one unknown ratio. Define rational functions $\psi_k(n, y)$ by

$$\psi_0 = 1, \quad \psi_1 = y, \quad \psi_{k+1} = \frac{P_{n+k}\psi_k + \tilde{Q}_{n+k}\psi_{k-1}}{(n+k+1)^2} \quad (k \geq 1). \quad (17.25)$$

At $y = \varrho_{n+1}$ induction gives $\psi_k = a_{n+k}/a_n$, so

$$\widehat{D}_r(n, y) = \det[\psi_{i+j}(n, y)]_{0 \leq i, j < r}, \quad D_r(n) = a_n^r \widehat{D}_r(n, \varrho_{n+1}). \quad (17.26)$$

Let $\text{Den}_r(n)$ be the positive denominator

r	$\text{Den}_r(n)$
2	$(n+2)^2$
3	$(n+2)^6(n+3)^4(n+4)^2$
4	$(n+2)^8(n+3)^8(n+4)^6(n+5)^4(n+6)^2$
5	$(n+2)^{10}(n+3)^{10}(n+4)^{10}(n+5)^8(n+6)^6(n+7)^4(n+8)^2$

For $r = 2, 3, 4, 5$ the expression

$$Q_r(n, u) = n^{11r} \text{Den}_r(n) \widehat{D}_r\left(n, \frac{\Pi_-(n) + 2Ku}{n^{11}}\right) \quad (17.27)$$

is a polynomial in n, u of degree at most r in u ; this identity is itself part of the certificate, not an unchecked simplification. Write $Q_r(n, u) = \sum_{j=0}^r Q_{r,j}(n)u^j$ and its Bernstein form

$$Q_r(n, u) = \sum_{k=0}^r Q_{r,k}^B(n) \binom{r}{k} u^k (1-u)^{r-k}, \quad Q_{r,k}^B(n) = \sum_{j=0}^k \frac{\binom{k}{j}}{\binom{r}{j}} Q_{r,j}(n). \quad (17.28)$$

Each $Q_{r,k}^B(5+x)$ has all coefficients strictly positive. This is exactly what is needed: for $0 \leq u \leq 1$ every Bernstein basis function is nonnegative and at least one is positive. Formula (17.28) and (17.25) specify every certificate polynomial without a symbolic factorization chosen by hand.

Polynomial family	Number	Degree after translation	Positive coefficients
Ξ_0	1	11	12
Ξ_-, Ξ_+	2	13	28
$Q_{2,k}^B, 0 \leq k \leq 2$	3	22	69
$Q_{3,k}^B, 0 \leq k \leq 3$	4	39	160
$Q_{4,k}^B, 0 \leq k \leq 4$	5	60	305
$Q_{5,k}^B, 0 \leq k \leq 5$	6	85	516
Total	21		1090

Proof of the positive part of Theorem 17.1. For $n \geq 5$, Lemma 17.7 places the true ratio at a unique $u \in (0, 1)$ in (17.27). Positivity of the translated coefficients and of the Bernstein basis gives $Q_r(n, u) > 0$; since n , $\text{Den}_r(n)$ and a_n are positive, (17.26) gives $D_r(n) > 0$ for $2 \leq r \leq 5$. For $0 \leq n \leq 4$ the twenty required determinants are computed exactly (Subsection 17.14). Fekete's finite-order criterion proves $\mathbf{H}^{(0)} \in \text{TP}_5$, and (17.3) makes the order sharp. Strict log-convexity is the case of the 2×2 minors (and is proved independently in Proposition 17.5). $\square$

Why the certificate replay is a proof, not a finite experiment. The data file records each polynomial as a positive rational scale times a list of positive integer coefficients in $x = n - 5$. A SymPy program regenerates these lists; separately, a Python-standard-library program checks the coefficient signs and verifies the claimed identities using only integers and exact fractions, without importing SymPy, mpmath or the generator. The degree argument that makes finite identity replay sufficient is this. For fixed r put $m = 2r - 2$ and

$$\text{Den}_r^*(n) = \prod_{j=2}^m (n+j)^2, \quad \deg \text{Den}_r^* = 4r - 6.$$

By (17.25), $\text{Den}_r^*(n)\psi_k(n, y)$ is polynomial in n, y , of n -degree at most $4r - 6$ and y -degree at most one, for $0 \leq k \leq m$. After substituting $y = (\Pi_-(n) + 2Ku)/n^{11}$, every entry

$$\mathbf{E}_{ij}(n, u) = n^{11} \text{Den}_r^*(n) \psi_{i+j}\left(n, \frac{\Pi_-(n) + 2Ku}{n^{11}}\right)$$

is polynomial of n -degree at most $4r + 5$ and u -degree at most one. The identity to check is

$$\det \mathbf{E}(n, u) = \frac{\text{Den}_r^*(n)^r}{\text{Den}_r(n)} \sum_{k=0}^r Q_{r,k}^B(n) \binom{r}{k} u^k (1-u)^{r-k}. \quad (17.29)$$

The quotient $\text{Den}_r^*(n)^r / \text{Den}_r(n)$ is a polynomial, as the displayed exponents show. Both sides have n -degree at most $4r^2 + 5r$ and u -degree at most r , so the checker verifies (17.29) at $4r^2 + 5r + 1$ distinct integers n and the $r + 1$ values $u = 0, 1/r, \dots, 1$; univariate polynomial uniqueness, applied in u and then in n , proves the full identity. That is $81 + 208 + 425 + 756 = 1470$ exact bivariate evaluation points. The barrier identities use degree bounds 11 and 24 before cancellation and are checked at 12 and 25 points. Thus the proof does not infer a universal sign from sampled signs: it verifies a polynomial identity of bounded degree, then proves its sign for every real $n \geq 5$ from its coefficients; the remaining induction concerns the actual integer-indexed sequence. [Write note, 2026-10-04, batch 97G: The intake re-derived this degree argument by hand ($\text{Den}_r^*(n)\psi_k$ has n -degree at most $2(k - 1)$ by induction) and read the checker in full: it checks the degree bound of every certificate polynomial against $4r^2 + 5r$, including the $\text{Den}_r^*/\text{Den}_r$ multiplier, before interpolating. Given Fekete's classical criterion, the order-five certificate is a proof. Here $\mathbf{E}$ is the checker's matrix, called M in manuscript 07.]

17.6 A recurrence-only route to the expansion of Theorem 2.1

Remark 17.8 (A second route, not a new result). [Write note, 2026-10-04, batch 97G: The expansion part of Theorem 2.1 (existence of $C > 0$ and of the full expansion with rational corrections) is proved in this report by Frobenius theory and singularity analysis, in Section 7 (manuscripts 58 and 56) and Subsection 16.4 (manuscript 32). Manuscript 07 proves it again by an elementary stable-ratio argument that uses only the recurrence; it is printed here as a marked second route of that kind. It does not determine the value of C , which is inherited from Theorem 2.1, and it gives $c_1, \dots, c_4$ equal to (5). Its formal ratio series extends the coefficients of Corollary 2.3 from order four to order eleven.]

This subsection makes the analytic input to Theorem 17.3 independent of the rest of the report, except for the value of C .

Formal coefficients and stability. Let t be a formal variable. Substituting $\varrho_{n+1} \sim \Omega(1/n)$ in the recurrence gives the formal equation

$$(1+t)^2 \Omega(t) \Omega\left(\frac{t}{1-t}\right) - (26 + 13t + 2t^2) \Omega\left(\frac{t}{1-t}\right) - (27 - 27t + 6t^2) = 0, \quad \Omega(t) = \sum_{j \geq 0} \omega_j t^j, \quad \omega_0 = 27. \quad (17.30)$$

For $m \geq 1$ the coefficient of ω_m in the t^m equation is 28, and all other terms involve earlier coefficients, so (17.30) determines $\omega_m \in \mathbb{Q}$ uniquely. Its first eleven coefficients are the table of Subsection 17.5; for example

$$\omega_{11} = -\frac{14905209005450163975}{75826372274028544}, \quad K = 2|\omega_{11}| + 1.$$

This explains how the barrier was found; its validity was proved separately, by the polynomial inequalities.

Lemma 17.9 (A stable error estimate). *If $|\epsilon_{n+1}| \leq \theta|\epsilon_n| + K'n^{-\gamma}$ for all sufficiently large n , with $0 < \theta < 1$ and $\gamma > 0$, then $\epsilon_n = O(n^{-\gamma})$.*

Proof. Iterate the inequality. In the resulting geometric convolution, the terms with indices at least $n/2$ contribute at most a constant times $n^{-\gamma}$; all earlier terms carry a factor at most $\theta^{n/2}$, and their total, with the initial error, is exponentially small compared with $n^{-\gamma}$. $\square$

Proposition 17.10 (All-orders ratio expansion; manuscript 07). *For every fixed $M \geq 0$,*

$$\varrho_{n+1} = \frac{a_{n+1}}{a_n} = \sum_{j=0}^M \omega_j n^{-j} + O_M(n^{-M-1}). \quad (17.31)$$

Proof. Let $\Omega_M(n) = \sum_{j=0}^M \omega_j n^{-j}$. Formal coefficient matching gives $\Omega_M(n+1) - f_{n+1}(\Omega_M(n)) = O_M(n^{-M-1})$. Both $\Omega_M(n)$ and ϱ_{n+1} tend to 27, the latter by Proposition 17.5. Uniformly between these two arguments, $|f'_{n+1}(x)| = \bar{Q}_{n+1}/((n+2)^2 x^2) \rightarrow 1/27$, so for large n it is bounded by a fixed $\theta < 1$. The mean value theorem and Lemma 17.9 applied to $\varrho_{n+1} - \Omega_M(n)$ prove the claim. $\square$

The normalization constant and the full expansion. Set $\mathfrak{z}_n = \log a_n - n\lambda + \frac{3}{2} \log n$ for $n \geq 1$. Because $\omega_1/27 = -3/2$, (17.31) gives

$$\mathfrak{z}_{n+1} - \mathfrak{z}_n = \log \varrho_{n+1} - \lambda + \frac{3}{2} \log(1 + 1/n) = O(n^{-2}).$$

The increments are absolutely summable, so $\mathfrak{z}_n$ converges to a finite limit $\log C$; in particular $C > 0$, and neither its positivity nor its existence is inferred from a numerical fit. At any finite order, subtract a polynomial in $1/n$ from $\mathfrak{z}_n$, solving its coefficients successively from the increment identity (the leading contribution of $\ell_j n^{-j}$ to the increment is $-j \ell_j n^{-j-1}$; the ℓ_j are those of (77)); the remaining increment is as small as desired by Proposition 17.10, and summing its tail proves the error bound for $\mathfrak{z}_n$. Exponentiation gives:

Theorem 17.11 (Recurrence-only all-orders expansion; manuscript 07). *There is a constant $C > 0$ such that, for every fixed $M \geq 0$,*

$$a_n = C 27^n n^{-3/2} \left(\sum_{j=0}^M c_j n^{-j} + O_M(n^{-M-1}) \right), \quad c_0 = 1, \quad (17.32)$$

with every c_j rational; $c_1, \dots, c_4$ are the values in (5).

For an explicit algorithm, with $S_{27}(t) = \sum_j c_j t^j$ as in (74), the ratio identity is

$$\Omega(t) S_{27}(t) = 27(1+t)^{-3/2} S_{27}\left(\frac{t}{1+t}\right); \quad (17.33)$$

in the coefficient of t^{m+1} , the unknown c_m has coefficient $27m$ once everything is moved to the left. This is the triangular algorithm of the supporting code, a third generator beside (75) and (16.15).

The argument proves a Poincaré expansion: each finite truncation has its stated remainder. It does not assert convergence of the formal series, an optimal truncation theorem, or any determination of exponentially small sectors. Exponentially small terms are irrelevant to the fixed-order sign argument but may matter when the determinant order grows with n .

17.7 A general all-orders theorem for shifted Hankel determinants

Consider a real sequence (s_n) with, for every fixed M ,

$$s_n = C \Lambda^n n^{-\alpha} \left(\sum_{j=0}^M a_j n^{-j} + O_M(n^{-M-1}) \right), \quad C, \Lambda, \alpha > 0, \quad a_0 = 1. \quad (17.34)$$

The coefficients may be real, and no moment representation is assumed. Put $\mathcal{D}_r(n) = \det[s_{n+i+j}]_{0 \leq i, j < r}$ and

$$\mathcal{V}_r(\alpha) = \prod_{j=0}^{r-1} j! (\alpha)_j. \quad (17.35)$$

Theorem 17.12 (All orders, with an explicit first correction; manuscript 07). *For fixed $r \geq 1$ the determinant has a full expansion*

$$\mathcal{D}_r(n) = C^r \Lambda^{rn+r(r-1)} n^{-r\alpha-r(r-1)} \mathcal{V}_r(\alpha) \left(1 + \frac{\mathfrak{b}_{r,1}}{n} + \frac{\mathfrak{b}_{r,2}}{n^2} + \dots \right), \quad (17.36)$$

in the finite-truncation sense, with first correction

$$\mathfrak{b}_{r,1} = r(\alpha + r - 1) \left(\frac{\mathfrak{a}_1}{\alpha} - (r - 1) \right). \quad (17.37)$$

All later coefficients are given by the finite determinant generator (17.39). If α and all $\mathfrak{a}_j$ are rational, so are all $\mathfrak{b}_{r,j}$.

The general method is classical finite-difference and Vandermonde algebra and is not claimed as new; the explicit generator and the formula for $\mathfrak{b}_{r,1}$ are manuscript 07's presentation. (The report's neighbour hankel-determinants/growth-and-runs/apery-hankel-determinant-growth lets the matrix order N grow with the shift $r = sN$ (its notation), a different regime from Theorem 17.12, whose order is fixed.)

Finite differences remove the degeneracy. Substituting only the leading terms in the determinant is not legitimate: the leading normalized matrix has all entries equal to one and rank one, so the proof must resolve the cancellations. Put $t = 1/n$. After factoring $C^r \Lambda^{rn+r(r-1)} n^{-r\alpha}$ (the factor Λ^{i+j} of each entry being absorbed by the row and column scalings Λ^i, Λ^j), the formal normalized entries are $\mathcal{Z}_{i+j}(t)$, where

$$\mathcal{Z}_k(t) = \sum_{m \geq 0} \mathfrak{a}_m t^m (1 + kt)^{-\alpha-m}.$$

Take successive forward differences Δ in the row and in the column index; the corresponding triangular transformations have determinant one, and the (i, j) entry becomes $\Delta^{i+j} \mathcal{Z}_0(t)$. Factoring $(-t)^i$ from row i and $(-t)^j$ from column j extracts $t^{r(r-1)}$, because $r(r-1)$ is even. The resulting entry depends only on $s = i + j$ and has the explicit expansion

$$\mathcal{X}_s(t) = \sum_{h \geq 0} t^h \sum_{m=0}^h \mathfrak{a}_m (-1)^{h-m} \frac{s! (\alpha + m)_{s+h-m}}{(s + h - m)!} \begin{Bmatrix} s + h - m \\ s \end{Bmatrix}, \quad (17.38)$$

with Stirling numbers of the second kind, $\begin{Bmatrix} 0 \\ 0 \end{Bmatrix} = 1$ and $\begin{Bmatrix} k \\ 0 \end{Bmatrix} = 0$ for $k > 0$; it follows from $\Delta^s k^d|_{k=0} = s! \begin{Bmatrix} d \\ s \end{Bmatrix}$.

For $s = 0$ it gives $\mathcal{X}_0(t) = \sum_h \mathfrak{a}_h t^h$, as it should. At $t = 0$ the transformed matrix is $\mathbf{P} = [(\alpha)_{i+j}]_{0 \leq i, j < r}$, with determinant exactly $\mathcal{V}_r(\alpha)$: factor $(\alpha)_i$ from row i ; the remaining entries $(\alpha + i)_j$ are monic polynomials of degree j evaluated at $\alpha, \alpha + 1, \dots, \alpha + r - 1$, whose determinant is the Vandermonde product $\prod_{j < r} j!$. Since $\alpha > 0$ this leading determinant is strictly positive.

The complete coefficient generator and the remainder. The all-orders generator is the identity of formal power series (not an assertion of convergence)

$$\sum_{j \geq 0} \mathfrak{b}_{r,j} t^j = \frac{\det[\mathcal{X}_{i+j}(t)]_{0 \leq i, j < r}}{\mathcal{V}_r(\alpha)}. \quad (17.39)$$

The coefficient of t^M needs only $\mathbf{a}_0, \dots, \mathbf{a}_M$ and finitely many Stirling numbers. To justify the remainder, start from (17.34) truncated much further than the desired order, say with normalized remainder $O(t^{M'})$ where $M' \geq M + 2r$. There are only finitely many shifts $0, \dots, 2r - 2$, so the bound is uniform over them; a fixed finite difference preserves it, and dividing the (i, j) entry by t^{i+j} changes it to at worst $O(t^{M'-2r+2})$, smaller than the requested error. The constant matrix $\mathbf{P}$ is nonsingular, so expanding the finite determinant preserves the order. This proves (17.36), and shows why entrywise substitution without additional terms would have been insufficient.

The first correction. Let $\mathbf{N} = \text{diag}(0, 1, \dots, r-1)$ and $\mathbf{P}_1 = [(\alpha)_{i+j+1}]$. By (17.38), $[t] [\mathcal{X}_{i+j}(t)] = \frac{\mathbf{a}_1}{\alpha} \mathbf{P}_1 - \frac{1}{2} (\mathbf{N} \mathbf{P}_1 + \mathbf{P}_1 \mathbf{N})$, so the derivative of the determinant gives

$$\mathbf{b}_{r,1} = \frac{\mathbf{a}_1}{\alpha} \text{tr}(\mathbf{P}^{-1} \mathbf{P}_1) - \frac{1}{2} \text{tr}(\mathbf{P}^{-1} (\mathbf{N} \mathbf{P}_1 + \mathbf{P}_1 \mathbf{N})). \quad (17.40)$$

Since $\mathbf{P}_1 = \alpha \mathbf{P} + \mathbf{N} \mathbf{P} + \mathbf{P} \mathbf{N}$, $\mathcal{T} := \text{tr}(\mathbf{P}^{-1} \mathbf{P}_1) = r\alpha + 2 \text{tr} \mathbf{N} = r(\alpha + r - 1)$. The first $r - 1$ rows of $\mathbf{P}_1 \mathbf{P}^{-1}$ are $e_1^\top, \dots, e_{r-1}^\top$, because those rows of $\mathbf{P}_1$ are the next rows of $\mathbf{P}$; its first $r - 1$ diagonal entries vanish, so $\text{tr}(\mathbf{N} \mathbf{P}_1 \mathbf{P}^{-1}) = (r - 1) \mathcal{T}$, and symmetry of $\mathbf{P}, \mathbf{P}_1$ gives the same value for the other trace. Thus $\mathbf{b}_{r,1} = \mathcal{T}(\mathbf{a}_1/\alpha - (r - 1))$, which is (17.37).

Specialization to A279619. Theorem 2.1 (or Theorem 17.11) with $\Lambda = 27$, $\alpha = 3/2$ and $\mathbf{a}_j = c_j$ proves the expansion part of Theorem 17.3; then $\alpha_r = \frac{3}{2}r + r(r - 1)$, and (17.37) with $c_1 = -215/1008$ is (17.8). The first two corrections produced by (17.39) are

r	$\mathbf{b}_{r,1}$	$\mathbf{b}_{r,2}$
1	$-215/1008$	$-1265/290304$
2	$-8635/1512$	$31343875/1524096$
3	$-3239/144$	$246722173/870912$
4	$-4751/84$	$10336793/6048$
5	$-344465/3024$	$1529842985/225792$
6	$-101075/504$	$31599315995/1524096$

(for $r = 1$ they are c_1, c_2). For instance

$$D_2(n) = \frac{3}{2} C^2 27^{2n+2} n^{-5} \left(1 - \frac{8635}{1512n} + \frac{31343875}{1524096n^2} + O(n^{-3}) \right). \quad (17.41)$$

The data file `data/97-07-hankel-asymptotic-coefficients.json` gives the corrections through order four for $1 \leq r \leq 6$; the algorithm works to any fixed order. No general closed expression for $\mathbf{b}_{r,j}$ as a function of both indices is asserted beyond the generator.

17.8 Arbitrary gaps, eventual total positivity, and nonuniformity

Let $0 \leq g_0 < \dots < g_{r-1}$ and $0 \leq g'_0 < \dots < g'_{r-1}$ be fixed integer offsets, and $\text{Vand}(g) = \prod_{i < j} (g_j - g_i)$, similarly for g' .

Proposition 17.13 (Fixed gaps; manuscript 07). *Under (17.34),*

$$\det[\mathbf{s}_{n+g_i+g'_j}]_{0 \leq i,j < r} \sim C^r \Lambda^{rn + \sum_i g_i + \sum_j g'_j} n^{-r\alpha - r(r-1)} \text{Vand}(g) \text{Vand}(g') \prod_{j=0}^{r-1} \frac{(\alpha)_j}{j!}, \quad (17.42)$$

and the leading constant is strictly positive.

Proof (sketch, as given in manuscript 07). Expand the normalized entry $(1 + (g_i + g'_j)t)^{-\alpha}$ in monomials in g_i and g'_j : the coefficient of $g_i^p g_j'^q$ is $(-1)^{p+q}(\alpha)_{p+q} t^{p+q} / (p!q!)$. In the determinant, the smallest nonvanishing row and column exponent sets are both $\{0, \dots, r-1\}$; finite Cauchy–Binet expansion gives the two Vandermonde factors and the determinant of $[(\alpha)_{p+q} / (p!q!)]$, whose value is $\prod_j (\alpha)_j / j!$. Every correction $\mathbf{a}_m t^m$, $m > 0$, contributes only at higher total order, and taking the initial expansion to sufficiently high order controls the remainder as in Subsection 17.7. $\square$

For $g = g' = (0, \dots, r-1)$, (17.42) is the leading term of (17.36), since then $\text{Vand}(g)^2 \prod_j (\alpha)_j / j! = \mathcal{V}_r(\alpha)$.

The fixed-offset proposition alone would not justify a statement about all offsets simultaneously; the contiguous-minor criterion supplies that step.

Corollary 17.14 (Every fixed order is eventually restored; manuscript 07). *Under (17.34), for each fixed r there is an integer N_r^{TP} such that $(\mathbf{s}_{N+i+j})_{i,j \geq 0}$ is TP_r whenever $N \geq N_r^{\text{TP}}$.*

Proof. For each $1 \leq k \leq r$ the leading constant in Theorem 17.12 is positive, so $\mathcal{D}_k(n) > 0$ for all large n . Choose N_r^{TP} beyond these finitely many thresholds. Every contiguous minor of order at most r of the shifted matrix is one of these determinants with shift at least N_r^{TP} , and Fekete’s criterion, applied to finite rectangles, proves the assertion. $\square$

For A279619, $N_r^{\text{TP}} = 0$ for $r \leq 5$ (Theorem 17.1). No optimal threshold is proved for $r \geq 6$; the finite experiments of Subsection 17.11 suggest specific small values but are not substituted for proof.

Remark 17.15 (Why the hypothesis and the order of quantifiers matter; manuscript 07). A leading equivalent alone does not force even eventual log-convexity. For $0 < \varepsilon < 1$ and large n the positive sequence $\mathbf{s}_n = n^{-\alpha} (1 + \varepsilon(-1)^n/n)$ has $\mathbf{s}_n \sim n^{-\alpha}$, but $\mathbf{s}_n \mathbf{s}_{n+2} - \mathbf{s}_{n+1}^2 = n^{-2\alpha} (4\varepsilon(-1)^n/n + O(n^{-2}))$, so its adjacent 2×2 determinants change sign indefinitely; the all-orders hypothesis rules out this algebraically large oscillation. Nor does Corollary 17.14 interchange its quantifiers: it says “for every fixed r , eventually in N ”, not “one N for all r ”. Even the first relative correction for A279619 grows cubically, $\mathbf{b}_{r,1} = -r^3 + O(r^2)$, so the fixed- r expansion gives no uniform accuracy as r grows with n ; it suggests r^3/n as a scale worth investigating, but is not a proof of a transition law. In particular no fixed-tail Stieltjes representation follows from the asymptotic theorem. [Write note, 2026-10-04, batch 97G: Theorem 17.17 below shows that none exists: for every N , $\mathbf{H}^{(N)}$ fails to be totally nonnegative at some finite order, which must tend to infinity with N by Corollary 17.14.]

17.9 Inverse scales for Hankel determinant growth

[Write note, 2026-10-04, batch 97G: The construction of this subsection is the mechanism of Section 11 (and of Remark 11.3) applied to $D_r(n)$ instead of a_n , with λ replaced by $r\lambda$ and $\beta = 3/2$ by α_r . Its Lambert- W_{-1} core and triangular corrections are the instance $a = r\lambda$, $\beta_{\text{CTI}} = -\alpha_r$, $q_j^{\text{CTI}} = \ell_j^{[r]}$ of Theorem `t2:thm:balanced-inverse` of [16]: its d_1, d_2 are the two corrections in (17.44) (checked by hand at this write), and its rounding statement has the form of the separation condition of `p0:thm:staircase` of [17]. Manuscript 07 cites neither, and no novelty is claimed for the method.] This subsection concerns a smooth asymptotic envelope; a discrete sequence has no unique continuous inverse. For fixed r the large solution of $Y = C_r^{\text{H}} e^{r\lambda x} x^{-\alpha_r}$ is

$$x_0^{[r]}(Y) = -\frac{\alpha_r}{r\lambda} W_{-1}\left(-\frac{r\lambda}{\alpha_r} \left(\frac{C_r^{\text{H}}}{Y}\right)^{1/\alpha_r}\right), \quad (17.43)$$

with W_{-1} the real lower branch of Lambert's function; this follows by multiplying $xe^{-r\lambda x/\alpha_r} = (C_r^H/Y)^{1/\alpha_r}$ by $-r\lambda/\alpha_r$. Put $\ell_1^{[r]} = \mathbf{b}_{r,1}$ and $\ell_2^{[r]} = \mathbf{b}_{r,2} - \mathbf{b}_{r,1}^2/2$, the first two coefficients of $\log \sum_j \mathbf{b}_{r,j} t^j$ (the analogues of the ℓ_j of (77)). Substituting into the logarithm of (17.6) gives, with $x_0 = x_0^{[r]}(Y)$, the formal inverse corrections

$$x^{[r]}(Y) = x_0 - \frac{\ell_1^{[r]}}{r\lambda x_0} - \frac{1}{x_0^2} \left(\frac{\alpha_r \ell_1^{[r]}}{(r\lambda)^2} + \frac{\ell_2^{[r]}}{r\lambda} \right) + O_r(x_0^{-3}). \quad (17.44)$$

At each later order the new coefficient enters linearly with nonzero coefficient $r\lambda$, so the inverse expansion continues to every fixed order; for fixed r its coefficients lie in $\mathbb{Q}(\log 27)$ once the leading scale x_0 is separated. This is an algebraic-order inverse expansion; it does not resolve exponentially small inverse sectors or Stokes data.

A rounding statement with the correct safeguard. The ratio $D_r(n+1)/D_r(n)$ tends to $27^r > 1$ by Theorem 17.3, so there is $n_*^{[r]}$ beyond which these determinants are positive and increasing. Define $N^{[r]}(Y) = \min\{n \geq n_*^{[r]} : D_r(n) \geq Y\}$. For fixed $M \geq 0$ let $x_M^{[r]}(Y)$ be the large root of the continuous truncation $\mathcal{F}_M^{[r]}(x) = C_r^H e^{r\lambda x} x^{-\alpha_r} \sum_{j=0}^M \mathbf{b}_{r,j} x^{-j} = Y$, which for large Y exists uniquely on the increasing branch. There is a constant $K_M^{[r]} > 0$ such that

$$\lceil x_M^{[r]} - K_M^{[r]} (x_M^{[r]})^{-M-1} \rceil \leq N^{[r]}(Y) \leq \lceil x_M^{[r]} + K_M^{[r]} (x_M^{[r]})^{-M-1} \rceil. \quad (17.45)$$

Indeed, the relative error in $D_r(n)/\mathcal{F}_M^{[r]}(n)$ is $O(n^{-M-1})$, while $(\log \mathcal{F}_M^{[r]})'(x) = r\lambda + O(1/x)$ is bounded away from zero; a displacement of the stated order therefore dominates the logarithmic error, and monotonicity gives the two integer bounds. [Write note, 2026-10-04, batch 97G: As in Proposition 11.2, the brackets hold for all sufficiently large Y , a qualifier manuscript 07 leaves implicit.] The constants are existential, not an explicit numerical stopping rule; a finite asymptotic truncation cannot be rounded with an unconditional single ceiling when its error interval straddles an integer.

17.10 The singularity at $z = -1$ and the absence of a Stieltjes tail (added at this write)

[Write note, 2026-10-04, batch 97G: This subsection is new at this write; manuscript 07 does not contain it. Its numerical starting point is the batch-97 intake's observation, by integrating (17) from $z = -0.02$ to $z = -1 + 10^{-6}$ at 40 digits, that $\sqrt{1+z} A'(z)$ approaches about -0.0975 ; the proof below makes that rigorous and does not use the integration. Dated note (5 October 2026): an independent adversarial check of Propositions 17.16 and 17.19, Theorem 17.17 and Corollary 17.18, made by the intake after this write, found no gap: it recomputed B_- by ODE integration matched to the two local solutions at $z = -1$ (agreement with the ${}_2F_1$ form to about 10^{-61}), the alternating-series bounds in exact fractions, $D_6(1)$ and the finite data behind Corollary 17.18. It is a careful reading, not a formal verification or an external review.] The analytic continuation of A along the segment $(-1, 0]$ exists by Section 3 (the only singular points of (17) are $0, \rho, -1$). Its exponents at $z = -1$ are 0 and $\frac{1}{2}$, so near -1 it has the form $U_- + B_- \sqrt{1+z} V_-$ with U_-, V_- analytic; the question is whether $B_- \neq 0$. Let $F(x) = {}_2F_1(\frac{1}{12}, \frac{5}{12}; 1; x)$ and P, J be as in (22).

Proposition 17.16 (The connection coefficient at -1 ; this write). *For $-1 < z \leq 0$ put $\check{w} = \sqrt{(1+z)/(1-27z)} \in (0, 1]$. Then the continuation of A along $(-1, 0]$ is*

$$A(z) = P(1/\check{w}) F(J(1/\check{w})), \quad (17.46)$$

where

$$P(1/\check{w}) = \left(\frac{7(1+27\check{w}^2)}{15+96\check{w}+85\check{w}^2} \right)^{1/4}, \quad J(1/\check{w}) = \frac{1728(\check{w}-1)^7(\check{w}+1)}{(1+27\check{w}^2)(15+96\check{w}+85\check{w}^2)^3}.$$

Consequently, with $s_- = 1 + z$,

$$A(z) = U_-(s_-) + B_- \sqrt{s_-} V_-(s_-), \quad V_-(0) = 1, \quad (17.47)$$

where U_- , V_- are analytic at 0,

$$U_-(0) = \left(\frac{7}{15}\right)^{1/4} F\left(-\frac{64}{125}\right), \quad B_- = \frac{4}{5\sqrt{7}} \left(\frac{7}{15}\right)^{1/4} \left(\frac{1008}{125} F'\left(-\frac{64}{125}\right) - F\left(-\frac{64}{125}\right)\right), \quad (17.48)$$

and $-0.2075 < B_- < -0.1754$. In particular A is singular at $z = -1$: its continuation has $A'(z) \rightarrow -\infty$ as $z \downarrow -1$.

Proof. In the variable $w = 1/\check{w} = \sqrt{(1-27z)/(1+z)}$ of (21), the segment $z \in (-1, 0]$ is $w \in [1, \infty)$, traversed by the real-analytic diffeomorphism $w \mapsto z = (1-w^2)/(27+w^2)$. For $w \geq 1$, $T(w) > 0$, so $P(w)$ is positive and analytic, and $J(w) \leq 0$ because $(1-w)^7 \leq 0$; as F is analytic on $\mathbb{C} \setminus [1, \infty)$, the function $z \mapsto P(w(z))F(J(w(z)))$ is analytic on a complex neighbourhood of $(-1, 0]$ (there $w(z)$ is analytic, its radicand being positive). By Proposition 4.1 it equals A near $z = 0$ (germs at $w = 1$), so by the identity theorem it is the continuation of A along $(-1, 0]$. Writing it in $\check{w} = 1/w$ gives (17.46).

The function $G(\check{w}) = P(1/\check{w})F(J(1/\check{w}))$ is analytic on a neighbourhood of $[0, 1]$: the displayed rational expressions are regular there, $15 + 96\check{w} + 85\check{w}^2 > 0$, and $J(1/\check{w}) \leq 0$ for $0 \leq \check{w} \leq 1$. Write $G(\check{w}) = G_e(\check{w}^2) + \check{w} G_o(\check{w}^2)$ with G_e, G_o analytic at 0. Since $1 - 27z = 28 - 27s_-$, we have $\check{w}^2 = s_-/(28 - 27s_-)$, analytic at $s_- = 0$, and $\check{w} = \sqrt{s_-} (28 - 27s_-)^{-1/2}$. Hence (17.47) holds with $U_-(s_-) = G_e(s_-/(28 - 27s_-))$ and $B_- = G_o(0)/\sqrt{28} = G'(0)/(2\sqrt{7})$, provided $G'(0) \neq 0$, which is shown next. At $\check{w} = 0$: $P(\infty) = (7/15)^{1/4}$ and $J(\infty) = -1728/3375 = -64/125$; logarithmic differentiation gives $(\log P(1/\check{w}))'|_{\check{w}=0} = \frac{1}{4} \cdot (-\frac{96}{15}) = -\frac{8}{5}$ and $(\log |J(1/\check{w})|)'|_{\check{w}=0} = -7 + 1 - \frac{3 \cdot 96}{15} = -\frac{126}{5}$, so $\frac{d}{d\check{w}} J(1/\check{w})|_{\check{w}=0} = \frac{8064}{625}$. Therefore

$$G'(0) = \left(\frac{7}{15}\right)^{1/4} \left(\frac{8064}{625} F'\left(-\frac{64}{125}\right) - \frac{8}{5} F\left(-\frac{64}{125}\right)\right),$$

which gives (17.48). For the sign, let $-1 < x < 0$. The series $F(x) = \sum_k h_k x^k$, $h_k = (\frac{1}{12})_k (\frac{5}{12})_k / k!^2$, alternates with terms decreasing in modulus, since $h_{k+1}/h_k = (k + \frac{1}{12})(k + \frac{5}{12})/(k+1)^2 < 1$; hence $1 - \frac{5}{144}|x| \leq F(x) \leq 1$. Likewise $F'(x) = \frac{5}{144} {}_2F_1(\frac{13}{12}, \frac{17}{12}; 2; x)$, and the last series alternates with decreasing terms because $(k + \frac{13}{12})(k + \frac{17}{12}) < (k+1)(k+2)$; hence $\frac{5}{144}(1 - \frac{221}{288}|x|) \leq F'(x) \leq \frac{5}{144}$. At $x = -\frac{64}{125}$ these give $\frac{221}{225} \leq F \leq 1$ and $\frac{5}{144} \cdot \frac{683}{1125} \leq F' \leq \frac{5}{144}$, so

$$-\frac{23344}{28125} \leq \frac{1008}{125} F'\left(-\frac{64}{125}\right) - F\left(-\frac{64}{125}\right) \leq \frac{7}{25} - \frac{221}{225} = -\frac{158}{225}.$$

Multiplying by $\frac{4}{5\sqrt{7}} (\frac{7}{15})^{1/4} = 0.24991 \dots$ gives the stated enclosure of B_- . Finally $A'(z) = \frac{1}{2} B_- s_-^{-1/2} + O(1)$ as $s_- \downarrow 0$. $\square$

Numerically (mpmath, 50 digits)

$$U_-(0) = 0.81408739831171111286 \dots, \quad B_- = -0.19550105237098325058 \dots,$$

in agreement with the intake's ODE integration ($\frac{1}{2} B_- = -0.09775 \dots$). The closed forms these values appear to have are recorded as a question in Subsection 17.13. The number $-B_-/(2\sqrt{\pi}) = 0.05515 \dots$ is the coefficient that the transfer (73) would attach to $(-1)^n n^{-3/2}$; whether it has a canonical meaning for a_n itself, whose dominant expansion is only asymptotic, is the open question of Section 13 on exponentially improved asymptotics, and nothing is claimed about it here.

Theorem 17.17 (No tail is a Stieltjes moment sequence; this write). *For no $N \geq 0$ is $(a_{N+n})_{n \geq 0}$ a Stieltjes moment sequence. Equivalently, by Stieltjes' theorem [26, Chapter 3], for every N at least one of $\mathbf{H}^{(N)}$, $\mathbf{H}^{(N+1)}$ is not positive semidefinite. In particular no $\mathbf{H}^{(N)}$ is totally nonnegative.*

Proof. Suppose $a_{N+n} = \int_{[0, \infty)} x^n d\mu(x)$ for all $n \geq 0$, with μ positive. Since $a_n \leq 27^n$ (Section 3), for every $\delta > 0$, $\mu((27 + \delta, \infty)) \leq 27^{N+n}(27 + \delta)^{-n} \rightarrow 0$, so μ is supported in $[0, 27]$. For $|z| < \rho$, dominated convergence gives

$$z^{-N} \left(A(z) - \sum_{k < N} a_k z^k \right) = \sum_{n \geq 0} a_{N+n} z^n = \int_{[0, 27]} \frac{d\mu(x)}{1 - xz}.$$

The right side is analytic on $\mathbb{C} \setminus [\rho, \infty)$, a connected set containing $[-1, 0]$. Hence the continuation of A along $(-1, 0]$ is the restriction of a function analytic at $z = -1$, and $A'(z)$ stays bounded as $z \downarrow -1$, contradicting Proposition 17.16. For the equivalent form: a Stieltjes moment sequence has both Hankel forms positive semidefinite (integrate $v(x)^2$ and $xv(x)^2$), and Stieltjes' theorem is the converse. If $\mathbf{H}^{(N)}$ were totally nonnegative, its principal minors and those of its submatrix $\mathbf{H}^{(N+1)}$ (rows from 1, columns from 0) would be nonnegative, so both would be positive semidefinite. $\square$

Corollary 17.18 (The order of first failure; this write). *Let $r^*(N)$ be the least r such that $\mathbf{H}^{(N)}$ is not TN_r . Then $r^*(N) < \infty$ for every $N \geq 0$, $r^*(0) = 6$, and $r^*(N) \rightarrow \infty$ as $N \rightarrow \infty$.*

Proof. Finiteness is Theorem 17.17, $r^*(0) = 6$ is Theorem 17.1, and $r^*(N) > r$ for $N \geq N_r^{\text{TP}}$ by Corollary 17.14. $\square$

The finite data of Subsection 17.11 fit this picture: the first negative contiguous minor $D_r(N)$ occurs at odd shifts $N = 1, 3, 5, 7, 9, 11$ for $r = 6, 9, 12, 15, 17, 20$, while all $D_r(N)$ with $r \leq 60$ are positive at even $N \leq 12$.

Proposition 17.19 (Where a Hamburger measure could live; this write). *If $a_n = \int_{\mathbb{R}} x^n d\mu(x)$ for a positive measure μ , then μ is unique, $\text{supp } \mu \subset [-1, 27]$, $\mu(\{-1\}) = 0$, and $-1 \in \text{supp } \mu$.*

Proof. From $a_{2n} \leq 27^{2n}$, as above, $\text{supp } \mu \subset [-27, 27]$; compact support makes the moment problem determinate, so μ is unique. The function $S_\mu(z) = \int d\mu(x)/(1 - xz)$ is analytic off $(-\infty, -\rho] \cup [\rho, \infty)$ and equals A for $|z| < \rho$. The ODE (17) has no singular point in the simply connected domain $\mathcal{U}_- = \mathbb{C} \setminus ((-\infty, -1] \cup [\rho, \infty))$ other than $z = 0$, where A is analytic; so A continues to a single-valued analytic function on $\mathcal{U}_-$, and $S_\mu = A$ on $\mathbb{C} \setminus ((-\infty, -\rho] \cup [\rho, \infty))$. In the variable $\varsigma = 1/z$, $S_\mu(1/\varsigma)/\varsigma = \int d\mu(x)/(\varsigma - x)$ is the Cauchy transform of μ , and it extends analytically across $\varsigma \in (-27, -1)$ (where $1/\varsigma \in (-1, -\rho) \subset \mathcal{U}_-$) with equal boundary values from both sides; by Stieltjes–Perron inversion μ vanishes on $(-27, -1)$. An atom at -27 or at -1 would make the Cauchy transform unbounded as $\varsigma \rightarrow -27$ or $\varsigma \rightarrow -1$ along the imaginary direction, whereas A is analytic at $z = -\rho$ and bounded near $z = -1$ by (17.47). So $\text{supp } \mu \subset [-1, 27]$ and $\mu(\{-1\}) = 0$. If -1 were not in the support, then $\text{supp } \mu \subset [-1 + \delta, 27]$ for some $\delta > 0$, S_μ would be analytic near $z = -1$, and the argument of Theorem 17.17 would contradict Proposition 17.16. $\square$

Proposition 17.19 does not assert that such a measure exists. It makes manuscript 07's suggested route concrete: the only candidate is the measure whose Cauchy transform is $A(1/\varsigma)/\varsigma$, determined by the jumps of A across $(-\infty, -1]$ and $[\rho, \infty)$, and the question is whether that candidate is positive. (The jumps cannot see an atom at $x = 0$, which corresponds to $z = \infty$; such an atom is excluded because the exponents of (17) at $z = \infty$ are $1/3$ and $2/3$, so $A(z) = O(|z|^{-1/3})$ there.)

17.11 Reproducibility, computations, and proof boundaries

The shipped programs. File names here are the shipped ones (the delivered names are listed in the report's README.md). Three programs accompany manuscript 07:

code/97-07-hankel-verify_certificates.py

Python standard library only. Checks all positive coefficient lists, the bounded-degree polynomial identities, the ratio base case, the twenty initial determinants, the negative minor, the polynomial moment witness and the exact repair; any failed equality or inequality terminates with an error. Its recorded output is data/97-07-hankel-verification_log.txt.

code/97-07-hankel-generate_certificates.py

Exact SymPy arithmetic regenerating the 21 polynomials and their positive-coefficient representation; not imported by the checker.

code/97-07-hankel-asymptotics_and_experiments.py

Generates the sequence and the Hankel corrections through order four, the exact finite determinant tables, and high-precision comparisons with the asymptotic formulas; these comparisons are illustrations, not proofs of sign. Its standard output is data/97-07-hankel-experiment_log.txt.

No script uses network access. The tested versions are data/97-07-hankel-requirements.txt (SymPy 1.14.0, mpmath 1.3.0, the same pins as data/58-gamma-requirements.txt). Each record of data/97-07-hankel-certificates.json has fields degree, shift, scale and coefficients and represents $\text{scale} \sum_j \text{coefficients}[j] (n-5)^j$, the scale a positive rational stored as an exact fraction and every coefficient a positive decimal integer; all coefficient vectors, not only hashes, are included. data/97-07-hankel-moment_obstruction.json holds the integer coefficients of $\mathcal{W}$, the value (17.15), the actual a_{11} , and the exact minimum and repair; data/97-07-hankel-asymptotic_coefficients.json records $c_0, \dots, c_4$ and $b_{r,0}, \dots, b_{r,4}$ for $1 \leq r \leq 6$; the two CSV files hold exact small determinants and the signed asymptotic errors; data/97-07-hankel-finite_sign_experiment.json records its tested range so that it cannot be mistaken for a universal sign theorem. The mathematical certificate is the combination of the complete coefficient data, the exact defining formulas, the degree bounds and the replay program; manuscript 07's checksum manifest was an integrity aid only and is not shipped.

Finite sign data beyond the proved global order. Using exact Dodgson condensation,

$$D_{r+1}(n)D_{r-1}(n+2) = D_r(n)D_r(n+2) - D_r(n+1)^2, \quad (17.49)$$

the supporting experiment computed determinants through order fifteen, with all divisions exact and all divisors nonzero. For $0 \leq n \leq 200$ the negative shifts were:

Orders r	Shifts n with $D_r(n) < 0$ in the tested range
2, 3, 4, 5	none
6, 7, 8	1
9, 10, 11	1, 3
12, 13, 14	1, 3, 5
15	3, 5, 7

These are finite exact data, not universal theorems; $D_{15}(1) > 0$ warns against an overly simple monotone picture in r . The order-five result does not rely on this experiment. [Write note, 2026-10-04, batch 97G:

The intake reproduced the table exactly by fraction-free (Bareiss) elimination instead of condensation, and extended it: $D_r(0) > 0$ for every $r \leq 120$; for $r \leq 60$, $D_r(N) > 0$ at every even $N \leq 12$, while the first nonpositive $D_r(N)$ at $N = 1, 3, 5, 7, 9, 11$ has $r = 6, 9, 12, 15, 17, 20$. This was rerun at this write. Finite data only.]

Numerical scale of the first corrections. Signed relative errors $(\text{approximation} - D_r(n))/D_r(n)$, with the exact determinant computed first and compared in 100-digit arithmetic; the leading constant uses the value of C from Theorem 2.1:

r	n	Leading term	Through n^{-1}	Through n^{-2}
2	100	$5.8326 \cdot 10^{-2}$	$-2.1144 \cdot 10^{-3}$	$6.2103 \cdot 10^{-5}$
2	1000	$5.7230 \cdot 10^{-3}$	$-2.0623 \cdot 10^{-5}$	$6.0402 \cdot 10^{-8}$
3	100	$2.4852 \cdot 10^{-1}$	$-3.2311 \cdot 10^{-2}$	$3.0587 \cdot 10^{-3}$
3	1000	$2.2717 \cdot 10^{-2}$	$-2.8705 \cdot 10^{-4}$	$2.6796 \cdot 10^{-6}$
5	100	2.0385	-1.4227	0.63607
5	1000	0.12033	$-7.2867 \cdot 10^{-3}$	$3.0403 \cdot 10^{-4}$

The order-five row at $n = 100$ is kept on purpose: it shows why a formal large- n expansion must not be treated as a small- n sign certificate. The Bernstein proof is valid there regardless of the quality of the truncation.

Trust and formalization boundary. The elementary ratio proof and the general asymptotic arguments are conventional mathematical proofs. The global TP₅ assertion also uses finite exact certificate arithmetic; the independent checker reduces trust in symbolic simplification to integer operations, fractions, determinant elimination, and polynomial uniqueness with the printed degree bounds. This is not a Lean proof and not a claim of proof-assistant verification. The statement depends on no numerical heuristic, but the executable proof relies on ordinary software execution and on the correctness of the accompanying reduction.

17.12 Scope and literature audit of manuscript 07

This report posed the positivity problem addressed here and supplied the gamma value and the rational corrections; those are not claimed as new by manuscript 07. It inspected the OEIS entry [1], on October 4, 2026, and this report’s text and research questions, and searched for A279619 together with “Hankel”, “positivity”, “total positive” and “log-convexity”, and for asymptotic positivity of polynomial-recursive sequences. No inspected source stated the sharp TP₅/not-TN₆ classification or the exact twelfth-moment repair. This is a limited novelty audit, not a proof that no equivalent result exists. The strict contiguous-minor criterion is classical [23]; asymptotic approaches to log-convexity of polynomial-recursive and Apéry-like sequences are established topics [24, 25]; the finite-difference derivation is given in full so that its use needs no priority claim for the general method. The principal new items are the sequence-specific sharp order, the optimal truncated-moment obstruction, the explicit global certificates, and the associated Hankel expansions and inverse scales. Manuscript 07 settles neither the Hamburger problem, nor the existence of a totally positive tail (settled negatively at this write, Theorem 17.17), nor a uniform growing-order asymptotic, nor the OEIS entry’s congruence questions. It made no OEIS edit and no repository change. Its delivery included a draft OEIS comment (shipped as `data/97-07-hankel-OEIS_NOTE.txt`, headed “UNSUBMITTED DRAFT — REQUIRES REVIEW BEFORE ANY OEIS EDIT”); it has not been submitted, and this report does not submit it.

17.13 Further questions and research

Items 1–10 are manuscript 07’s ten questions, each with what this write adds; item 11 is new at this write, and item 12 turns manuscript 07’s restriction of its finite sign data into a question. They are separated from the proved results above.

1. A positive Hamburger measure (manuscript 07). Is there a positive measure on $\mathbb{R}$ with moments a_n , that is, are the unshifted Hankel matrices positive semidefinite in every size? The order-six shifted obstruction does not answer this. Manuscript 07 suggests identifying the Cauchy transform $z^{-1}A(1/z)$ and proving its sign in the upper half-plane, rather than inferring positivity from real Taylor coefficients, and checking any proposed density globally, not only near the dominant endpoint. Proposition 17.19 narrows the question: the measure, if it exists, is unique, lives on $[-1, 27]$ and charges every neighbourhood of -1 , and it is the Stieltjes–Perron inverse of the jumps of A on its two cuts; what is missing is a proof that this inverse is positive. Evidence (finite, not proof): $D_r(0) > 0$ for every $r \leq 120$, and $D_r(N) > 0$ for $r \leq 60$ at every even $N \leq 12$, as positive semidefiniteness of $\int x^{2k} x^{i+j} d\mu$ would require, while odd shifts fail, as a measure with mass on $[-1, 0)$ allows.

2. Optimal tail thresholds (manuscript 07). For $r \geq 6$ determine the least N_r^{TP} . Manuscript 07 reads its finite data as suggesting $N_6^{\text{TP}} = 2$ and the same value for orders seven and eight. Read with the same definition, the table of Subsection 17.11 suggests $N_r^{\text{TP}} = 4$ for $r = 9, 10, 11$, 6 for $r = 12, 13, 14$, and at least 8 for $r = 15$ (data only; contiguous minors with $n \leq 200$). These are targets for extending the rational ratio barriers and Bernstein certificates; an asymptotic expansion alone gives existence, not the optimal integer.

3. Can a fixed tail be a Stieltjes moment sequence? (manuscript 07; answered). Manuscript 07 asked whether some $(a_{N+n})_{n \geq 0}$ is a Stieltjes moment sequence, noting that this is strictly stronger than eventual TP_r . Theorem 17.17 answers it negatively. Still open: the exact order of first failure $r^*(N)$ of Corollary 17.18, its growth rate, and whether the failing minors can always be taken contiguous at odd shifts, as the data suggest.

4. The two-parameter sign boundary (manuscript 07). Study $D_r(n)$ when both r and n grow. The cubic first correction suggests testing regimes such as $r^3/n = O(1)$, but a uniform expansion needs estimates beyond fixed-order asymptotics. The change of sign of $D_r(1)$ as r increases also suggests studying the inertia, not only the determinant, of shifted Hankel blocks.

5. Smaller symbolic certificates (manuscript 07). Can the TP_5 proof be shortened to elementary rational inequalities with low-degree numerators? The width- $O(n^{-11})$ ratio interval was chosen for a clean coefficientwise certificate, not proved optimal: one could lower the truncation order, move the centre, use a rational rather than polynomial approximation, or subdivide the ratio interval.

6. Structure of the moment witness (manuscript 07). The polynomial $\mathcal{W}$ comes from a finite moment problem. Is it the first member of a natural family of orthogonal or quasi-orthogonal polynomials tied to the differential equation of A ? Do its coefficients and $a_{11}^{\min}$ have a conceptual expression? The finite spectral construction is a starting point but does not explain why the first failure occurs at five positive nodes.

7. Nearby recurrences and positivity phase diagrams (manuscript 07). Vary the coefficients of $(n+1)^2 u_{n+1} = (an^2 + bn + c)u_n + (dn^2 + en + f)u_{n-1}$ with positive initial data and identify regions with

a prescribed largest global TP_r order; the objective is a rigorous phase diagram rather than a collection of examples. Manuscript 07 asserts in passing that “the strict certificate inequalities persist under small perturbations at fixed normalization when appropriately controlled”. That sentence is unproved as stated: it gives no norm, no control and no argument. What is missing is a quantitative statement of how the barrier polynomials, the 1,090 coefficients and the twenty base-case determinants depend on the recurrence coefficients (the barriers themselves are built from the recurrence’s own ratio expansion), together with the observation that the asymptotic exponent and the best barriers change with the coefficients.

8. Convolution powers and the square sequence (manuscript 07). Determine the Hankel positivity order of the coefficients of A^m for positive integers m (for $m = 2$, A183204), or of suitable real powers. Squaring a generating function convolves coefficients and does not square a moment sequence pointwise, so a preservation theorem must be proved, not assumed. Theorem 10.1 and Remark 10.2 supply the fixed-power asymptotic input for Theorem 17.12.

9. Closed correction formulas and inverse bounds (manuscript 07). Find explicit expressions for $b_{r,j}$ as functions of r for more j , identify denominator patterns, and obtain rigorous remainder constants; these would turn the existential brackets (17.45) into certified algorithms (compare Section 13.1 for a_n).

10. A proof-assistant certificate pipeline (manuscript 07). Formalize the recurrence, the ratio induction, the positive-coefficient criterion on $[0, \infty)$, Bernstein positivity on $[0, 1]$ and exact polynomial interpolation, with a separate library theorem for the finite-order Fekete reduction. A formalization could import the positive coefficient vectors as integer data and prove (17.29) by exact interpolation. The target is a small reusable checker for other OEIS recurrences, not a transcription of one large coefficient list. (Compare Section 13.12.)

11. Closed forms at $z = -1$ (this write). Numerically, to 200 digits and with an unrelated canary constant receiving coefficient zero in an integer-relation search,

$$U_-(0) = \frac{\mathcal{G}_7}{4\pi^2} = \frac{2}{3}A_\star, \quad B_- = -\frac{2\pi}{\mathcal{G}_7} = \frac{B}{\sqrt{3}}, \quad U_-(0)B_- = -\frac{1}{2\pi},$$

the analogues of $A_\star = 3\mathcal{G}_7/(8\pi^2)$, $B = -2\sqrt{3}\pi/\mathcal{G}_7$ and $BA_\star = -3\sqrt{3}/(4\pi)$ at ρ ; equivalently $F(-\frac{64}{125}) = (\frac{15}{7})^{1/4}\mathcal{G}_7/(4\pi^2)$, where $-\frac{64}{125} = 1728/j$ with $j = -3375$ the j -invariant of discriminant -7 . A suggested route, not carried out: the point $\tau_- = \frac{1}{2} + \frac{i}{2\sqrt{7}}$ satisfies, numerically, $w_7(\tau_-) = -\frac{1}{7}$, hence $X_7(\tau_-) = -1$ by (40) (where $X_7 = -1$ is the double root $w_7 = -\frac{1}{7}$), and $z_7(\tau_-) = \mathcal{G}_7/(4\pi^2)$; the computation of Section 16.2 at τ_- , with the branch continued along $\Re \tau = \frac{1}{2}$, should give both values. Proving them would add the point -1 to the connection data asked for in question R5 of Section 13.13.

12. Finite sign data (manuscript 07). The sign table of Subsection 17.11 covers $r \leq 15$ and $0 \leq n \leq 200$ only; nothing outside that range is asserted. A proof for $r = 6$ and all $n \geq 2$ (which would give $N_6^{TP} = 2$, given $D_k(n) > 0$ for $k \leq 5$) is the natural first target for item 2.

17.14 All initial determinants needed in the global certificate

The $r = 1$ cases are positive by the recurrence. The twenty remaining base values for $0 \leq n \leq 4$, all regenerated by the independent exact checker:

r	n	$D_r(n)$	r	n	$D_r(n)$
2	0	18	4	0	504704088
2	1	188	4	1	132810611472
2	2	19236	4	2	1249180621287984
2	3	3356892	4	3	19147848994682842896
2	4	813648528	4	4	578868348472601428779120
3	0	14132	5	0	839827568468592
3	1	777000	5	1	780401205934374816
3	2	729706824	5	2	99981705616145401812192
3	3	1259077127616	5	3	12149812258132224263957487360
3	4	3565362281719392	5	4	3730588839202741103021612486642400

A Checking the classical period normalization

The table in Zucker's singular-value notes [8, Table 1] uses $K[N] = K(k_N)$ with $K(k'_N)/K(k_N) = \sqrt{N}$. At $N = 7$ it gives

$$K[7] = \frac{2^{5/7} 7^{3/4}}{14} \frac{\beta(1/7)\beta(2/7)}{\beta(1/14)}, \quad k_7^2 = \frac{8 - 3\sqrt{7}}{16}, \quad (103)$$

where $\beta(t) = \Gamma(t)^2/\Gamma(2t)$. We spell out the simplification because confusing the elliptic modulus with its square would give an incorrect constant.

Set $g_1 = \Gamma(1/7)$, $g_2 = \Gamma(2/7)$, $g_4 = \Gamma(4/7)$, and $g_{1/2} = \Gamma(1/14)$. Then

$$\frac{\beta(1/7)\beta(2/7)}{\beta(1/14)} = \frac{g_1^3 g_2}{g_4 g_{1/2}^2}.$$

The gamma duplication formula at $1/14$ is

$$g_{1/2} g_4 = 2^{6/7} \sqrt{\pi} g_1.$$

It follows that the beta quotient is $g_1 g_2 g_4 / (2^{12/7} \pi) = \mathcal{G}_7 / (2^{12/7} \pi)$. Substitution into (103) gives

$$K[7] = \frac{7^{3/4} \mathcal{G}_7}{28\pi} = \frac{\mathcal{G}_7}{4\pi 7^{1/4}},$$

which is exactly (34). The classical table evaluation is an external input; the algebraic conversion here checks that its normalization matches the one used in the proof.

B Two derivations from manuscript 56

B.1 The recurrence-to-ODE identity

For completeness, write $\vartheta = z d/dz$. Then

$$\sum_{n \geq 0} n a_n z^n = \vartheta A, \quad \sum_{n \geq 0} n^2 a_n z^n = \vartheta^2 A.$$

The left side of (1), summed over $n \geq 0$, is $A' + zA''$. The first term on the right gives $26z^2 A'' + 39zA' + 2A$, and the shifted term gives $27z^3 A'' + 54z^2 A' + 6zA$. Their difference is exactly (17).

B.2 A compact derivation of the coefficient algorithm

Assume

$$a_n = C27^n n^{-3/2} \mathcal{A}_n, \quad \mathcal{A}_n \sim S_{27}(1/n).$$

Divide (1) by $C27^n n^{1/2}$ and put $t = 1/n$. The three terms become

$$27(1+t)^{1/2} S_{27}\left(\frac{t}{1+t}\right), \quad (26+13t+2t^2)S_{27}(t), \\ \left(1-t+\frac{2}{9}t^2\right)(1-t)^{-3/2} S_{27}\left(\frac{t}{1-t}\right),$$

which is $\mathcal{E}_{27}[S_{27}] = 0$ in (74). The coefficient of t^2 is $-215/36 - 28c_1$, giving $c_1 = -215/1008$; successive coefficients proceed triangularly: after the automatic cancellations in degrees 0 and 1, each next coefficient determines one new c_j linearly. This proves that all c_j are rational and supplies a fast exact algorithm.

C Possible OEIS formula updates

Both manuscripts propose an update; neither was submitted. The following is suggested mathematical content for an update after independent review. No edit to OEIS has been made as part of this work.

C.1 Manuscript 58's proposal

Let $G = \Gamma(1/7)\Gamma(2/7)\Gamma(4/7)$. Then

$$a(n) \sim \frac{\sqrt{3\pi}}{G} \frac{27^n}{n^{3/2}}, \quad \sum_{n \geq 0} \frac{a(n)}{27^n} = \frac{3G}{8\pi^2}.$$

More precisely, for every fixed truncation order there is an asymptotic expansion with a remainder of the next order. Its first corrections are

$$1 - \frac{215}{1008n} - \frac{1265}{290304n^2} + \frac{4683055}{877879296n^3} + O(n^{-4}).$$

The first three corrections were already calculated formally by O'Brien [3].

The separate Lucas-congruence statement is not affected by this report.

C.2 Manuscript 56's proposal

A compact statement suitable for adaptation to the OEIS entry after independent review (its plain-text form is `data/56-exact-OEIS_update.txt`):

Let $\mathcal{G}_7 = \Gamma(1/7)\Gamma(2/7)\Gamma(4/7)$. Then

$$a(n) \sim \frac{\sqrt{3\pi}}{\mathcal{G}_7} 27^n n^{-3/2}.$$

More precisely,

$$a(n) = \frac{\sqrt{3}\pi}{\mathcal{G}_7} 27^n n^{-3/2} \left(1 - \frac{215}{1008n} - \frac{1265}{290304n^2} + \frac{4683055}{877879296n^3} + O(n^{-4}) \right).$$

For the generating function $A(x) = \sum_{n \geq 0} a(n)x^n$,

$$A(1/27) = \frac{3\mathcal{G}_7}{8\pi^2}, \quad A(1/125) = \frac{5\mathcal{G}_7}{16\pi^2}.$$

At the dominant singularity,

$$A(x) = \frac{3\mathcal{G}_7}{8\pi^2} - \frac{2\pi\sqrt{3}}{\mathcal{G}_7} \sqrt{1-27x} + O(1-27x).$$

[Merge note, batch 7302: The first three corrections were calculated formally by O'Brien (2016 thesis, §8.4) [3]. Manuscript 56's proposal and its plain-text file do not say so; any submission must add this credit.]

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